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AI VIET NAM – AI COURSE 2025

Cloud Practitioner Quiz - Week 4

Lớp AWS (AIO 2025)

1. Which AWS service is specifically designed to automatically scan and evaluate security vulnerabilities based on CVE databases and check network reachability?
   A. Amazon GuardDuty
   B. Amazon Inspector
   C. AWS Shield
   D. AWS WAF
   Answer: B

2. To implement a strong identity security posture, which IAM feature requires users to provide an additional authentication factor from an external device (e.g., a code from a phone) after entering their password?
   A. Permission Boundaries
   B. Active Directory Federation
   C. IAM Roles
   D. Multi-Factor Authentication (MFA)
   Answer: D

3. A financial company stores a large amount of customer records on Amazon S3. The security team is concerned about accidentally exposing Personally Identifiable Information (PII) such as names, emails, and credit card numbers. They need a service to automatically scan S3 buckets to alert them of these exposure risks. Which AWS service is most appropriate?
   A. Amazon GuardDuty
   B. AWS Key Management Service (KMS)
   C. Amazon Macie
   D. AWS Security Hub
   Answer: C

4. An e-commerce web application is experiencing attacks using SQL injection and Cross-Site Scripting (XSS) methods. The IT team needs to block these malicious requests before they reach the application servers. Which AWS solution is best suited to solve this problem?
   A. AWS Shield Advanced
   B. AWS WAF (Web Application Firewall)
   C. Amazon Inspector
   D. AWS Artifact

   Answer: B

5. A developer is writing an application running on Amazon EC2. This application needs permission to retrieve some configuration data from Amazon S3. According to AWS security "Best Practices", what is the safest way to grant S3 access to this EC2 instance?
   A. Open public access (Public Read) for that S3 bucket so EC2 can access it easily.
   B. Create an IAM User, grant S3 permissions, and embed the Access Key/Secret Key directly into the application's source code.
   C. Embed the AWS root account credentials in a hidden file on the EC2 instance.
   D. Create an IAM Role with a policy allowing S3 read access, then attach this Role to the EC2 instance.

   Answer: D

6. In the context of AWS Cost Breakdown, how is data transfer generally billed?
   A. Both inbound and outbound data transfers are completely free.
   B. Both inbound and outbound data transfers are always charged at the same rate.
   C. Inbound data transfer (data into AWS) is generally free, while outbound data transfer (data out to the internet) is charged.
   D. Outbound data transfer is free, while inbound data transfer is charged.

   Answer: C

7. Your company plans to run a steady-state database on Amazon EC2 continuously for the next 3 years. To optimize costs and get the highest possible discount compared to On-Demand pricing, which purchasing option should you apply?
   A. Spot Instances
   B. Reserved Instances (or Savings Plans)
   C. Dedicated Hosts
   D. On-Demand Instances

   Answer: B

8. A startup is planning to migrate their on-premises infrastructure to AWS. Before migrating, management wants to estimate the total expected monthly AWS bill for their designed architecture (including EC2, S3, and RDS). Which tool is best suited for this scenario?

A. AWS Pricing Calculator

B. AWS Cost Explorer

C. AWS Billing and Cost Management Console

D. AWS Inspector

Answer: A

9. A data processing application runs on EC2 and processes batch jobs. These jobs are flexible, meaning they can start and stop at any time, and can be interrupted without affecting the final result. Which EC2 pricing model provides the most cost-effective solution for this specific scenario?

   A. Spot Instances

   B. Reserved Instances

   C. On-Demand Instances

   D. Savings Plans

   Answer: A

10. An enterprise wants to simplify its billing process by combining the AWS usage of its 10 different departments into a single monthly invoice. They also want to take advantage of volume pricing discounts across all departments. Which AWS feature should they use to achieve this?

    A. AWS Cost Categories

    B. Consolidated Billing (AWS Organizations)

    C. AWS Resource Access Manager (RAM)

    D. AWS Budgets

    Answer: B

11. A company uses AWS Organizations to manage multiple accounts. They have a central "production"account and a member account for training. A user in the training account attempts to create an S3 bucket inside the "production"account but receives an "Access Denied"error. What is the primary cause of this issue?

    A. S3 bucket names must be globally unique across all AWS accounts.

    B. The user lacks the appropriate cross-account IAM permissions/permission sets to create resources in the production account.

    C. The production account has reached its maximum billing limit.

    D. Consolidated billing is not enabled.

    Answer: B

12. Your FinOps team wants to implement a proactive cost monitoring strategy. They need to receive an email alert automatically whenever their forecasted monthly AWS spending exceeds $1,000. Which service should they configure?

A. AWS Cost Explorer

B. AWS Trusted Advisor

C. AWS Budgets

D. AWS Organizations

Answer: C

13. Your DevSecOps team is setting up a CI/CD pipeline. They want to ensure that container images do not contain known software vulnerabilities before being deployed to Amazon Elastic Container Registry (ECR). Which of the following solutions best meets this requirement automatically?

    A. Enable Amazon Inspector to automatically scan container images (Amazon ECR).

    B. Configure AWS WAF on the Application Load Balancer.

    C. Set up Network Firewall for the Docker environment.

    D. Run Amazon Macie to analyze source code inside the container.

    Answer: A

14. To meet data retention and compliance requirements in DevSecOps, a company needs to record the entire history of API calls made within their AWS account. Which service should you apply?

    A. Amazon Inspector

    B. Amazon Macie

    C. AWS Systems Manager

    D. AWS CloudTrail

    Answer: D

15. According to the Shared Responsibility Model, which of the following tasks is the responsibility of the CUSTOMER when using Amazon EC2?

    A. Physical security of AWS data centers.

    B. OS patching and installing antivirus software on the instance.

    C. Maintaining AWS network hardware and routers.

    D. Safely disposing of broken server hard drives.

    Answer: B

16. A security analyst wants a service that monitors for unusual API calls, suspicious network traffic, and DNS-related signs of compromise. Which AWS service best matches this need?

    A. Amazon Macie fits best.

    B. Amazon GuardDuty fits best.

    C. AWS Secrets Manager fits best.

    D. Amazon Inspector fits best.

    Answer: B

17. According to AWS's Shared Responsibility Model, which of the following components falls under AWS's security responsibility (Security of the Cloud)?
    A. Managing the Guest OS and software patches
    B. Physical security for the data center infrastructure
    C. Configuring firewall rules (Security Groups) for EC2 instances
    D. Client-side data encryption

    Answer: B

18. Which type of AWS KMS key allows the public key to be downloaded and shared with external parties for encryption, while the private key never leaves the service?
    A. Symmetric KMS Key
    B. AWS Managed Key
    C. AWS Owned Key
    D. Asymmetric KMS Key

    Answer: D

19. Which statement best describes the AWS Shared Responsibility Model?
    A. AWS is responsible for everything in the cloud, including customer data and IAM
    B. Security and compliance are shared between AWS and the customer
    C. Customers are responsible only for physical data centers
    D. AWS is responsible only for IAM users and groups

    Answer: B

20. An EC2 application only needs to read files from one S3 bucket. A junior engineer suggests attaching AdministratorAccess to the EC2 instance "because it is easier and avoids errors.". Which solution best follows the least privilege principle?
    A. Attach admin access to the instance role
    B. Store access keys directly inside the code
    C. Create a role with only required S3 read
    D. Give full access and enable MFA later

    Answer: C

21. What is the core difference between IAM Roles and IAM Users in Identity and Access Management services?
    A. IAM Users can contain multiple IAM Groups, but IAM Roles cannot.
    B. IAM Roles are region-bound resources, while IAM Users are global resources.
    C. IAM Users have long-term login credentials, while IAM Roles use temporary security credentials.
    D. IAM Users are for humans only, while IAM Roles are for AWS services only.

Answer: C

22. Why are Managed Services (like Amazon S3 or DynamoDB) considered to have "Less Customer Responsibility"compared to Infrastructure Services (like Amazon EC2)?
    A. The customer does not need to use IAM for managed services
    B. Customers are not responsible for client-side encryption in managed services
    C. AWS handles the management of the underlying OS, network configuration, and platform software for managed services
    D. Managed services do not require any configuration from the customer

    Answer: C

23. Under the shared responsibility model, which of the following is a customer responsibility for Amazon EC2?
    A. Physical data centers
    B. Global infrastructure such as Regions and Availability Zones
    C. The operating system and applications running on the instance
    D. The virtualization layer (hypervisor)

    Answer: C

24. A company has three problems at the same time: (1) It wants to find PII inside S3 buckets. (2) It wants to detect CVEs and package vulnerabilities in ECR images. (3) It wants alerts for suspicious API activity and possible DNS-based exfiltration. Which mapping is the most correct?
    A. Macie / Inspector / GuardDuty
    B. GuardDuty / Macie / Inspector
    C. Inspector / GuardDuty / Macie
    D. Macie / GuardDuty / Inspector

    Answer: A

25. An organization needs to automatically scan millions of files in Amazon S3 buckets to detect sensitive information such as credit card numbers and personal identifiers (PIIs). Which of the following services performs this task best?
    A. Amazon Macie
    B. AWS Inspector
    C. Amazon Security Tasks
    D. AWS WAF

    Answer: A

26. Which list correctly shows the main components in the AWS cost breakdown?
    A. Compute, Managed Services, Storage, Network Traffic Data Transfer, Misc
    B. Compute, IAM, DNS, Storage, Support

C. EC2, Lambda, S3, VPC, CloudTrail

D. CPU, RAM, Disk, API, Backup

Answer: A

27. According to the slide, which formula is used to calculate compute cost?

    A. Compute Cost = Instance Type × Instance Amount × Runtime

    B. Compute Cost = Storage × Requests × Region

    C. Compute Cost = Runtime × Bandwidth × Users

    D. Compute Cost = Instance Type × Availability Zone × Storage

    Answer: A

28. Which of the following is NOT one of the four storage cost drivers shown in the slide?

    A. Storage class tiers

    B. PUT/GET request volume

    C. Lifecycle transitions

    D. CPU clock speed

    Answer: D

29. Which of the following is listed as an advantage of the Pay-as-you-go model?

    A. Revenue is unpredictable and volatile

    B. Pricing is always simple

    C. Lower upfront costs attract users

    D. It is difficult to retain customers

    Answer: C

30. Which option correctly lists the four AWS pricing models shown in the document?

    A. On-Demand, Spot Instances, Reserved Instances, Savings Plans

    B. Free Tier, Premium Tier, Enterprise Tier, Custom Tier

    C. Compute Plan, Storage Plan, Network Plan, Security Plan

    D. Basic, Standard, Advanced, Professional

    Answer: A

31. Which service should you use to protect your web applications from common exploits such as SQL Injection or Cross-Site Scripting (XSS)?

    A. AWS WAF (Web Application Firewall)

    B. AWS Shield

    C. Network ACL

    D. Amazon Inspector

Answer: A

32. Regarding networking costs in AWS, which of the following actions typically does NOT incur data transfer charges?
    A. Data transfer between two different Regions.
    B. Data transfer from AWS to the Internet (Outbound Data Transfer).
    C. Data transfer from the Internet to AWS (Inbound Data Transfer).
    D. Data transfer between two different Availability Zones (AZs) within the same Region.

    Answer: C

33. A company wants to allow an application running on an EC2 instance to access S3 securely without embedding access keys in the code. What is the BEST solution?
    A. Create an IAM user and store credentials in the application
    B. Use an IAM role attached to the EC2 instance
    C. Store credentials in environment variables
    D. Store credentials in environment variables

    Answer: B

34. A company needs to manage permissions for multiple developers who require the same level of access to AWS resources. What is the MOST efficient approach?
    A. Assign policies individually to each IAM user
    B. Create IAM roles for each developer
    C. Create an IAM group and attach policies to the group
    D. Share one IAM user among developers

    Answer: C

35. A company wants to enforce that all data stored in S3 is encrypted using customer-managed keys with full control over key rotation. Which service should they use?
    A. AWS Certificate Manager
    B. AWS Key Management Service
    C. Amazon Inspector
    D. AWS Shield

    Answer: B

36. A company wants to analyze EC2 instances for vulnerabilities and unintended network exposure. Which AWS service should they use?
    A. Amazon Inspector
    B. AWS Shield
    C. AWS WAF

D. AWS IAM

Answer: A

37. A company uses multiple AWS accounts and wants to centrally manage security policies and billing. What should they use?

    A. IAM roles

    B. IAM groups

    C. AWS Organizations

    D. AWS Config

    Answer: C

38. A company wants to encrypt sensitive data stored in an RDS database and ensure that encryption keys are rotated automatically. Which solution should they choose?

    A. Use IAM policies

    B. Use AWS KMS with automatic key rotation

    C. Use AWS Shield

    D. Use AWS Organizations

    Answer: B

39. A company is running a web application on Amazon EC2 instances that needs to upload files to an Amazon S3 bucket. The security team requires that no long-term AWS credentials be stored on the instances. Which solution provides secure access while following AWS best practices?

    A. Create an IAM user with S3 permissions and store the access keys in a configuration file on the EC2 instance.

    B. Create an IAM role with the required S3 permissions and attach it to the EC2 instance.

    C. Store AWS access keys in AWS Systems Manager Parameter Store and retrieve them at runtime.

    D. Create an S3 bucket policy that allows public write access from the EC2 instance's public IP address.

    Answer: B

40. A company deploys an Application Load Balancer across two public subnets in different Availability Zones. The load balancer distributes traffic to EC2 instances running in private subnets. What is the PRIMARY advantage of this architecture?

    A. It reduces the cost of running EC2 instances by eliminating the need for Auto Scaling.

    B. It ensures high availability and fault tolerance by distributing traffic across multiple Availability Zones.

    C. It allows EC2 instances in private subnets to have direct internet access.

    D. It removes the need to configure security groups for the EC2 instances.

    Answer: B

41. A company needs to provision a logically isolated virtual network in AWS where it can define its own IP address range, create subnets, configure route tables, and control inbound and outbound traffic. Which AWS service should the company use?
    A. Amazon EC2
    B. Amazon VPC
    C. AWS Direct Connect
    D. Amazon Route 53

    Answer: B

42. A company wants to create an isolated network in AWS that behaves like its own private data center. Which AWS service should be used?
    A. Amazon Route 53
    B. Amazon VPC
    C. Amazon CloudFront
    D. Amazon S3

    Answer: B

43. A company deploys an Application Load Balancer across two public subnets in different Availability Zones and routes traffic to web servers in private subnets. What is the main benefit of this design?
    A. It eliminates the need for Security Groups
    B. It provides direct internet access to private instances
    C. It improves availability and distributes traffic across targets
    D. It replaces Route 53 completely

    Answer: C

44. According to the AWS Shared Responsibility Model, which of the following is the responsibility of the customer?
    A. Maintaining physical hardware and global network infrastructure
    B. Managing the guest operating system, applications, and customer data
    C. Operating the virtualization layer (hypervisor) on host servers
    D. Controlling environmental and physical security at data centers

    Answer: B

45. What does the "Least Privilege"principle in AWS IAM mean?
    A. Granting only the minimum permissions necessary for a user to perform their job
    B. Giving every user in a group full administrative privileges to ensure they can work
    C. Only allowing access to AWS resources from the corporate internal network
    D. Allowing users to freely grant permissions to other accounts when needed

Answer: A

46. Which AWS pricing model is best suited for unpredictable, short-term workloads that cannot be interrupted?
    A. Reserved Instances (RI)
    B. Spot Instances
    C. On-Demand Instances
    D. Savings Plans

    Answer: C

47. Which AWS tool helps users plan budgets and receive notifications when actual costs exceed a pre-defined threshold?
    A. AWS Cost Explorer
    B. AWS Budgets
    C. AWS Trusted Advisor
    D. Consolidated Billing

    Answer: B

48. Which security service uses Machine Learning to automatically discover and protect sensitive data (such as PII) stored in Amazon S3?
    A. Amazon GuardDuty
    B. Amazon Inspector
    C. Amazon Macie
    D. AWS WAF

    Answer: C

49. Which of the following is not a Core Component of AWS Identity and Access Management (IAM)?
    A. IAM User
    B. IAM Role
    C. IAM Group
    D. IAM Tree

    Answer: D

50. Under the shared responsibility model, which of the following is the responsibility of AWS?
    A. Client-side encryption
    B. Protecting infrastructure devices
    C. Server-side encryption
    D. Filtering traffic with Security Groups

Answer: B

51. What is the primary fundamental charging principle of AWS?
    A. Pay-per-storage
    B. Pay-as-you-go
    C. Fixed monthly payment
    D. Free for all services for new users

    Answer: B

52. According to the "Volume Discounts"principle, what happens as your usage of AWS services increases?
    A. The unit cost decreases
    B. The unit cost increases to compensate for resources
    C. The unit cost remains unchanged
    D. Your resources will be limited

    Answer: A

53. Which EC2 pricing model is most suitable for unpredictable workloads that require the flexibility to be turned on or off at any time?
    A. Reserved Instances
    B. On-Demand
    C. Spot Instances
    D. Savings Plans

    Answer: B

54. A government agency is getting started with managing their AWS Cloud costs. They want to set up notifications to alert them when costs exceed a certain threshold for any given month. Which AWS service would they use to achieve this task?
    A. AWS Cost Explorer
    B. AWS Budgets
    C. AWS Organizations
    D. AWS Pricing Calculator

    Answer: B

55. What do you gain from setting up consolidated billing for five different AWS accounts under another master account?
    A. AWS services' costs will be reduced to half the original price.
    B. The consolidated billing feature is just for organizational purpose.
    C. Each AWS account gets volume discounts.

D. Each AWS account gets five times the free-tier services capacity.

Answer: C

56. According to Finops best practices, at what increment should you set AWS Budget alerts to effectively catch early overspend patterns?

    A. $500 increments$

    A. $50 increments$

    A. $5,000 increments$

    A. $10,000 increments$

    Answer: A

57. Under the AWS Shared Responsibility Model, which of the following is an AWS responsibility?

    A. Configuring AWS Identity and Access Management (IAM) policies.

    B. Encrypting customer data stored in Amazon S3.

    C. Patching the guest operating system on EC2 instances

    D. Securing the physical data centers and hardware

    Answer: D

58. What action does an IAM Policy take if a specific API request is neither explicitly allowed nor explicitly denied?

    A. It results in an implicit deny.

    B. It allows the request but flags it in CloudTrail.

    C. It results in an explicit deny.

    D. It prompts the user for MFA.

    Answer: A

59. What is the most secure way to grant an Amazon EC2 instance permissions to access an Amazon S3 bucket?

    A. Embed the IAM user's access keys directly into the application code.

    B. Store the access keys in an S3 bucket and have the EC2 instance download them.

    C. Attach an IAM Role to the EC2 instance.

    D. Create a new IAM User specifically for the EC2 instance and share the password.

    Answer: C

60. What is not listed as one of the four pricing models of AWS?

    A. On-Demand

    B. Spot Instances

    C. Saving Plans

D. Premium

Answer: D

61. A large organization is trying to decide the level of AWS Support that best suits their needs. The company has the following requirements: Basic AWS Trusted Advisor security checks; 15-minute response time for business-critical system down scenarios; Full set of AWS Trusted Advisor checks and prioritized recommendations by an AWS account team; A dedicated Technical Account Manager (TAM) for consultative architectural and operational guidance. Which AWS Support Plan should the company choose based on these requirements?

    A. Basic Support

    B. Developer Support

    C. Business Support

    D. Enterprise Support

    Answer: D

62. What are the two main types of KMS keys, and how do they differ in usage?

    A. Symmetric keys use one key for encryption only, while asymmetric keys use one key for decryption only.

    B. Symmetric keys use a single key for encryption and decryption, while asymmetric keys use a public/private key pair.

    C. Symmetric keys use public/private pairs, while asymmetric keys use a single shared key.

    D. Both symmetric and asymmetric keys require two keys for encryption and decryption.

    Answer: B

63. A compliance requirement mandates that all API calls made within an AWS account must be recorded for auditing purposes. Which AWS service fulfills this requirement?

    A. Amazon CloudWatch

    B. AWS CloudTrail

    C. AWS Config

    D. Amazon SNS

    Answer: B

64. Which pricing model is best for long-term stable workload?

    A. On-Demand

    B. Savings Plans

    C. Reserved Instances

    D. Spot Instances

    Answer: C

65. Which services does Amazon Inspector integrate with to send findings?

A. Amazon S3 Amazon DynamoDB

B. AWS Security Hub Amazon EventBridge

C. Amazon RDS AWS Lambda

D. Amazon CloudFront AWS Shield

Answer: B

66. Which AWS service is specifically designed to allow you to centrally govern and manage multiple AWS accounts?

    A. AWS Organizations

    B. AWS Cost Explorer

    C. IAM Identity Center

    D. Amazon EventBridge

    Answer: A

67. Which AWS service allows a user to set custom alerts that will notify them via email or SNS when their AWS usage or spending exceeds a predefined threshold?

    A. AWS Cost Explorer

    B. AWS Trusted Advisor

    C. AWS Budgets

    D. AWS Cost Anomaly Detection

    Answer: C

68. Which is the correct workflow for identifying sensitive data using Amazon Macie?

    A. Amazon S3 → Amazon Macie (analyze sensitive data) → Amazon EventBridge (notify integrate)

    B. Amazon EC2 → AWS Shield → Amazon SNS

    C. Amazon RDS → AWS WAF → Amazon CloudWatch

    D. AWS Lambda → Amazon GuardDuty → AWS Security Hub

    Answer: A

69. You need to encrypt application data at rest in Amazon S3 using AWS Key Management Service (KMS) with fine-grained control over who can use the key. Which key type is the best fit?

    A. Customer managed KMS key that you create and control

    B. Default S3 server-side encryption with S3-managed keys (SSE-S3)

    C. KMS key for AWS CloudHSM only, with no IAM policies

    D. AWS managed key aws/s3 used transparently by S3

    Answer: A

70. An organization with many AWS accounts wants to centrally define and enforce AWS WAF rules and Shield Advanced protections across all accounts in the organization. Which service is designed to simplify this centralized management?
    A. Amazon Macie
    B. AWS Config
    C. AWS GuardDuty
    D. AWS Firewall Manager

    Answer: D

71. You have set up consolidated billing for several AWS accounts. One of the accounts has purchased a number of reserved instances for 3 years. Which of the following is true regarding this scenario?
    A. The Reserved Instance discounts can only be shared with the master account
    B. All accounts can receive the hourly cost benefit of the Reserved Instances
    C. The purchased instances will have better performance than On-demand instances
    D. There are no cost benefits from using consolidated billing; It is for informational purposes only

    Answer: B

72. Which statement best describes the main difference between AWS Certificate Manager (ACM) and AWS Key Management Service (KMS)?
    A. ACM manages SSL/TLS certificates for securing network connections, while KMS manages encryption keys for protecting data at rest and in transit
    B. ACM is used only for database encryption, while KMS is used solely for logging and monitoring
    C. KMS automatically deploys SSL/TLS certificates to Elastic Load Balancers, while ACM only stores private keys in hardware modules
    D. Both ACM and KMS are used only for generating client-side encryption keys that never leave the customer's environment

    Answer: A

73. Which service natively supports automatic rotation of database credentials?
    A. AWS Secrets Manager
    B. AWS Certificate Manager
    C. SSM Parameter Store
    D. AWS KMS

    Answer: A

74. In the Shared Responsibility Model of Amazon Web Services, which of the following responsibilities belongs to AWS?
    A. Configuring firewall rules for EC2 instances

B. Encrypting data at the application level

C. Maintaining the physical hardware in data centers

D. Managing IAM users and permissions

Answer: C

75. You want to enhance security for the root account and IAM users in AWS. Which of the following is a best practice?

A. Use strong passwords only, without additional factors

B. Use AWS Identity and Access Management roles instead of users for all access

C. Enable Multi-Factor Authentication for the root account and critical IAM users

D. Share access keys among developers for easier management

Answer: C

76. Which of the following is a common threat that AWS WAF protects against?

A. Unencrypted EBS volumes

B. SQL Injection

C. SYN Floods

D. Unauthorized OS logins

Answer: B

77. You want to track AWS costs per project and receive alerts when budgets are exceeded. Which combination of services should you use?

A. Amazon CloudWatch + Trusted Advisor

B. AWS Cost Explorer + AWS Budgets

C. AWS Shield + AWS WAF

D. AWS Organizations + IAM

Answer: B

78. A customer is using multiple AWS accounts with separate billing. How can the customer take advantage of volume discounts with minimal impact to the AWS resources?

A. Create one global AWS account and move all AWS resources to that account

B. Sign up for three years of Reserved Instance pricing up front

C. Use the consolidated billing feature from AWS Organizations

D. Sign up for the AWS Enterprise support plan to get volume discounts

Answer: C

79. Which of the following services uses machine learning to continuously monitor for malicious activity and unauthorized behavior?

A. AWS Macie

B. AWS KMS

C. Amazon GuardDuty

D. AWS CloudTrail

Answer: C

80. Which of the following is an advantage of AWS consolidated billing?

A. The ability to receive one bill for multiple accounts

B. Service limits increasing by default in all accounts

C. A fixed discount on the monthly bill

D. The automatic extension of the master account's AWS support plan to all accounts

Answer: A

81. Which service automatically scans Amazon S3 buckets to discover and protect sensitive data like Personally Identifiable Information (PII)?

A. Amazon Inspector

B. Amazon GuardDuty

C. Amazon Macie

D. AWS Secrets Manager

Answer: C

82. Which service enables automated vulnerability scanning of Amazon EC2 instances and container images in Amazon ECR?

A. Amazon GuardDuty

B. Amazon Inspector

C. AWS Config

D. AWS WAF

Answer: B

83. According to the principle of "Least Privilege", how should permissions be granted in AWS?

A. Grant users full administrator access to avoid permission errors.

B. Grant users only the specific permissions necessary to perform their required tasks.

C. Apply the same permissions to all users in the account using a single IAM Group.

D. Allow all users access to the root account for emergencies.

Answer: B

84. What is described as a 'non-negotiable' requirement to accurately track and attribute costs to different business units or projects?

A. Running all workloads in a single VPC

B. Using Amazon S3 Intelligent- Tiering

C. Enabling AWS CloudTrail

D. Implementing Cost Allocation Tags

Answer: D

85. A company wants to implement detailed tracking of its cloud costs by department and project. Which AWS feature or service should the company use?

    A. AWS Marketplace

    B. AWS Budgets

    C. Cost allocation tags

    D. Consolidated billing

    Answer: C

86. Which of the following security measures protect access to an AWS account?

    A. Activate multi-factor authentication (MFA) for privileged users

    B. Enable AWS CloudTrail

    C. Create one IAM user and share with many developers and users

    D. Enable Amazon CloudFront

    Answer: A

87. How can a customer increase security to AWS account logons?

    A. Configure AWS Certificate Manager

    B. Enable Multi-Factor Authentication (MFA)

    C. Use Amazon Cognito to manage access

    D. Enable AWS Organizations

    Answer: B

88. Which statement is not true about Server-side Encryption at Rest?

    A. Data is encrypted after being received by the server

    B. Data is decrypted after being sent

    C. It is stored in an encrypted form thanks to a key (usually a data key)

    D. The encryption / decryption keys must be managed somewhere, and the server must have access to it

    Answer: B

89. A developer wants to store credentials securely and rotate them automatically. Which service should be used for this, as opposed to encrypting large volumes of data?

    A. AWS KMS

    B. AWS IAM

    C. AWS Artifact

D. AWS Secrets Manager

Answer: D

90. What is the primary purpose of AWS KMS in the context of security?
    A. Managing network security groups
    B. Storing user passwords for IAM users
    C. Creating and managing encryption keys
    D. Auditing infrastructure spend

    Answer: C

91. Which is not a type of KMS Key?
    A. Middle-man Managed Keys
    B. AWS Managed Keys
    C. AWS Owned Keys
    D. Customer Managed Keys

    Answer: A

92. Which pricing model is most suitable for a developer who needs to run a short-term, unpredictable testing workload for only a few hours?
    A. Savings Plans
    B. Reserved Instances
    C. Spot Instances
    D. On-Demand Instances

    Answer: D

93. Which commitment-based pricing model provides the highest level of flexibility by covering compute usage across EC2, AWS Lambda, and AWS Fargate regardless of instance family or region?
    A. Standard Reserved Instances
    B. EC2 Instance Savings Plans
    C. Compute Savings Plans
    D. Convertible Reserved Instances

    Answer: C

94. According to AWS AI case studies, what is typically the primary cause for the observed average monthly budget gap of $3,400-13{,}800$?
    A. Rapid scaling of On-Demand instance counts
    B. Invisible costs like uncleaned storage and unmodeled data transfer
    C. High premium charges for Enterprise Support plans

D. Inaccurate pricing forecasts from AWS Cost Explorer

Answer: B

95. According to the "Pay-as-you-go"principle, which of the following is considered a primary driver of S3 storage costs?

    A. IAM User count

    B. Data Transfer OUT

    C. Instance Runtime

    D. Number of AWS accounts

Answer: B

96. For non-customer-facing AI workloads like historical fraud scoring, which strategy is recommended to reduce compute costs by 60-70

    A. Utilizing Spot Instances for all inference tasks

    B. Enabling HNSW-PQ compression for vector storage

    C. Using Batch Transform jobs instead of real-time endpoints

    D. Implementing automated instance scheduling for off-hours

Answer: C

97. Which AWS tool allows you to visualize and forecast your monthly spend while grouping costs by service or linked accounts?

    A. AWS Budgets

    B. AWS Organizations

    C. AWS Cost Explorer

    D. AWS CloudWatch

Answer: C

98. Which AWS service enables you to quickly purchase and deploy SSL/TLS certificates?

    A. Amazon GuardDuty

    B. Amazon Detective

    C. AWS WAF

    D. AWS ACM

Answer: D

99. In a production RAG deployment, which specific technique can reduce annual vector storage costs from $75,000 downto approximately$10,000?

    A. HNSW-FP16 compression

    B. HNSW-PQ compression

    C. Implementing lifecycle transitions to S3 Glacier

D. Consolidating multiple OpenSearch clusters

Answer: B

100. Which AWS pricing principle suggests that customers should "pay less by using more,"as seen in tiered S3 storage pricing?

A. Economies of Scale

B. Pay-as-you-go

C. On-Demand Pricing

D. Volume Discounts

Answer: D

101. When accessing AWS services like S3 or DynamoDB within the same region, which architectural pattern incurs both a per-hour service charge and a per-GB data processing charge?

A. Pattern 1: Internet Gateway

B. Pattern 2: NAT Gateway

C. Pattern 3: VPC Peering

D. Pattern 4: Gateway Load Balancer

Answer: B

102. Account A wants to share an Amazon Machine Image (AMI) backed by an encrypted EBS volume with Account B. Which combination of actions is strictly REQUIRED for Account B to successfully launch an EC2 instance from this shared AMI?

A. Only share the underlying KMS key with Account B, because the AMI is inherently linked to it.

B. Copy the AMI to an S3 bucket and grant Account B cross-account access to the bucket.

C. Share the AMI launch permissions with Account B AND share the underlying KMS key with Account B.

D. Make the AMI public and temporarily disable KMS encryption during the launch process.

Answer: C

103. In an AWS Organization, a Service Control Policy (SCP) is applied to an Organizational Unit (OU). The SCP explicitly allows access to Amazon S3. However, an IAM user within a member account of that OU receives an 'Access Denied' error when trying to upload a file to S3. What is the most likely reason?

A. SCPs only monitor billing limits and do not actually affect resource access.

B. The IAM user does not have an IAM policy attached that explicitly grants them S3 permissions.

C. The master account must perform an S3 action first to activate the SCP for the member accounts.

D. The SCP must be applied directly to the IAM user instead of the Organizational Unit.

Answer: B

104. A security team wants to audit which IAM users in the account have not used their access keys in the past 90 days. Which IAM tool should they use?

    A. IAM Access Advisor

    B. AWS CloudTrail

    C. IAM Credentials Report

    D. AWS Config

    Answer: C

105. Both AWS Secrets Manager and AWS Systems Manager (SSM) Parameter Store can be used to securely store database credentials. What is a key capability unique to AWS Secrets Manager that justifies its higher cost?

    A. It uses a hierarchical path structure to organize configuration data.

    B. It natively supports automatic rotation of secrets on a schedule using a Lambda function.

    C. It provides free public TLS/SSL certificates for your applications.

    D. It is the only service that integrates with AWS KMS for encryption at rest.

    Answer: B

106. A company stores sensitive documents in Amazon S3 and needs to encrypt them using keys that they can fully control, rotate manually, and audit via IAM policies. Which AWS KMS key type should they use?

    A. AWS Owned Keys

    B. AWS Managed Keys

    C. Symmetric Imported Keys

    D. Customer Managed Keys (CMK)

    Answer: D

107. Under the AWS Shared Responsibility Model, which of the following is a responsibility specifically managed by the CUSTOMER when running an application on Amazon EC2 ?

    A. Disposal of decommissioned hard drives.

    B. Patching the guest operating system and the application software.

    C. Maintenance of the underlying virtualization software (Hypervisor).

    D. Physical security of the data centers.

    Answer: B

108. A company's web application hosted behind an Application Load Balancer is experiencing a large-scale SQL injection attack. Which AWS service should be deployed to block these malicious HTTP requests?

    A. AWS Shield Standard

B. AWS Network Firewall

C. AWS WAF with a Web ACL attached to the ALB

D. Amazon GuardDuty

Answer: C

109. To satisfy a regulatory audit, a company needs to provide a history of all API calls made within their AWS account, including who made the call and from which IP address. Which service provides this record?

A. Amazon CloudWatch Logs.

B. AWS Config.

C. AWS CloudTrail.

D. Amazon Inspector.

Answer: C

110. Your application needs to securely store a third-party API key that rarely changes. You want the most cost-effective solution that supports encryption at rest via KMS and allows hierarchical organization of the configuration data. Which service should you choose ?

A. AWS Systems Manager (SSM) Parameter Store

B. AWS Certificate Manager (ACM)

C. AWS Key Management Service (KMS)

D. AWS Secrets Manager

Answer: A

111. A security engineer needs to automatically discover and report any Personally Identifiable Information (PII) that has been accidentally uploaded to the company's Amazon S3 buckets. Which AWS service is purpose-built for this task?

A. Amazon GuardDuty

B. AWS Trusted Advisor

C. Amazon Inspector

D. Amazon Macie

Answer: D

112. A data analytics team is running a batch processing job that can be interrupted and resumed at any time. Which EC2 instance type offers the deepest discount (up to 90

A. Convertible Reserved Instances.

B. On-Demand Instances.

C. Standard Reserved Instances.

D. Spot Instances.

Answer: D

113. According to the AWS Compute Cost calculation formula, which three variables determine the total cost?
    A. On-demand Rate, Reserved Capacity, and Throughput
    B. Storage Tier, Data Transfer, and Region
    C. vCPU Count, Memory Size, and API Requests
    D. Instance Type, Instance Amount, and Runtime

    Answer: D

114. What is identified as a primary drawback for a provider using the 'Pay-as-you-go' pricing model?
    A. It is too simple for customers to understand
    B. Customers are locked into long-term contracts
    C. It attracts fewer users due to high entry costs
    D. Revenue is unpredictable and volatile

    Answer: D

115. For an unpredictable workload where you do not know how long the system will run, which pricing model is most appropriate?
    A. Compute Savings Plans
    B. On-demand Pricing
    C. Convertible Reserved Instances
    D. Standard Reserved Instances

    Answer: B

116. When accessing AWS services within the same Region, what is the cost impact of using a NAT Gateway compared to an Internet Gateway?
    A. NAT Gateway has no data transfer charge, while Internet Gateway charges per GB.
    B. Both have a flat monthly fee regardless of usage.
    C. NAT Gateway incurs both a per-hour service charge and a per-GB data processing charge.
    D. Internet Gateway incurs a per-hour service charge.

    Answer: C

117. For advanced FinOps in AI deployments, why is it highly recommended to use Batch Inference instead of Real-Time Endpoints whenever latency allows?
    A. Batch Inference runs 24/7 to ensure zero latency.
    B. Real-Time SageMaker endpoints run 24/7 whether they serve requests or not, leading to idle waste.
    C. Batch Inference utilizes more expensive GPU types.
    D. Real-Time Endpoints do not support modern AI models.

Answer: B

118. Which AWS principle is described as "Pay less by using more,"where the unit price decreases as the total usage increases?

    A. Economies of Scale

    B. Pay-as-you-go

    C. Volume Discounts

    D. Savings Plans

Answer: C

119. In the On-Demand pricing model, which of the following is considered a primary characteristic regarding its usage and commitment?

    A. It requires a 1-year or 3-year commitment for discounted rates.

    B. It is best for unpredictable workloads with no upfront commitment.

    C. It provides the highest discount, up to 90

    D. It offers lower costs per hour compared to Reserved Instances.

Answer: B

120. Regarding network architecture cost impact, which of the following scenarios typically incurs "No data transfer charge"?

    A. Transferring data between two different AWS Regions.

    B. Transferring data between Workload components across two different Availability Zones.

    C. Accessing AWS services in the same Region via an Internet Gateway.

    D. Data processing and service charges when using a NAT Gateway.

Answer: C

121. A company is deploying a non-customer-facing AI workload for tasks such as document processing and demand forecasting. To reduce compute costs by an estimated 60-70

    A. Compress vector embeddings using HNSW-PQ to drop storage costs.

    B. Set budget alerts at $5,000 incrementstomonitoroverspendpatterns.$

    B. Switch from real-time SageMaker endpoints to Batch Inference (Batch transform jobs).

    C. Implement Cost Allocation Tags to identify who is responsible for the spend.

Answer: C

122. Which AWS tool is used to estimate the cost of using a specific set of AWS services before any resources are actually provisioned?

    A. AWS Pricing Calculator

    B. AWS Trusted Advisor

    C. AWS Cost Explorer

D. AWS Budgets

Answer: A

123. Under the AWS Shared Responsibility Model, which of the following is a responsibility of the customer regarding billing?

A. Determining the base price for EC2 instances

B. Calculating the tax rates for all global regions

C. Configuring AWS Budgets and alerts

D. Maintaining the billing infrastructure hardware

Answer: C

124. Which feature of AWS Organizations allows a company to receive a single bill for all of its AWS accounts?

A. Unified Cost management

B. Resource Groups

C. Consolidated Billing

D. Control Tower

Answer: C

125. Which of the following determines the cost of using Amazon S3 storage?

A. Which of the following determines the cost of using Amazon S3 storage?

B. Storage class used and the total amount of data stored

C. The type of data (e.g images, text,...) being stored

D. The operating system used to upload the files

Answer: B

126. Which AWS billing tool provides a visual representation of your cost and usage trends over time and allows you to forecast future costs?

A. AWS Marketplace

B. AWS Purchase Order Management

C. AWS License Manager

D. AWS Cost Explorer

Answer: D

127. A user wants to save money on their compute costs and is willing to commit to a consistent amount of usage (measured in $/hour) for a 1-year term. Which option should they choose$?

A. Pay-as-you-go pricing

B. Volume Discounts

C. On-demand Capacity Reservations

D. Saving plans

Answer: D

128. Which AWS service is used to manage user identities and their access levels to AWS resources and services?

A. AWS Secrets Manager

B. AWS Directory Service

C. AWS Identity and Access Management (IAM)

D. AWS Resource Access Manager (RAM)

Answer: C

129. A company needs to manage encryption keys and control their use across a wide range of AWS services. Which service should they use?

A. AWS Cloud HSM

B. AWS Systems Manager (SSM)

C. AWS Key Management Service (KMS)

D. AWS Certificate Manager (ACM)

Answer: C

130. Which AWS service helps protect web applications from common web exploits like SQL injection or cross-site scripting (XSS)?

A. AWS Shield

B. Amazon GuardDuty

C. AWS WAF (Web Application Firewall)

D. AWS Network Firewall

Answer: C

131. A user needs to provision, manage, and deploy public and private SSL/TLS certificates for use with AWS services. Which service should they use?

A. AWS Certificate Manager (ACM)

B. AWS IAM

C. AWS Artifact

D. AWS Secrets Manager

Answer: A

132. Which task falls under the customer's responsibility when utilizing managed services like Amazon S3 or Amazon DynamoDB?

A. Maintaining the physical hardware and data centers

B. Managing the host operating system and hypervisor

C. Configuring client-side data encryption

D. Securing the global infrastructure and edge locations

Answer: C

133. How does AWS handle key rotation for Customer Managed Keys (CMKs) where the key material was manually imported by the user?

A. Keys are rotated automatically every 365 days

B. Keys are rotated automatically every 90 days

C. AWS Support must rotate the key upon receiving a support ticket

D. There is no automatic rotation; it must be done manually

Answer: D

134. A development team wants to store database credentials securely and needs a service that natively supports automatic rotation of these secrets. Which AWS service is best suited for this requirement?

A. AWS Systems Manager Parameter Store

B. AWS Secrets Manager

C. AWS Key Management Service (AWS KMS)

D. AWS Certificate Manager (ACM)

Answer: B

135. A system administrator has provisioned a public TLS certificate via AWS Certificate Manager (ACM). Which of the following resources can this certificate be directly deployed on to provide in-flight encryption?

A. Amazon EC2 instances

B. Application Load Balancers (ALB)

C. Amazon S3 buckets

D. Amazon RDS instances

Answer: B

136. An architecture requires AWS WAF for Layer 7 protection and a fixed, static IP address for external whitelisting. Since an Application Load Balancer (ALB) does not offer a static IP by default, what is the recommended architecture?

A. Apply AWS WAF directly to a Network Load Balancer (NLB)

B. Attach an Elastic IP directly to the AWS WAF Web ACL

C. Place AWS Global Accelerator in front of the ALB

D. Use Amazon Route 53 to generate a static IP for WAF

Answer: C

137. A startup is running a machine learning training job that can be interrupted and resumed later without affecting the final model. The team needs to minimize compute costs as much as possible. Which pricing model should they choose?

    A. On-Demand Instances

    B. Reserved Instances

    C. Spot Instances

    D. Savings Plans

    Answer: C

138. A company has deployed two Amazon EC2 instances within an Amazon Virtual Private Cloud (VPC). To ensure no data transfer charges occur when these instances communicate with each other, how must the instances be deployed?

    A. In different AWS Regions using VPC Peering.

    B. In the same Availability Zone.

    C. In different Availability Zones within the same AWS Region.

    D. Behind an AWS Transit Gateway in the same AWS Region.

    Answer: B

139. Which of the following is NOT one of the primary cost drivers for Amazon S3 storage?

    A. Storage Class Tiers

    B. PUT/GET Request Volume

    C. Number of IAM Users accessing the bucket

    D. Lifecycle Transitions

    Answer: C

140. What is a significant risk for companies that do NOT implement cost allocation tags on their AI models?

    A. They cannot choose an AWS Region.

    B. They cannot attribute spend or identify wasted resources.

    C. They will be unable to use Amazon S3.

    D. Their AWS Support Plan will be automatically downgraded.

    Answer: B

141. An enterprise is deploying an AI document processing tool that analyzes historical financial records overnight. There is no requirement for instant results. Which approach will be the most cost-effective for running this compute workload?

    A. Deploying a real-time Amazon SageMaker endpoint.

    B. Using Amazon EC2 On-Demand instances to run real-time inference 24/7.

    C. Utilizing batch inference jobs.

D. Provisioning Amazon DynamoDB with maximum read capacity.

Answer: C

142. A company wants to track and organize its AWS costs by individual departments, such as "Marketing"and "Development". What is the most effective way to achieve this?

A. Create separate AWS Regions for each department.

B. Apply User-Defined Cost Allocation Tags to resources.

C. Use AWS Cost Anomaly Detection to filter traffic.

D. Enable Amazon CloudWatch Metrics Dashboard.

Answer: B

143. Which AWS service can manage permissions for AWS resources by using policies?

A. AWS Identity and Access Management (IAM)

B. Amazon Detective

C. Amazon GuardDuty

D. Amazon Inspector

Answer: A

144. A company needs to block SQL injection attacks. Which AWS service or feature provides this functionality?

A. AWS WAF

B. Network ACLs

C. Security groups

D. AWS Trusted Advisor

Answer: A

145. Which AWS offering can analyze a company's AWS environment to discover security vulnerabilities on Amazon EC2 instances?

A. Security groups

B. AWS Shield Standard

C. Amazon Macie

D. Amazon Inspector

Answer: D

146. A company has multiple AWS accounts. The company needs to receive a consolidated bill from AWS and must centrally manage security compliance. Which AWS service or feature should the company use to meet these requirements?

A. AWS Security Hub

B. AWS Config

C. AWS Organizations

D. AWS Cost and Usage Report

Answer: C

147. Which AWS service integrates with other AWS services to provide the ability to encrypt data at rest?

A. AWS Security Hub

B. AWS Identity and Access Management (IAM)

C. AWS Certificate Manager (ACM)

D. AWS Key Management Service (AWS KMS)

Answer: D

148. Which AWS service is used to temporarily provide federated security credentials to access AWS resources?

A. AWS Certificate Manager

B. AWS Secrets Manager

C. AWS Simple Token Service (AWS STS)

D. Amazon GuardDuty

Answer: C

149. A company has an AWS Business Support plan. The company needs to gain access to the AWS DDoS Response Team (DRT) to help mitigate DDoS events. Which AWS service or resource must the company use to meet these requirements?

A. AWS Shield Standard

B. AWS Enterprise Support

C. AWS WAF

D. AWS Shield Advanced

Answer: D

150. An AWS user wants to proactively detect when an instance or account might be compromised or if there are threats from attacks. Which AWS service should the user choose?

A. Amazon Inspector

B. AWS WAF

C. Amazon GuardDuty

D. AWS Shield

Answer: C

151. A company plans to launch an ecommerce website that contains many images for a product catalog. The company wants to keep the cost of running the website within a specific budget. Which AWS service or tool should the company use to monitor the ongoing costs of the website?

A. AWS CloudFormation

B. AWS Cost Explorer

C. AWS SDKs

D. EC2 Image Builder

Answer: B

152. A company wants to visualize and manage AWS Cloud costs and usage for a specific period of time. Which AWS service or feature will meet these requirements?

A. Cost Explorer

B. AWS Budgets

C. Consolidated billing

D. AWS Organizations

Answer: A

153. A company has multiple AWS accounts. The company needs to receive a consolidated bill from AWS and must centrally manage security and compliance. Which AWS service or feature should the company use to meet these requirements?

A. AWS Security Hub

B. AWS Config

C. AWS Organizations

D. AWS Cost and Usage Report

Answer: C

154. A company has moved all its infrastructure to the AWS Cloud. To plan ahead for each quarter, the finance team wants to track the cost and usage data of all resources from previous months. The finance team wants to automatically generate reports that contains the data. Which AWS service or feature should the finance team use to meet these requirements?

A. Amazon Detective

B. AWS Pricing Calculator

C. AWS Budgets

D. AWS Savings Plans

Answer: C

155. Which AWS service or tool provides a visualization of historical AWS spending patterns and projections of future AWS costs?

A. Amazon Cloud Watch

B. AWS Budgets

C. AWS Cost and Usage Report

D. Cost Explorer

Answer: D

156. A company wants to track the monthly cost and usage of all Amazon EC2 instances in a specific AWS environment. Which AWS service or tool will meet these requirements?
    A. AWS Trusted Advisor
    B. AWS Compute Optimizer
    C. AWS Cost Anomaly Detection
    D. AWS Budgets

Answer: D

157. A company is using multiple AWS accounts for different business teams. The finance team wants to receive one bill for all of the company's accounts. Which AWS service or tool should the finance team use to meet this requirement?
    A. AWS Budgets
    B. Cost Explorer
    C. AWS Trusted Advisor
    D. AWS Organizations

Answer: D

158. Which of the following can the AWS Pricing Calculator do?
    A. Provide users with access to their monthly bills.
    B. Project monthly AWS costs.
    C. Provide in-depth information about AWS pricing strategies.
    D. Calculate historical AWS costs

Answer: B

159. A company wants to implement detailed tracking of its cloud costs by department and project. Which AWS feature or service should the company use?
    A. AWS Marketplace
    B. AWS Budgets
    C. Cost allocation tags
    D. Consolidated billing

Answer: C

160. Which AWS service or tool helps users visualize, understand, and manage spending and usage over time?
    A. AWS Organizations
    B. AWS Pricing Calculator
    C. AWS Service Catalog

D. AWS Cost Explorer

Answer: D

161. A company has multiple AWS accounts that include compute workloads that cannot be interrupted. The company wants to obtain billing discounts that are based on the company's use of AWS services. Which AWS feature or purchasing option will meet these requirements?

A. Resource tagging

B. Consolidated billing

C. Pay-as-you-go pricing

D. Spot Instances

Answer: B

162. A company plans to migrate to AWS and wants to create cost estimates for its AWS use cases. Which AWS service or tool can the company use to meet these requirements?

A. AWS Pricing Calculator

B. Amazon CloudWatch

C. AWS Cost Explorer

D. AWS Budgets

Answer: A

163. What is the best practice for granting permissions in AWS IAM?

A. Share the AWS account root user credentials with all administrators

B. Apply the principle of "least privilege"by granting only the permissions required for a task

C. Grant "AdministratorAccess"to every user to avoid permission issues

D. Use a single IAM user for multiple employees to simplify management

Answer: B

164. Which AWS service or feature provides an additional layer of security by requiring a user to provide a unique code from a device in addition to their password?

A. AWS Shield

B. Multi-Factor Authentication (MFA)

C. AWS CloudTrail

D. AWS Key Management Service (AWS KMS)

Answer: B

165. A company stores sensitive customer data including credit card numbers and personal information in Amazon S3 buckets. The security team wants to automatically identify and classify this sensitive data. Which AWS service should they use?

A. Amazon Macie

B. WS Security Hub

C. Amazon Inspector

D. Amazon GuardDuty

Answer: A

166. A company uses AWS Organizations. They want to prevent all accounts in a specific department from launching EC2 instances. Which feature should they use?

A. IAM Permission Boundary

B. AWS Config Rules

C. Service Control Policies (SCP)

D. AWS Security Hub

Answer: C

167. Which IAM component is used to grant temporary security credentials to AWS services like EC2 or Lambda?

A. IAM User

B. IAM Group

C. IAM Role

D. IAM Policy

Answer: C

168. Which AWS service is used to easily provision, manage, and deploy SSL/TLS certificates for encryption in transit?

A. AWS KMS

B. AWS Secrets Manager

C. AWS Certificate Manager (ACM)

D. SSM Parameter Store

Answer: C

169. Which type of KMS Key allows you to download and share the Public Key with others?

A. Symmetric KMS Keys

B. AWS Owned Keys

C. Asymmetric KMS Keys

D. AWS Managed Keys

Answer: C

170. Which AWS Cloud service can send alerts to customers if custom spending thresholds are exceeded?

A. AWS Cost Explorer

B. AWS Cost Allocation Tags

C. AWS Organizations

D. AWS Budgets

Answer: D

171. A company is planning to migrate its on-premises workloads to AWS. They want to estimate their potential monthly costs for a specific set of services (such as Amazon EC2, Amazon RDS, and Amazon S3) based on their expected usage before actually launching any resources. Which AWS tool should they use?

A. AWS Cost Explorer

B. AWS Pricing Calculator

C. AWS Budgets

D. AWS Trusted Advisor

Answer: B

172. Under the AWS Shared Responsibility Model, which of the following is a responsibility of the customer regarding billing and cost management?

A. Maintaining the physical billing infrastructure.

B. Managing cost allocation tags to track expenses by department.

C. Ensuring the global availability of the AWS Billing console

D. Determining the underlying hardware costs for EC2 instances.

Answer: B

173. A large enterprise has multiple AWS accounts for different business units. The finance department wants to receive a single, combined invoice for all accounts to simplify the payment process and benefit from volume-based discounts. Which feature of AWS Organizations should they implement?

A. Service Control Policies (SCPs)

B. Resource Groups

C. AWS Cost and Usage Report

D. Consolidated Billing

Answer: D

AI VIET NAM – AI COURSE 2025

Cloud Practitioner Quiz - Week 3

Lớp AWS (AIO 2025)

1. A company is looking at real-time processing of streaming big data for their ad-tech platform. Which of the following AWS services is the right choice for this requirement?
   A. Amazon Kinesis Data Streams
   B. Amazon Simple Queue Service (Amazon SQS)
   C. Amazon Redshift
   D. Amazon EMR

   Answer: A

2. An e-commerce application sends out messages to a downstream application whenever an order is created. The downstream application processes the messages and updates its own systems. Currently, the two applications directly communicate with each other. Which service will you use to decouple this architecture, without any communication loss between the two systems?
   A. Amazon Simple Queue Service (SQS)
   B. Amazon Kinesis Data Streams
   C. Amazon Simple Notification Service (Amazon SNS)
   D. AWS Lambda

   Answer: A

3. Which of the following is an AWS database service?
   A. AWS Database Migration Service (AWS DMS)
   B. Amazon Redshift
   C. AWS Glue
   D. AWS Storage Gateway

   Answer: B

4. An e-commerce company has deployed an RDS database in a single Availability Zone (AZ). The engineering team wants to ensure that in case of an AZ outage, the database should continue working on the same endpoint without any manual administrative intervention. Which of the following solutions can address this use-case?
   A. Configure the database in RDS Multi-AZ deployment with automatic failover to the standby
   B. Configure the database in RDS read replica mode with automatic failover to the standby
   C. Provision the database via AWS CloudFormation
   D. Deploy the database via AWS Elastic Beanstalk

   Answer: A

5. An e-commerce company wants to store data from a recommendation engine in a database. As a Cloud Practioner, which AWS service would you recommend to provide this functionality with the LEAST operational overhead for any scale?
   A. Amazon Neptune
   B. Amazon Simple Storage Service (Amazon S3)
   C. Amazon Relational Database Service (Amazon RDS)
   D. Amazon DynamoDB

   Answer: D

6. Which of the following AWS services allows a database to have flexible schema and supports document data models?
   A. Amazon DynamoDB
   B. Amazon Relational Database Service (Amazon RDS)
   C. Amazon Aurora
   D. Amazon Redshift

   Answer: A

7. What is the primary benefit of deploying an Amazon Relational Database Service (Amazon RDS) database in a Read Replica configuration?
   A. Read Replica reduces database usage costs
   B. Read Replica protects the database from a regional failure
   C. Read Replica enhances database availability
   D. Read Replica improves database scalability

   Answer: D

8. Which AWS service can be used for online analytical processing?
   A. Amazon Relational Database Service (Amazon RDS)
   B. Amazon ElastiCache

C. Amazon Redshift

D. Amazon DynamoDB

Answer: C

9. Which of the following is correct regarding the Amazon Relational Database Service (Amazon RDS) service?

   A. You can use read replicas for disaster recovery only and multi-AZ deployment for improved read performance only

   B. You can use both read replicas and multi-AZ deployment for disaster recovery

   C. You can use read replicas for improved read performance only and multi-AZ deployment for disaster recovery only

   D. You can use both read replicas and multi-AZ deployment having single standby for improved read performance

   Answer: B

10. According to the AWS Shared Responsibility Model, which of the following are responsibilities of the customer for Amazon RDS?

    A. Database encryption

    B. Applying patches to the Amazon Relational Database Service (Amazon RDS) database

    C. Applying patches to the underlying OS

    D. Managing the underlying server hardware on which Amazon Relational Database Service (Amazon RDS) runs

    Answer: A

11. A company has an Amazon EC2 instance in a private subnet. The company wants to initiate a connection to the internet to pull operating system updates while preventing traffic from the internet from accessing the EC2 instance. Which AWS managed service allows this?

    A. VPC endpoint

    B. NAT gateway

    C. Amazon PrivateLink

    D. VPC peering

    Answer: B

12. A company wants to securely access an Amazon S3 bucket from an Amazon EC2 instance without accessing the internet. What should the company use to accomplish this goal?

    A. VPN connection

    B. Internet gateway

    C. VPC endpoint

    D. NAT gateway

Answer: C

13. A network engineer needs to build a hybrid cloud architecture connecting on premises networks to the AWS Cloud using AWS Direct Connect. The company has a few VPCs in a single AWS Region and expects to increase the number of VPCs to hundreds over time. Which AWS service or feature should the engineer use to simplify and scale this connectivity as the VPCs increase in number?

    A. VPC endpoints

    B. AWS Transit Gateway

    C. Amazon Route 53

    D. AWS Secrets Manager

    Answer: B

14. Which task can a company perform by using security groups in the AWS Cloud?

    A. Allow access to an Amazon EC2 instance through only a specific port.

    B. Deny access to malicious IP addresses at a subnet level.

    C. Protect data that is cached by Amazon CloudFront.

    D. Apply a stateless firewall to an Amazon EC2 instance.

    Answer: A

15. A company hosts its website on Amazon EC2 instances. The company needs to ensure that the website reaches a global audience and provides minimum latency to users. Which AWS service should the company use to meet these requirements?

    A. Amazon Route 53

    B. Amazon CloudFront

    C. Elastic Load Balancing

    D. AWS Lambda

    Answer: B

16. A company wants to monitor and block malicious HTTP and HTTPS requests that its Amazon CloudFront distributions receive. Which AWS service should the company use to meet these requirements?

    A. Amazon GuardDuty

    B. Amazon Inspector

    C. AWS WAF

    D. Amazon Detective

    Answer: C

17. Which AWS service or feature requires an internet service provider (ISP) and a colocation facility to be implemented?

A. AWS VPN

B. Amazon Connect

C. AWS Direct Connect

D. Internet gateway

Answer: C

18. Which AWS service or feature gives users the ability to capture information about network traffic in a VPC?

A. VPC Flow Logs

B. Amazon Inspector

C. VPC route tables

D. AWS CloudTrail

Answer: A

19. Which of the following accurately describes a Public Subnet within an AWS Virtual Private Cloud (VPC)?

A. It is a subnet where Security Groups cannot be applied.

B. It is a logically isolated section of the AWS cloud with no external access.

C. It is a subnet that relies solely on a NAT Gateway for outbound internet access.

D. It is a subnet that has a route to the internet via an Internet Gateway.

Answer: D

20. When configuring network security in AWS, what is a key difference between Security Groups and Network ACLs?

A. Security Groups are stateful firewalls, while Network ACLs are stateless firewalls.

B. Security Groups are stateless, while Network ACLs are stateful.

C. Security Groups support 'Deny' rules, while Network ACLs only support 'Allow' rules.

D. Security Groups operate at the subnet level, while Network ACLs operate at the instance level.

Answer: A

21. A startup wants a managed relational database for a steady, predictable workload using standard MySQL with minimal administration. High availability in a single Region is required, but they do not need global replication. Which service best fits their needs?

A. Amazon RDS for MySQL

B. Amazon DynamoDB

C. Amazon Aurora Global Database

D. Amazon Neptune

Answer: A

22. You notice that requests to your application fronted by an Application Load Balancer (ALB) sometimes fail when one Availability Zone experiences issues, even though other Zones are healthy. Which configuration change can improve resilience to AZ failures?
    A. Disable health checks on the ALB so that it continues sending traffic to all targets evenly.
    B. Register targets with the ALB only in a single Availability Zone to simplify routing.
    C. Enable cross-zone load balancing on the ALB and ensure there are healthy targets in multiple Availability Zones.
    D. Use Network ACLs to block traffic from the affected Availability Zone.

    Answer: C

23. An e-commerce company is experiencing order loss during peak traffic because the Order service directly calls the Payment service, and they want to decouple the system while ensuring no data loss, which solution is most appropriate?
    A. Amazon SQS
    B. Amazon SNS
    C. Amazon EC2
    D. Amazon RDS

    Answer: A

24. An AdTech company needs to process clickstream data from millions of users in real-time with very high throughput, which service is the best fit?
    A. Amazon SQS
    B. Amazon SNS
    C. Amazon Kinesis
    D. Amazon S3

    Answer: C

25. A student is learning about managed database services on AWS. Which service is specifically designed to manage relational databases and automate tasks like backups and patching?
    A. Amazon SQS for messaging between distributed applications.
    B. Amazon RDS for managing relational database instances.
    C. Amazon SNS for broadcasting messages to subscribers.
    D. Amazon S3 for storing unstructured object data.

    Answer: B

26. Which of the following database engines is not currently supported by Amazon RDS?
    A. PostgreSQL
    B. MariaDB
    C. MongoDB

D. IBM DB2

Answer: C

27. When using Amazon SQS, what happens to a message after a consumer retrieves it from the queue, but before it is explicitly deleted?
    A. The message is immediately removed from the queue to prevent duplicate processing.
    B. The message remains visible to all other consumers immediately.
    C. The message enters a visibility timeout period where it is hidden from other consumers.
    D. The message is moved to a Dead Letter Queue (DLQ) automatically.

    Answer: C

28. Which AWS service is used to send one message to multiple subscribers using a publish/subscribe model?
    A. Amazon SQS
    B. Amazon SNS
    C. Amazon RDS
    D. Amazon ElastiCache

    Answer: B

29. An e-commerce application experiences heavy read traffic during sales events. The primary database becomes a bottleneck, but write operations must remain consistent. What is the best solution?
    A. Enable Multi-AZ deployment to distribute read traffic.
    B. Replace RDS with SQS to queue database operations.
    C. Use SNS to broadcast database queries to all services.
    D. Create Read Replicas and redirect read queries there.

    Answer: D

30. What is the fundamental difference in replication mechanism between RDS Read Replicas and RDS Multi-AZ?
    A. ead Replicas uses synchronous replication (SYNC), while Multi-AZ uses asynchronous replication (ASYNC).
    B. Both features use synchronous replication (SYNC) to ensure no data loss.
    C. Read Replicas uses asynchronous replication (ASYNC), while Multi-AZ uses synchronous replication (SYNC).
    D. Read Replicas only supports replication within the same Availability Zone, while Multi-AZ

    Answer: C

31. A developer needs to ensure that a single event, such as 'New Order', is delivered to multiple independent downstream applications, and that each application receives its own copy of the event for separate processing. Which AWS architecture is most appropriate?

A. Use a single SQS queue that all processing applications poll from.

B. Use an SNS topic to publish the event, with multiple SQS queues subscribed to that topic.

C. Use RDS Proxy to broadcast the event to multiple database tables.

D. Use ElastiCache Redis to store the event and have applications check for updates.

Answer: B

32. Which statement is correct about Amazon SQS Standard Queue?

A. It guarantees strict message ordering

B. It can send only one message at a time

C. A message may be delivered more than once

D. It can only be used with EC2, not with Lambda

Answer: C

33. A company needs to build an order processing system where, whenever a new order is placed, information needs to be sent simultaneously to three independent systems: Warehouse, Accounting, and Shipping. Which AWS service is best suited for implementing this model?

A. Amazon ElastiCache

B. Amazon SNS

C. AWS Database Migration Service (DMS)

D. Amazon SQS

Answer: B

34. What is the primary difference between a Public Subnet and a Private Subnet in an AWS VPC?

A. A Public Subnet has a route table connected directly to an Internet Gateway, while a Private Subnet does not.

B. A Private Subnet can be accessed directly from the Internet, while a Public Subnet cannot.

C. Public Subnets only support IPv4, while Private Subnets only support IPv6.

D. A Public Subnet is always located in the first Availability Zone (AZ), while Private Subnets are in the others.

Answer: A

35. In Amazon RDS, which feature is primarily designed for High Availability and Disaster Recovery rather than read scaling?

A. Read Replicas

B. Multi-AZ Deployment

C. RDS Proxy

D. Vertical Scaling

Answer: B

36. Which of the following engines are supported by Amazon ElastiCache?
    A. MySQL and PostgreSQL
    B. Oracle and MariaDB
    C. Redis and Memcached
    D. DynamoDB and MongoDB

    Answer: C

37. Which component in the AWS network security model acts as a firewall for Subnets (Subnet-level) and supports both "Allow"and "Deny"rules?
    A. Security Groups
    B. Internet Gateway
    C. Route Table
    D. Network ACLs (NACL)

    Answer: D

38. Which of the following is a key benefit of using Amazon SQS (Simple Queue Service) in system integration?
    A. It enables real-time, synchronous communication between services.
    B. It allows for the "decoupling"of application components to improve scalability.
    C. It automatically sends email notifications to customers when a new message arrives.
    D. It stores data permanently until the user manually deletes it.

    Answer: B

39. In the CIDR notation 174.16.0.0/24, how many IP addresses are available in total?
    A. 65536
    B. 256
    C. 16
    D. 16.7 million

    Answer: B

40. A company wants to host a backend database that should NOT be accessible from the internet. Which subnet type is most suitable?
    A. Internet Gateway Subnet
    B. Edge Location
    C. Private Subnet
    D. Public Subnet

    Answer: C

41. Which component is required to allow resources in a public subnet to communicate with the internet?
    A. Customer Gateway
    B. Internet Gateway
    C. Virtual Private Gateway
    D. NAT Gateway

    Answer: B

42. What is the primary difference between a Security Group and a Network ACL (NACL)?
    A. Security Groups support "Allow"rule only; NACLs support both "Allow"and "Deny"rule
    B. Security Groups are stateless; NACLs are stateful
    C. Security Groups work at the subnet level; NACLs work at the instance level
    D. There is no functional difference; they are redundant service

    Answer: A

43. When configuring a Security Group for your Web Server in the Lab, what should you 'Allow' to ensure the Application Load Balancer (ALB) can reach it?
    A. All outbound traffic to the VPC CIDR
    B. SSH traffic from the Internet
    C. HTTP traffic from the Security Group ID of the ALB
    D. HTTP traffic from source 0.0.0.0/0

    Answer: C

44. A web application is hosted on EC2 instances in a Private Subnet. You have successfully created a NAT Gateway in a Public Subnet. However, the instances still cannot access the internet to download updates. What is the most likely missing step?
    A. The EC2 instances do not have a Public IP address assigned.
    B. The Route Table associated with the Private Subnet does not have a route pointing 0.0.0.0/0 to the NAT Gateway.
    C. The Security Group of the NAT Gateway is blocking inbound traffic from the Private Subnet
    D. The NAT Gateway needs to be attached to an Internet Gateway directly.

    Answer: B

45. You have two VPCs (VPC A and VPC B) in the same region. You want instances in VPC A to communicate with instances in VPC B using private IP addresses. Which AWS feature should you implement?
    A. NAT Gateway
    B. VPC Peering

C. Internet Gateway

D. AWS CloudFront

Answer: B

46. You want to perform a "Blue/Green"deployment where you shift 10

A. Simple Routing Policy.

B. Weighted Routing Policy

C. Failover Routing Policy.

D. Latency Routing Policy.

Answer: B

47. A security admin wants to prevent a specific range of malicious IP addresses from even reaching the subnets in a VPC. Which tool provides this 'Deny' capability at the network boundary?

A. Network ACLs

B. Security Groups

C. Application Load Balancer

D. NAT Gateway

Answer: A

48. A solutions architect is troubleshooting a connectivity issue. An EC2 instance in a public subnet can reach the internet, but an instance in the private subnet cannot, even though a NAT Gateway is present. What is the most likely cause?

A. The instance is missing an Elastic IP address

B. The NAT Gateway is located in the private subnet

C. The Route Table for the private subnet points 0.0.0.0/0 to the NAT Gateway

D. The private Instance does not have a public IP address

Answer: B

49. A company wants to implement a 'Defense in Depth' strategy. They need to block a specific list of known malicious IP addresses at the subnet level. Which feature should they use?

A. Route 53

B. Internet Gateway

C. Security Groups

D. Network ACLs

Answer: D

50. An Application Load Balancer (ALB) is configured to sit in front of several EC2 instances. To ensure maximum security, what should be the source of the inbound HTTP rule on the EC2 instances?

A. The Security Group ID of the ALB

B. The public IP of the ALB

C. The ID of the Internet Gateway

D. 0.0.0.0/0

Answer: A

51. You are hosting a static website on Amazon S3. You want to use a custom domain name (e.g., www.myapp.com). Which AWS service is required to map your domain name to the S3 bucket?

A. AWS CloudMap

B. Amazon VPC

C. AWS Direct Connect

D. Amazon Route 53

Answer: D

52. A company needs a dedicated, private connection between their on-premises data center and AWS that does not go over the public internet. Which service should they choose?

A. Amazon CloudFront

B. AWS Direct Connect

C. AWS VPN

D. VPC Peering

Answer: B

53. Which component is necessary to allow a VPC to communicate with an S3 bucket without using a NAT Gateway or an Internet Gateway?

A. Customer Gateway

B. VPC Peering

C. VPC Endpoint

D. AWS Transit Gateway

Answer: C

54. A company runs a read-heavy analytics workload that is overloading its Amazon RDS production database. Which solution should a cloud architect recommend to offload the read traffic while keeping the production database unaffected?

A. Enable Multi-AZ deployment on the RDS instance

B. Create an RDS Read Replica and direct analytics queries to it

C. Increase the RDS instance size to handle more connections

D. Enable RDS Storage Auto Scaling to expand capacity

Answer: B

55. Which Amazon S3 storage class replicates data within a single Availability Zone and is NOT resilient to the physical loss of an entire AZ?
    A. Glacier Deep Archive
    B. One Zone-IA
    C. Intelligent-Tiering
    D. Standard-IA

    Answer: B

56. Which AWS service allows you to create a private, logically isolated virtual network in the cloud?
    A. AWS Direct Connect
    B. Amazon VPC (Virtual Private Cloud)
    C. Amazon Route 53
    D. Amazon CloudFront

    Answer: B

57. In a VPC, what is the core difference between a 'Public Subnet' and a 'Private Subnet'?
    A. A Public Subnet has "Allow All"rules in its Security Group
    B. A Public Subnet has a Route Table that points to an Internet Gateway (IGW)
    C. A Private Subnet does not support internal IP addresses
    D. A Public Subnet is always in a different Availability Zone than a Private Subnet.

    Answer: B

58. What is the name of AWS's DNS service, and what is the default port for DNS services?
    A. Amazon Route 53, Port 53
    B. Amazon CloudFront, Port 80
    C. AWS Direct Connect, Port 53
    D. Amazon VPC, Port 22

    Answer: A

59. Which service is best for storing common database query results, which helps to alleviate database access load?
    A. Amazon Machine Learning.
    B. Amazon SQS.
    C. Amazon ElastiCache.
    D. Amazon EC2 Instance Store.

    Answer: C

60. A Cloud Practitioner must determine if any security groups in an AWS account have been provisioned to allow unrestricted access for specific ports. What is the SIMPLEST way to do this?
    A. Review the inbound rules for each security group in the Amazon EC2 management console to check for port 0.0.0.0/0.
    B. Run AWS Trusted Advisor and review the findings.
    C. Open the AWS IAM console and check the inbound rule filters for open access.
    D. In AWS Config, create a custom rule that invokes an AWS Lambda function to review firewall rules for inbound access.

    Answer: B

61. A company is considering using AWS for a self-hosted database that requires a nightly shutdown for maintenance and cost-saving purposes. Which service should the company use?
    A. Amazon Redshift.
    B. Amazon DynamoDB.
    C. Amazon Elastic Compute Cloud (Amazon EC2) with Amazon EC2 instance store.
    D. Amazon EC2 with Amazon Elastic Block Store (Amazon EBS).

    Answer: D

62. Which of the following scenarios CAN'T trigger an automatic failover in Amazon RDS Multi-AZ deployment?
    A. Availability Zone (AZ) failure
    B. Network failure
    C. Instance or storage failure
    D. High read traffic on the primary database

    Answer: D

63. A company is upgrading its Amazon RDS database from Single-AZ to Multi-AZ deployment to improve high availability. Which of the following correctly describes the process AWS performs during this upgrade?
    A. A snapshot of the primary DB is created → A standby DB is restored in the same AZ → Asynchronous replication is established
    B. A snapshot of the primary DB is created → A standby DB is restored in another AZ → Asynchronous replication is established
    C. A backup of the primary DB is created → A standby DB is restored in another AZ → Synchronous replication is established
    D. A snapshot of the primary DB is created → A standby DB is restored in another AZ → Synchronous replication is established

    Answer: D

64. A company is planning to enable Amazon RDS Multi-AZ deployment for its production database. Which of the following is a primary advantage of using Multi-AZ?
    A. It improves read performance by distributing read traffic across multiple instances
    B. It provides automatic failover and high availability in case of infrastructure failure
    C. It reduces storage costs by compressing data across Availability Zones
    D. It allows manual promotion of standby instances for scaling purposes

    Answer: B

65. We have an RDS database that struggles to keep up with the demand of requests from our website. Our million users mostly read news, and we don't post news very often. Which solution is NOT adapted to this problem?
    A. An ElastiCache Cluster
    B. RDS Multi-AZ
    C. RDS Read Replicas
    D. Use Amazon CloudFront

    Answer: B

66. Which statement best describes the difference between Aurora replicas and RDS read replicas?
    A. Aurora replicas are synchronous, while RDS read replicas are always asynchronous
    B. Aurora replicas can only be used for failover, while RDS read replicas can only be used for read scaling
    C. Aurora replicas share the same underlying storage as the writer, while RDS read replicas maintain independent storage
    D. Aurora replicas are available for all RDS engines, while RDS read replicas are Aurora-only

    Answer: C

67. A company wants to run a relational database in AWS without managing most of the underlying operational work. Which service best fits this need?
    A. Amazon EC2
    B. Amazon RDS
    C. AWS Lambda
    D. Amazon VPC

    Answer: B

68. A team wants a relational database service in AWS that is designed for high availability and runs in a VPC with multiple subnets across Availability Zones. Which service is the best fit?
    A. Amazon Aurora
    B. Amazon SQS
    C. Amazon Route 53

D. Amazon CloudFront

Answer: A

69. You are designing a new OLTP application that requires high availability across two Availability Zones with automatic failover. You want minimal application changes and support for MySQL. Which managed database option best fits these requirements?

A. Amazon RDS for MySQL with Multi-AZ deployment

B. Single-AZ Amazon RDS for MySQL instance

C. Self-managed MySQL on EC2 instances in each Availability Zone

D. Amazon Aurora Serverless v2 (MySQL-compatible) in a single Region

Answer: A

70. A company is migrating a critical OLTP workload to AWS and requires a relational database that provides high availability across multiple Availability Zones, automatic failover, and minimal replication lag for read scalability. Which solution best meets these requirements?

A. Deploy Amazon RDS with Multi-AZ enabled and create Read Replicas

B. Deploy Amazon Aurora with Aurora Replicas across multiple AZs

C. Deploy Amazon DynamoDB with Global Tables enabled

D. Deploy Amazon Redshift with cross-region snapshot replication

Answer: B

71. An application processes incoming orders and must ensure that each order is processed exactly once, even in the presence of retries. The system should also support decoupling between services. Which AWS service and configuration best meet these requirements?

A. Use Amazon SNS with multiple subscribers

B. Use Amazon SQS Standard Queue with visibility timeout

C. Use Amazon SQS FIFO Queue with content-based deduplication

D. Use AWS DMS with ongoing replication enabled

Answer: C

72. Which statement about subnets is correct?

A. A subnet can span multiple AWS Regions

B. A subnet can span multiple Availability Zones

C. A subnet must reside in one Availability Zone

D. A subnet is used only for databases

Answer: C

73. A company is designing a highly secure VPC architecture. They want to ensure that only specific inbound traffic is allowed at the subnet level, while also maintaining instance-level security controls. Additionally, they need a solution that can explicitly deny certain IP ranges. Which combination of AWS features should be used?

A. Use Security Groups only

B. Use Network ACLs only

C. Use both Security Groups and Network ACLs

D. Use Route 53 with health checks

Answer: C

74. Which option describes a customer responsibility when designing a VPC for databases?

A. Choosing IP address ranges and subnets

B. Maintaining AWS data centers

C. Replacing failed physical servers

D. Managing the AWS global backbone network

Answer: A

75. A learner is studying DNS and wants to understand its main job in web access. What is the main purpose of DNS?

A. It translates private keys into IAM policies

B. It translates domain names into IP addresses

C. It translates packets into cached objects

D. It translates routes into subnet addresses

Answer: B

76. When configuring a Network Access Control List (NACL) for a subnet, which of the following characteristics distinguishes it from a Security Group?

A. It is stateful, meaning return traffic is automatically allowed.

B. It operates at the instance level rather than the subnet level.

C. It supports both "Allow"and "Deny"rules.

D. It only filters inbound traffic, while outbound traffic is always permitted.

Answer: C

77. Which AWS service acts as a virtual firewall at the instance level?

A. Route Table

B. Security Group

C. Network ACL

D. Internet Gateway

Answer: B

78. A global website has users in many countries, and pages load slowly because all content comes from one origin server.

A. Use one deeper route table association

B. Use one larger subnet with wider CIDR

C. Use a CDN with nearby edge caching

D. Use one stricter NACL with deny rules

Answer: C

79. You have a web server in a public subnet. You have opened port 80 in the Security Group's inbound rules. However, users still cannot reach the website. If the issue lies within the Network ACL, what is the most likely reason?

A. The NACL does not have an outbound rule allowing traffic back to the users.

B. The NACL automatically trusts the Security Group's "Allow"rule.

C. NACLs do not support HTTP traffic.

D. The Internet Gateway is not attached to the NACL.

Answer: A

80. Which type of subnet has a direct route to the internet through an Internet Gateway?

A. Private subnet

B. Isolated subnet

C. Public subnet

D. Internal subnet

Answer: C

81. A company is deploying a web application that uses Amazon RDS as its database. To ensure optimal security and the ability to perform software patching, which architecture is the MOST suitable?

A. Deploy the RDS instance in a public subnet so that the AWS technical team can access and patch the database directly via the internet.

B. Deploy the RDS instance in a private subnet, use a NAT gateway to allow outbound connections for software updates, and configure a security group to allow traffic only from the web server.

C. Use an internet gateway to provide a direct connection between end-users and the RDS instance to reduce query latency.

D. Deploy both the RDS instance and the web server in the same public subnet and use a Network ACL to block all inbound traffic from external sources.

Answer: B

82. In an Amazon VPC, which of the following conditions characterizes a subnet as a 'Public Subnet'?

A. It contains EC2 instances with the most powerful configurations.

B. It is configured to deny all inbound traffic from external sources.

C. It is located within a default Availability Zone.

D. It has a route in its route table that points to an Internet Gateway.

Answer: D

83. Which VPC component allows resources in a private subnet to connect outbound to the internet (for example, to download software updates,....) but prevents the internet from initiating a connection to those resources?

A. Security Group

B. Internet Gateway

C. NAT Gateway

D. Virtual Private Gateway

Answer: C

84. A subnet has route table with route to Internet Gateway, and its instances do NOT have public IPs Is this subnet considered public or private?

A. Public subnet

B. Private subnet

C. Hybrid subnet

D. Depends on Security Group

Answer: B

85. A web application is experiencing high latency due to read-heavy traffic on its primary Amazon RDS database. The development team wants to implement an in-memory caching layer to reduce the load on the database and improve response times. Which AWS service is the best choice for this use case?

A. Amazon S3

B. Amazon SQS

C. Amazon ElastiCache

D. Amazon Aurora Serverless

Answer: C

86. An architecture includes a front-end web application and a back-end processing application. Traffic spikes frequently cause the tightly coupled back-end to become overwhelmed and fail. Which AWS service can be used to decouple these applications and asynchronously store messages until the back-end is ready to process them?

A. Amazon SNS

B. Amazon SQS

C. Amazon ElastiCache

D. Amazon API Gateway

Answer: B

87. Which of the following is a characteristic of AWS Security Groups?
    A. They operate at the subnet level.
    B. They support both "Allow"and "Deny"rules.
    C. They are stateless and require explicit rules for both inbound and outbound return traffic.
    D. They are stateful, meaning they automatically allow return traffic.

    Answer: D

88. A company needs to migrate its on-premises Oracle database to an Amazon Aurora PostgreSQL-compatible cluster. They require the source database to remain fully operational during the migration and need to convert the database schema before moving the data. Which combination of AWS tools should they use?
    A. AWS Snowball and Amazon S3
    B. AWS Backup and Amazon RDS Snapshot
    C. AWS Schema Conversion Tool (SCT) and AWS Database Migration Service (DMS)
    D. Amazon RDS Proxy and AWS Lambda

    Answer: C

89. A media company wants to collect and store large-scale, real-time streaming data from IoT devices for advanced analytics. They need the ability to "replay"or reprocess the data for up to 365 days if necessary. Which service should they choose?
    A. Amazon Data Firehose
    B. Amazon SQS FIFO Queue
    C. Amazon Kinesis Data Streams
    D. Amazon MQ

    Answer: C

90. Which of the following is NOT a supported database engine in Amazon RDS?
    A. PostgreSQL
    B. MySQL
    C. MongoDB
    D. Oracle

    Answer: C

91. An EC2 instance is launched in a private subnet and needs to download updates from the internet, but must not be directly reachable from the internet. Which solution should be used?
    A. Attach an Internet Gateway to the subnet
    B. Use a NAT Gateway in a public subnet and update the route table
    C. Assign a public IP address to the instance
    D. Use a VPC Peering connection

Answer: B

92. Which statement correctly describes the difference between Security Groups and Network ACLs?
    A. Security Groups are stateless, while NACLs are stateful
    B. Security Groups and NACLs are both stateless
    C. Security Groups are stateful, while NACLs are stateless
    D. Both Security Groups and NACLs are stateful

    Answer: C

93. A packet is allowed by a Security Group but denied by a Network ACL. What happens?
    A. Allowed (Security Group wins)
    B. Allowed (first match wins)
    C. Denied
    D. Depends on subnet type

    Answer: C

94. According to the presentation, what is the primary purpose of an Amazon VPC (Virtual Private Cloud)?
    A. To deliver content to users with low latency.
    B. To deliver content to users with low latency.
    C. To translate human-readable domain names into IP addresses.
    D. To automatically distribute incoming network traffic across multiple targets.

    Answer: B

95. How does Amazon Aurora Database Cloning differ from a traditional snapshot-and-restore approach?
    A. Cloning requires a full copy of all data at creation time, making it slower but more reliable
    B. Cloning uses a copy-on-write protocol, sharing the same storage volume initially, making it much faster and cost-efficient
    C. Cloning creates an independent storage volume immediately, which doubles the storage cost
    D. Cloning is only available for cross-region duplication and cannot be used within the same region

    Answer: B

96. What is the defining characteristic of a Public Subnet compared to a Private Subnet?
    A. It requires a NAT Gateway to allow outbound internet access.
    B. It does not have a route to the internet at all.

C. It has a direct route to the internet via an Internet Gateway.

D. It is primarily designed to host backend databases and internal APIs.

Answer: C

97. A cloud engineer has deployed several backend database instances in a private subnet within a VPC. These instances need to securely download software patches from the internet, but they must not be directly accessible from the public internet. Which AWS component is required to allow this outbound traffic?

A. Internet Gateway

B. NAT Gateway

C. Network ACL

D. Route 53

Answer: B

98. According to the presentation, what is the relationship between an AWS Subnet and an Availability Zone (AZ)?

A. A single subnet can span across multiple Availability Zones.

B. An Availability Zone is a small slice of a subnet's IP range.

C. A subnet is a slice of your VPC's IP range that lives in a specific Availability Zone.

D. A subnet automatically creates new Availability Zones for disaster recovery.

Answer: C

99. A company runs a production MySQL database on Amazon RDS that handles heavy read/write traffic. They now need to run complex analytics and reporting queries without impacting production performance. Additionally, they want to ensure the database remains available even if an entire Availability Zone fails. Which combination of RDS features should they implement?

A. Enable Multi-AZ deployment for read scaling, and use RDS Proxy for analytics queries

B. Create a Read Replica for the analytics workload, and enable Multi-AZ deployment for high availability

C. Enable Aurora Serverless for analytics, and use ElastiCache for high availability

D. Create multiple Read Replicas in the same AZ for both analytics and disaster recovery

Answer: B

100. Which of the following statements is true regarding Security Groups in AWS?

A. They operate at the subnet level to control inbound and outbound traffic.

B. They are stateless and require explicit rules for return traffic.

C. They operate at the instance level and are stateful.

D. They evaluate network traffic rules strictly in numbered order.

Answer: C

101. A company wants to improve the read performance of their Amazon RDS database because a new reporting application is causing high load. Which feature should they use to offload this traffic without affecting the primary database?
    A. Multi-AZ deployment
    B. Amazon RDS Read Replicas
    C. Amazon SQS
    D. Automated Backups

    Answer: B

102. By default, how do Security Groups handle outbound network traffic?
    A. They block all outbound traffic completely.
    B. They allow outbound traffic only to resources located within the same VPC.
    C. They only allow outbound traffic on standard ports like 80 and 443.
    D. They allow all outbound traffic.

    Answer: D

103. Which disaster recovery strategy has the lowest RTO (Recovery Time Objective) and RPO (Recovery Point Objective) by keeping a fully functional copy of the environment running in another region?
    A. Pilot Flight
    B. Warm Standby
    C. Active-Active (Multi-site)
    D. ackup and Restore

    Answer: C

104. A startup is developing a new application with infrequent and unpredictable workloads. They want a database that can automatically start, stop, and scale capacity up or down based on the application's needs to optimize costs. Which AWS database solution is the best fit?
    A. Amazon RDS Multi-AZ
    B. Amazon Aurora Serverless
    C. Amazon Aurora Global Database
    D. Amazon RDS Read Replicas

    Answer: B

105. Which network security feature allows you to create explicit "Deny"rules?
    A. Security Groups
    B. Network ACLs

C. Route Tables

D. Internet Gateways

Answer: B

106. How many copies of your data does Aurora automatically replicate across Availability Zones?

A. 6 copies across 3 AZs

B. 1 copy per AZ in every region

C. 3 copies in a single AZ

D. 2 copies across 2 AZs

Answer: A

107. Which component is required for resources in a Private Subnet to initiate outbound internet traffic?

A. Internet Gateway

B. Route 53

C. NAT Gateway

D. Bastion Host

Answer: C

108. When migrating an on-premises database to AWS, which service helps you migrate data quickly and securely while the source database remains operational?

A. AWS Snowball

B. AWS Database Migration Service (AWS DMS)

C. AWS DataSync

D. AWS Elastic File System (EFS)

Answer: B

109. Which AWS service is best suited for running a managed relational database such as MySQL or PostgreSQL without managing the underlying infrastructure?

A. Amazon DynamoDB

B. Amazon RDS

C. Amazon SQS

D. Amazon SNS

Answer: B

110. Which RDS feature provides high availability and automatic failover by maintaining a synchronous secondary copy of the database in a different Availability Zone?

A. Database Snapshots

B. Multi-AZ Deployment

C. Read Replicas

D. RDS Proxy

Answer: B

111. In the 'Pilot Light' disaster recovery strategy, what is typically kept running in the recovery region?

A. The web server, but not the database

B. Nothing; everything is turned off

C. Only the database/stores with live replication

D. A full-sized clone of the application stack

Answer: C

112. What is a key difference between Amazon Aurora and standard Amazon RDS database engines?

A. Aurora is a messaging service for event-driven systems

B. Aurora is AWS's proprietary high-performance relational database compatible with MySQL and PostgreSQL

C. Aurora is a NoSQL database designed for millisecond latency at any scale

D. Aurora is used only for data warehousing

Answer: B

113. A company wants to block a specific malicious IP address from accessing their application. Which option is the MOST effective?

A. Modify the Security Group to deny the IP

B. Modify the Network ACL to deny the IP

C. Remove inbound rules from Security Group

D. Add a route table rule to block the IP

Answer: B

114. Which of the following is a benefit of choosing Amazon Aurora over standard RDS MySQL?

A. it allows manual management of the underlying OS

B. It features is a distributed, fail-tolerant, self-healing storage system

C. It cheaper than RDS in every small workloads

D. It served SQL Server

Answer: B

115. Which AWS database service is fully managed, NoSQL, and designed for single-digit millisecond performance at scale?

A. Amazon Redshift

B. Amazon Aurora

C. Amazon DynamoDB

D. Amazon RDS for SQL Server

Answer: C

116. Which service is the best choice for large-scale analytical queries across massive structured datasets?

A. Amazon Redshift

B. Amazon SQS

C. Amazon SNS

D. Amazon DMS

Answer: A

117. A security architect is configuring a virtual firewall for an Amazon EC2 instance to control inbound and outbound traffic. The firewall must be stateful, meaning that if an inbound request is allowed, the return traffic is automatically permitted. Which AWS feature provides this instance-level security?

A. Security Groups

B. Network ACLs

C. Route Tables

D. Internet Gateway

Answer: A

118. What type of replication does Amazon RDS Multi-AZ use, and what is its primary purpose?

A. Asynchronous replication, used for scaling read traffic

B. Synchronous replication, used for disaster recovery and high availability

C. Asynchronous replication, used for disaster recovery and high availability

D. Synchronous replication, used for scaling read traffic

Answer: B

119. What is the main purpose of AWS Database Migration Service (DMS)?

A. Sending notifications to subscribers

B. Migrating and replicating databases with minimal downtime

C. Storing relational data for online transaction processing

D. Running large-scale data warehouse queries

Answer: B

120. Which AWS service follows a publish/subscribe model and can push messages to multiple subscribers?

A. Amazon SQS

B. Amazon SNS

C. Amazon RDS

D. Amazon DynamoDB

Answer: B

121. How does Amazon Aurora Database Cloning differ from a traditional snapshot-and-restore approach?

A. Cloning requires a full copy of all data at creation time, making it slower but more reliable

B. Cloning uses a copy-on-write protocol, sharing the same storage volume initially, making it much faster and cost-efficient

C. Cloning creates an independent storage volume immediately, which doubles the storage cost

D. Cloning is only available for cross-region duplication and cannot be used within the same region

Answer: B

122. Which AWS service is primarily used to decouple application components by storing messages in a queue until they are processed?

A. Amazon Aurora

B. Amazon SQS

C. Amazon Redshift

D. Amazon SNS

Answer: B

123. Which statement best describes the difference between Amazon SNS and Amazon SQS?

A. SNS stores messages for long-term analytics, while SQS runs SQL queries

B. SNS pushes messages to subscribers, while SQS queues messages for consumers to pull

C. SNS is relational, while SQS is NoSQL

D. SNS and SQS are both database migration tools

Answer: B

124. Which type of replication does an Amazon RDS Read Replica use?

A. Synchronous replication

B. Multi-region synchronous replication

C. Snapshot-based replication

D. Asynchronous replication

Answer: D

125. Between the two engines supported by Amazon ElastiCache, which one supports Multi-AZ with Auto-Failover and data persistence?
    A. Redis
    B. Neither
    C. Both Memcached and Redis
    D. Memcached

    Answer: A

126. Which of the following can be described as a global content delivery network (CDN) service?
    A. AWS VPN
    B. AWS Direct Connect
    C. AWS Regions
    D. Amazon CloudFront

    Answer: D

127. You need a message queue that guarantees messages are processed in the exact order they were sent and prevents duplicates. Which queue type should you use?
    A. Priority Queue
    B. Standard Queue
    C. Dead-Letter Queue
    D. FIFO Queue

    Answer: D

128. What is the AWS service that provides a virtual network dedicated to your AWS account?
    A. AWS VPN
    B. AWS Subnets
    C. AWS Dedicated Hosts
    D. Amazon VPC

    Answer: D

129. You are working on two projects that require completely different network configurations. Which AWS service or feature will allow you to isolate resources and network configurations?
    A. Internet Gateways
    B. Virtual Private Cloud (VPC)
    C. Security Groups
    D. Amazon CloudFront

    Answer: B

130. A company needs to offload read traffic from their primary database to improve performance for a reporting application. Which RDS feature should they use?
    A. Multi-AZ Deployment
    B. Read Replicas
    C. RDS Proxy
    D. Storage Autoscaling

    Answer: B

131. Which term below has the role of simplifying routing by auto-adding VPN routes?
    A. Propagation
    B. Edge Association
    C. Local Route
    D. Destination

    Answer: A

132. Which of the below options is true of Amazon VPC?
    A. Amazon VPC allows customers to control user interactions with all other AWS resources
    B. AWS Customers have complete control over their Amazon VPC virtual networking environment
    C. AWS is responsible for all the management and configuration details of Amazon VPC
    D. Amazon VPC helps customers to review their AWS architecture and adopt best practices

    Answer: B

133. Which feature below does not belong to Private Subnet?
    A. No requirement of NAT Gateway Usage for internet access
    B. Not directly accessible from the internet
    C. Route table does not include a route to the Internet Gateway
    D. Used for Databases, Internal Application Servers, or Batch Processing

    Answer: A

134. Which feature allows an Aurora database to be restored to a specific point in time without using backups or creating a new cluster?
    A. Point-in-Time Restore (PITR)
    B. Backtrack
    C. Snapshot Restore
    D. Database Cloning

    Answer: B

135. A developer is planning to build a two-tier web application that has a MySQL database layer. Which of the following AWS database services would provide automated backups for the application?

A. A MySQL database installed on an EC2 instance

B. Amazon DynamoDB

C. Amazon Aurora

D. Amazon Neptune

Answer: C

136. Which service is best for storing common database query results, which helps to alleviate database access load?

A. Amazon Machine Learning

B. Amazon SQS

C. Amazon ElastiCache

D. Amazon EC2 Instance Store

Answer: C

137. Amazon Relational Database Service (Amazon RDS) offers which of the following benefits over traditional database management?

A. AWS manages the data stored in Amazon RDS tables

B. AWS manages the maintenance of the operating system

C. AWS automatically scales up instance types on demand

D. AWS manages the database type

Answer: B

138. What is one of the advantages of the Amazon Relational Database Service (Amazon RDS)?

A. It simplifies relational database administration tasks

B. It provides 99.99

C. It automatically scales databases for loads

D. It enabled users to dynamically adjust CPU and RAM resources

Answer: A

139. A customer needs to run a MySQL database that easily scales. Which AWS service should they use?

A. Amazon Aurora

B. Amazon Redshift

C. Amazon DynamoDB

D. Amazon ElastiCache

Answer: A

140. What is the AWS service that provides five times the performance of a standard MySQL database?

A. Amazon Redshift

B. Amazon DynamoDB

C. Amazon Aurora

D. Amazon Neptune

Answer: C

141. Which of the following Amazon RDS features facilitates offloading of database read activity?

A. Database Snapshots

B. Multi-AZ Deployments

C. Automated Backups

D. Read Replicas

Answer: D

142. You are running a financial services web application on AWS. The application uses a MySQL database to store the data. Which of the following AWS services would improve the performance of your application by allowing you to retrieve information from fast in-memory caches?

A. Amazon EFS

B. Amazon Neptune

C. Amazon ElastiCache

D. Amazon SQS

Answer: C

143. You want to decouple two microservices so that the "Order Service"can continue to function even if the "Shipping Service"is temporarily unavailable. Which service should you use?

A. Amazon SNS

B. Amazon RDS Proxy

C. Amazon SQS

D. Amazon Route 53

Answer: C

144. Which of the following is a feature of Amazon RDS that performs automatic failover when the primary database fails to respond?

A. RDS Single-AZ

B. RDS Write Replica

C. RDS Snapshots

D. RDS Multi-AZ

Answer: D

145. What is the primary benefit of deploying an Amazon RDS database with a Read Replica configuration?

A. Provide synchronous replication for Disaster Recovery.

B. Offload read-heavy traffic from the primary database instance.

C. Automatically scale storage capacity when space runs out.

D. Improve the write performance (INSERT, UPDATE) of the database.

Answer: B

146. A company needs to ensure their database can automatically failover to another Availability Zone with zero manual intervention in case of an outage. Which RDS feature should they enable?

A. Read Replicas

B. Multi-AZ Deployment

C. Auto Scaling Group

D. Database Snapshotting

Answer: B

147. A company wants to send a single alert that triggers an email notification to the admin and simultaneously starts a Lambda function to log the event. Which architecture is best?

A. SNS Topic with Email and Lambda as subscribers

B. SQS Queue with Email and Lambda as consumers

C. Amazon CloudWatch Logs

D. SES (Simple Email Service)

Answer: A

148. Which CIDR suffix allows for the largest number of total IP addresses?

A. /32

B. /24

C. /16

D. /8

Answer: D

149. An engineering team wants to ensure that in case of an Availability Zone (AZ) outage, their RDS database should continue working on the same endpoint without any manual administrative intervention. Which of the following solutions addresses this use-case?

A. Configure the database in RDS Multi-AZ deployment

B. Configure the database in RDS Read Replica mode

C. Provision the database via AWS CloudFormation

D. Enable RDS Storage Auto Scaling

Answer: A

150. Which database engine is a "cloud-native"AWS proprietary relational database that offers up to 5x the performance of standard MySQL?
    A. Amazon Redshift
    B. Amazon DynamoDB
    C. Amazon Aurora
    D. MariaDB

    Answer: C

151. A solutions architect is migrating a legacy SQL Server database to AWS. The company wants to retain full control over the underlying Operating System to install custom third-party software. Which option should they choose?
    A. Amazon RDS for SQL Server
    B. Amazon Aurora
    C. Amazon EC2 with SQL Server installed
    D. Amazon DynamoDB

    Answer: C

152. Which of the following is a key difference between Kinesis Data Streams and Kinesis Data Firehose?
    A. Data Streams is serverless, while Firehose requires manual scaling of shards.
    B. Data Streams can store data for up to 365 days, while Firehose does not have a data retention period.
    C. Firehose is used for real-time processing, while Data Streams is used for batch loading.
    D. Data Streams only supports S3 as a destination, while Firehose supports any database.

    Answer: B

153. A startup wants a fully managed NoSQL database with millisecond latency and minimal operational overhead. Which service is BEST?
    A. Amazon RDS
    B. Amazon Redshift
    C. Amazon DynamoDB
    D. Amazon Aurora

    Answer: C

154. Which AWS service is a managed relational database service that supports engines like MySQL, PostgreSQL, and Oracle?
    A. Amazon DynamoDB
    B. Amazon RDS
    C. Amazon Redshift

D. Amazon ElastiCache

Answer: B

155. 4. What is the main purpose of an Internet Gateway (IGW) in a VPC?

A. To connect two different VPCs together.

B. To allow communication between your VPC and the internet.

C. To encrypt data stored in S3 buckets.

D. To balance traffic across multiple EC2 instances.

Answer: B

156. A company needs to improve read performance of its production database without affecting write operations. Which solution is MOST appropriate?

A. Enable Multi-AZ

B. Create Read Replicas

C. Enable automated backups

D. Use snapshots

Answer: B

157. Which RDS feature uses synchronous replication and automatically fails over without manual intervention?

A. Multi-AZ

B. Read Replica

C. Backup Restore

D. Auto Scaling

Answer: A

158. An application needs extremely fast data access with very low latency for frequently accessed data. Which service should be used?

A. Amazon RDS

B. Amazon Aurora

C. Amazon ElastiCache

D. Amazon SQS

Answer: C

159. A system requires guaranteed message ordering and no duplicate processing. Which combination should be used?

A. SNS FIFO + SQS FIFO

B. SNS + SQS Standard

C. SQS Standard only

D. Amazon MQ

Answer: A

160. A company wants the lowest-cost disaster recovery strategy and can tolerate long recovery time. Which approach should they choose?

A. Multi-site

B. Warm Standby

C. Pilot Light

D. Backup Restore

Answer: D

161. Which statement correctly describes Aurora compared to standard RDS?

A. Aurora does not support replication

B. Aurora is slower than RDS

C. Aurora automatically scales storage up to large capacity

D. Aurora only supports one Availability Zone

Answer: C

162. A company wants to deploy its infrastructure in AWS while keeping full control over IP addressing, internal communication rules, and internet access configuration. Which AWS service provides this capability?

A. Security Group

B. Subnet

C. VPC

D. Route Table

Answer: C

163. A company creates a subnet that has a route to an Internet Gateway and is directly accessible from the internet. What type of subnet is this?

A. Private Subnet

B. Public Subnet

C. Isolated Subnet

D. Internal Subnet

Answer: B

164. A private subnet needs to access the internet for downloading updates but must not be directly accessible from the internet. Which component is required?

A. Internet Gateway

B. Route Table

C. NAT Gateway

D. Security Group

Answer: C

165. Which statement correctly describes a subnet in AWS?

A. It spans multiple regions

B. It is a physical server

C. It is a logical subdivision of a VPC within a specific AZ

D. It replaces VPC

Answer: C

166. A network engineer wants to control traffic routing for specific subnets in a VPC. Which component should they configure?

A. Security Group

B. Route Table

C. Internet Gateway

D. DNS

Answer: B

167. Which AWS network security component operates at the subnet level and supports both allow and deny rules?

A. Security Group

B. Network ACL

C. Route Table

D. NAT Gateway

Answer: B

168. A system uses a firewall that automatically allows return traffic without requiring explicit rules. What type of firewall is this?

A. Stateless firewall

B. Stateful firewall

C. NAT firewall

D. Route-based firewall

Answer: B

169. A user types a domain name into a browser and receives a web page. Which process is responsible for translating the domain name into an IP address?

A. Routing

B. NAT

C. DNS

D. CIDR

Answer: C

170. A company wants to divide its VPC IP range into smaller manageable segments for better organization and control. Which concept is being applied?

A. Route Table

B. CIDR

C. Internet Gateway

D. Subnetting

Answer: D

171. A company wants to host its public-facing web servers while keeping its database servers completely isolated from the internet. Which AWS VPC architecture best satisfies this requirement?

A. Place all servers in a single public subnet and use Security Groups to block database traffic

B. Place web servers in a public subnet and database servers in a private subnet within the same VPC

C. Create two separate VPCs: one for web servers and one for databases

D. Place all servers in a private subnet and attach an Internet Gateway directly to each EC2 instance

Answer: B

172. A developer in a private subnet needs to download OS security patches from the internet, but the company policy forbids assigning public IP addresses to backend servers. Which AWS service enables this outbound-only internet access?

A. Internet Gateway (IGW)

B. VPC Peering

C. AWS Direct Connect

D. NAT Gateway

Answer: D

173. A global e-commerce company wants to reduce latency for users worldwide when they access static content such as images and videos stored in Amazon S3. Which AWS service is the MOST appropriate solution?

A. Amazon CloudFront

B. AWS Global Accelerator

C. AWS Route 53 with latency-based routing

D. Elastic Load Balancing (ELB)

Answer: A

174. Which AWS service or tool can be used to set up a firewall to control traffic going into and coming out of an Amazon VPC subnet?

    A. Security group

    B. AWS WAF

    C. AWS Firewall Manager

    D. Network ACL

    Answer: D

175. A company needs stateless network filtering for its VPC. Which AWS service, tool, or feature will meet this requirement?

    A. AWS PrivateLink

    B. Security group

    C. Network access control list (ACL)

    D. AWS WAF

    Answer: C

176. To reduce costs, a company is planning to migrate a NoSQL database to AWS. Which AWS service is fully managed and can automatically scale throughput capacity to meet database workload demands?

    A. Amazon Redshift

    B. Amazon Aurora

    C. Amazon DynamoDB

    D. Amazon RDS

    Answer: C

177. Which networking component acts as a firewall for associated subnets, controlling both inbound and outbound traffic at the subnet level?

    A. NAT Gateway

    B. Internet Gateway

    C. Security Group

    D. Network Access Control List (NACL)

    Answer: D

178. You are designing a Hub-and-Spoke architecture with 50 VPCs that need to communicate with a central hub. You want to allow these VPCs to communicate with each other without creating hundreds of individual Peering connections. Which AWS service is the most optimal solution?

    A. AWS VPC Endpoint Services (PrivateLink)

B. AWS Transit Gateway

C. Full-mesh VPC Peering

D. AWS Software VPN

Answer: B

179. A company has an Amazon EC2 instance in a private subnet. The company wants to initiate a connection to the internet to pull operating system updates while preventing traffic from the internet from accessing the EC2 instance. Which AWS managed service allows this?

A. VPC endpoint

B. NAT gateway

C. Amazon PrivateLink

D. VPC peering

Answer: B

180. A company wants to securely access an Amazon S3 bucket from an Amazon EC2 instance without accessing the internet. What should the company use to accomplish this goal?

A. VPN connection

B. nternet gateway

C. VPC endpoint

D. NAT gateway

Answer: C

181. A network engineer needs to build a hybrid cloud architecture connecting on-premises networks to the AWS Cloud using AWS Direct Connect. The company has a few VPCs in a single AWS Region and expects to increase the number of VPCs to hundreds over time. Which AWS service or feature should the engineer use to simplify and scale this connectivity as the VPCs increase in number?

A. VPC endpoints

B. AWS Transit Gateway

C. Amazon Route 53

D. Amazon Route 53 D. AWS Secrets Manager

Answer: B

182. Which of the following AWS database services is best described as an in-memory data store used to improve application response time?

A. Amazon DynamoDB

B. Amazon RDS

C. Amazon ElastiCache

D. Amazon Redshift

Answer: C

183. What type of replication does Amazon RDS Multi-AZ use between the primary and standby instance?
   A. Asynchronous replication to minimize latency
   B. Synchronous replication to ensure data consistency
   C. Eventual consistency replication
   D. Snapshot-based replication every 5 minutes

   Answer: B

184. A company is migrating a legacy banking application with complex JOIN queries and strict ACID transaction requirements. Which AWS database is the most appropriate?
   A. Amazon DynamoDB
   B. Amazon Redshift
   C. Amazon RDS
   D. Amazon Neptune

   Answer: C

185. Which AWS service is a managed relational database service that supports engines like MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server?
   A. Amazon ElastiCache
   B. Amazon SQS
   C. Amazon RDS
   D. Amazon S3

   Answer: C

186. What is the primary use case for RDS Read Replicas?
   A. To provide synchronous replication for disaster recovery.
   B. To scale read operations by offloading traffic from the primary database.
   C. To improve the performance of database write operations.
   D. To automatically increase database storage when running low.

   Answer: B

187. Which statement accurately describes Amazon Aurora's storage reliability and write availability requirements?
   A. It maintains 3 copies of data in 2 AZs and requires a 2/3 quorum for writes.
   B. It maintains 6 copies of data in 3 AZs and requires a 3/6 quorum for writes.
   C. It maintains 6 copies of data in 3 AZs and requires a 4/6 quorum for writes.
   D. It uses asynchronous replication across 3 AZs to achieve a write quorum of 4/6.

Answer: C

188. When comparing ElastiCache engines, which feature is exclusive to Redis (not supported by Memcached)?

A. Multi-threaded architecture for simple caching.

B. Multi-AZ with Auto-Failover and Backup/Restore capabilities.

C. Data partitioning through sharding.

D. Non-persistent storage where data is lost on restart.

Answer: B

189. To implement a "Fan-out"architecture that preserves strict message ordering and deduplication across multiple worker services, which combination is required?

A. SNS Standard Topic + SQS FIFO Queues.

B. SNS FIFO Topic + SQS FIFO Queues.

C. SQS FIFO Queue + Multiple Lambda consumers.

D. Kinesis Data Streams + Amazon MQ.

Answer: B

190. A Web Server in a Public Subnet cannot receive external traffic. The Security Group allows inbound Port 80, and the Network ACL allows inbound Port 80. What is the most likely issue?

A. Security Groups are stateless and need an outbound rule for Port 80.

B. Network ACLs are stateless and need an outbound rule for ephemeral ports.

C. The NAT Gateway is not configured to allow inbound traffic.

D. Route 53 cannot resolve the instance's private IP address.

Answer: B

191. Which Disaster Recovery (DR) strategy involves running a functional but scaled-down version of the full system in the cloud to achieve a lower RTO?

A. Backup and Restore.

B. Pilot Light.

C. Warm Standby.

D. Multi-Site Active-Active.

Answer: C

192. A company is building an e-commerce application using a relational database. They require: - Automatic read scaling - Fast failover (under 30 seconds) - Minimal operational overhead

Which solution is the most appropriate?

A. Amazon RDS Multi-AZ with Read Replica

B. Amazon Aurora with Aurora Replicas

C. Amazon DynamoDB with DAX

D. Amazon RDS Single-AZ with automated backup

Answer: B

193. When migrating an on-premises Oracle database to Amazon Aurora MySQL, what is the primary role of the AWS Schema Conversion Tool (SCT)?

A. It provides continuous data replication using Change Data Capture (CDC).

B. It transports physical data volumes to AWS via Snowball devices.

C. It converts database objects (tables, indexes, views) to be compatible with a different engine.

D. It serves as the primary replication instance for AWS DMS.

Answer: C

194. An order processing system needs to ensure: - Independent service processing (decoupling) - No message loss - Ability to retry on failure

Which service is the most appropriate?

A. Amazon SNS

B. Amazon SQS Standard

C. Amazon SQS FIFO

D. Amazon Kinesis Data Streams

Answer: C

195. A company wants to implement a disaster recovery (DR) strategy with the following requirements: - Lowest possible cost - Recovery can take a few hours

Which strategy is the most appropriate?

A. Backup Restore

B. Pilot Light

C. Warm Standby

D. Multi-site Active-Active

Answer: A

196. An application is using Amazon RDS but is experiencing a bottleneck with a read-heavy workload. Which is the most optimal solution?

A. Increase instance size

B. Add Read Replica

C. Switch to Multi-AZ

D. Enable backup

Answer: B

197. A video processing pipeline requires: - Very high throughput - Processing in small batches - No need for strict global ordering

Which service should be chosen?

A. Amazon SQS FIFO

B. Amazon SQS Standard

C. Amazon Kinesis Data Streams

D. Amazon SNS

Answer: B

198. A public-facing web server must be accessible from the internet. Which condition makes a subnet a public subnet?

A. It contains an EC2 instance with a public IP only

B. It has a route to an Internet Gateway

C. It has a Security Group that allows HTTP

D. It is placed in Availability Zone A

Answer: B

199. A private application server in a private subnet needs outbound internet access to download updates, but it must not be directly reachable from the internet. Which solution is most appropriate?

A. Internet Gateway

B. NAT Gateway

C. Security Group

D. Route 53

Answer: B

200. A global website needs lower latency and faster delivery of static assets such as images, videos, and HTML by caching content closer to users. Which service is the best choice?

A. Amazon Route 53

B. Amazon CloudFront

C. Amazon RDS

D. Amazon EC2 Auto Scaling

Answer: B

201. A company wants to create an isolated network in AWS that behaves like its own private data center. Which AWS service should be used?

A. Amazon Route 53

B. Amazon VPC

C. Amazon CloudFront

D. Amazon S3

Answer: B

202. A company deploys an Application Load Balancer across two public subnets in different Availability Zones and routes traffic to web servers in private subnets. What is the main benefit of this design?

A. It eliminates the need for Security Groups

B. It provides direct internet access to private instances

C. It improves availability and distributes traffic across targets

D. It replaces Route 53 completely

Answer: C

203. A web application experiences heavy read traffic, causing the primary Amazon RDS for MySQL database to become a performance bottleneck. The architecture also requires automated failover in case of an Availability Zone outage. Which combination of features should a Solutions Architect implement?

A. Create a Read Replica in the same AZ and enable Multi-AZ on the Read Replica.

B. Enable Multi-AZ on the primary instance and offload read queries to the standby instance.

C. Migrate the database to Amazon DynamoDB for infinite read scaling.

D. Enable Multi-AZ on the primary instance and create one or more Read Replicas to handle the read traffic

Answer: D

204. A financial company requires a database solution with a Recovery Point Objective (RPO) of less than 1 second and a Recovery Time Objective (RTO) of less than 1 minute in the event of a full AWS Region outage. Which service is best suited for this requirement?

A. Amazon Redshift cross-region snapshots

B. Amazon Aurora Global Database

C. Amazon RDS with Cross-Region Read Replicas

D. Amazon DynamoDB Global Tables

Answer: B

205. A gaming company is launching a new multiplayer game that expects highly unpredictable traffic spikes during launch week, followed by a steady but unknown baseline of users. Which Amazon DynamoDB capacity mode is the most cost-effective and operationally efficient for this launch?

A. Provisioned capacity with maximum read/write capacity units allocated upfront.

B. DynamoDB Accelerator (DAX) with minimum provisioned capacity.

C. On-Demand capacity mode.

D. Provisioned capacity with Auto Scaling configured to target 50

Answer: C

206. A data analytics team needs to run complex, long-running SQL queries combining historical sales data stored in an Amazon S3 data lake with recent transactional data. They want to avoid loading all the S3 data into the database compute nodes. Which AWS service and feature should they use?

   A. Amazon DynamoDB with Amazon EMR

   B. Amazon RDS with S3 Select

   C. Amazon Redshift Spectrum

   D. Amazon Aurora with Federated Query

   Answer: C

207. A company is migrating its legacy on-premises Oracle database to Amazon Aurora PostgreSQL. The schema structures and stored procedures differ significantly between the two database engines. Which approach should the migration team take?

   A. Take an Oracle RMAN backup and restore it directly to Amazon Aurora.

   B. Use the AWS Schema Conversion Tool (AWS SCT) to convert the schema and code objects, then use AWS DMS to migrate the data.

   C. Use AWS Database Migration Service (AWS DMS) to automatically convert the schema and migrate the data.

   D. Export data to CSV files and use the Aurora command to import data and schemas.

   Answer: B

208. An application reads messages from an Amazon SQS Standard queue and processes video files. Processing a video usually takes 3 minutes. However, developers notice that the same video is occasionally being processed multiple times by different worker instances. What is the most likely cause of this issue?

   A. The Visibility Timeout of the SQS queue is set lower than the time required to process the message.

   B. The SQS queue is configured as a Standard queue instead of a FIFO queue.

   C. The worker instances are failing to send a ReceiveMessage API call.

   D. The message retention period is shorter than the processing time.

   Answer: A

209. A startup wants to build a serverless REST API that automatically scales based on traffic and requires minimal operational overhead. Which AWS services should be used?

   A. Amazon EC2 + Auto Scaling

   B. Amazon API Gateway + AWS Lambda

   C. Amazon RDS

   D. AWS Elastic Beanstalk

Answer: B

210. A company wants to store frequently accessed data in memory to reduce database load and improve application performance. Which AWS service is the best fit?

A. Amazon DynamoDB

B. Amazon ElastiCache

C. Amazon S3

D. Amazon Redshift

Answer: B

211. A company wants to analyze log data stored in Amazon S3 using standard SQL queries without managing infrastructure. Which AWS service should they use?

A. Amazon Athena

B. Amazon RDS

C. Amazon EC2

D. AWS Glue

Answer: A

212. A company has two applications running in separate Amazon VPCs in the same region. They need private communication between the VPCs with: low latency, no internet exposure, simple setup. Which solution is the best fit?

A. Use VPC Peering

B. Use AWS Transit Gateway

C. Use Internet Gateway

D. Use AWS Site-to-Site VPN

Answer: A

213. A company hosts its application in two regions for high availability. They want: automatic failover if one region becomes unhealthy, minimal downtime. Which Amazon Route 53 routing policy should they use?

A. Simple routing

B. Weighted routing

C. Latency-based routing

D. Failover routing

Answer: D

214. An application runs on EC2 instances in a private subnet and needs to access data in Amazon S3. The company wants: no internet exposure, secure and private access. Which solution is the best choice?

A. Attach an Internet Gateway and use public S3 endpoint

B. Use a NAT Gateway

C. Configure an VPC Endpoint for S3

D. Assign public IP to EC2 instances

Answer: C

215. A company runs a production MySQL database on Amazon RDS. The database must remain available with automatic failover if the primary database instance or its Availability Zone experiences an outage. The company does not want to perform any manual intervention during a failure event. Which solution meets these requirements?

A. Create an Amazon RDS read replica in a different Availability Zone and manually promote it if the primary fails.

B. Enable Amazon RDS Multi-AZ deployment

C. Create an Amazon RDS read replica in a different AWS Region

D. Take automated snapshots and restore to a new instance when failure occurs

Answer: B

216. A company needs a relational database solution on AWS that is compatible with PostgreSQL. The solution must provide automatic storage scaling, support up to 15 read replicas with sub-10-millisecond replica lag, and offer failover that completes in less than 30 seconds. Which AWS service meets ALL of these requirements?

A. Amazon RDS for PostgreSQL with Multi-AZ enabled

B. Amazon Aurora PostgreSQL-Compatible Edition

C. Amazon DynamoDB

D. Amazon Redshift

Answer: B

217. A company operates an e-commerce application where a front-end web tier receives customer orders and a back-end processing tier fulfills them. During peak traffic events, the back-end tier becomes overwhelmed, causing orders to be lost. The company wants to ensure that no orders are lost and that each component can scale independently. Which architecture meets these requirements?

A. Increase the instance size of the back-end processing tier.

B. Place an Amazon Simple Queue Service (Amazon SQS) queue between the front-end and back-end tiers, and use an Auto Scaling group for the back-end consumers.

C. Use Amazon Simple Notification Service (Amazon SNS) to send orders directly to the back-end tier.

D. Deploy Amazon ElastiCache between the two tiers to temporarily store orders.

Answer: B

218. Which AWS service is a fully managed relational database service that supports engines such as MySQL, PostgreSQL, Oracle, and SQL Server?

A. Amazon DynamoDB

B. Amazon RDS

C. Amazon ElastiCache

D. Amazon Redshift

Answer: B

219. What type of replication do RDS Read Replicas use?

A. Synchronous (SYNC)

B. Asynchronous (ASYNC)

C. Semi-synchronous

D. Real-time streaming

Answer: B

220. What is the primary purpose of RDS Multi-AZ deployment?

A. Improve read performance by distributing queries

B. Provide disaster recovery and high availability

C. Reduce storage costs

D. Enable cross-region data access

Answer: B

221. How many copies of data does Amazon Aurora store, and across how many Availability Zones?

A. 3 copies across 2 AZs

B. 4 copies across 2 AZs

C. 6 copies across 3 AZs

D. 9 copies across 3 AZs

Answer: C

222. Which workload type is Aurora Serverless best suited for?

A. Stable, predictable workloads

B. Infrequent or unpredictable workloads

C. Workloads requiring sustained high performance

D. Write-heavy OLTP workloads only

Answer: B

223. Which caching engines does Amazon ElastiCache support?

A. MySQL and PostgreSQL

B. Redis and Memcached

C. MongoDB and Cassandra

D. DynamoDB and DocumentDB

Answer: B

224. Which ElastiCache engine supports Multi-AZ with Auto-Failover and data persistence?
   A. Memcached only
   B. Redis only
   C. Both Redis and Memcached
   D. Neither

   Answer: B

225. What delivery behavior does an Amazon SQS Standard Queue provide?
   A. Exactly-once delivery with strict ordering
   B. At-least-once delivery with best-effort ordering
   C. At-most-once delivery with FIFO ordering
   D. Exactly-once delivery with no ordering guarantee

   Answer: B

226. What communication model does Amazon SNS use?
   A. Point-to-point queue model
   B. Streaming model
   C. Publish/Subscribe (Pub/Sub) model
   D. Request/Response model

   Answer: C

227. Which disaster recovery strategy provides the fastest RTO but at the highest cost?
   A. Backup and Restore
   B. Pilot Light
   C. Warm Standby
   D. Multi Site / Hot Site

   Answer: D

228. What does RPO (Recovery Point Objective) measure?
   A. The maximum acceptable downtime after a disaster
   B. The maximum acceptable amount of data loss measured in time
   C. The number of servers needed for disaster recovery
   D. The total cost of the recovery solution

   Answer: B

229. What is the primary benefit of Amazon RDS Proxy

A. Automatically increases database storage capacity

B. Provides connection pooling and reduces failover time by up to 66

C. Encrypts all data at rest automatically

D. Creates read replicas on demand

Answer: B

230. A developer notices that some messages in an SQS Standard Queue are being processed twice by different EC2 instances, even though the instances are not failing. The average task processing time is 50 seconds, and the queue's visibility timeout is set to 30 seconds. What is the most likely cause?

A. The producer/user is sending the same message twice due to networking latency (lagging).

B. The EC2 instances are not using Long Pooling

C. The message was not moved to the Dead Letter Queue

D. The visibility timeout is too short for the task duration.

Answer: D

231. A global healthcare provider must comply with regulations stating they cannot lose more than 15 minutes of patient data (RPO) and must be back online within 4 hours (RTO) if a primary AWS Region fails. They want to minimize costs while ensuring the database is always synchronized. Which strategy is best ?

A. Pilot Light

B. Multi-site Active Active

C. Backup and Restore

D. Multi-AZ Deployment

Answer: A

232. A global financial application requires a relational database that can handle massive read traffic across three continents. Additionally, the company needs a Disaster Recovery (DR) plan that allows them to promote a secondary database in a different geographic region if the primary region fails. Which configuration meets both needs?

A. Amazon DynamoDB with a flexible schema.

B. Amazon Redshift with columnar storage.

C. RDS Multi-AZ deployment with a single Standby instance.

D. RDS with Cross-Region Read Replicas.

Answer: D

233. An application needs to perform three independent tasks whenever an 'Order Placed' event occurs: 1) Send a confirmation email, 2) Update the Inventory database, 3) Send data to a Machine Learning model for fraud detection.

Each task has different processing speeds and failure rates. What is the most 'Simple , Resilient and Scalable' architecture?

A. RDS Read Replicas for each device

B. A single SQS Queue that al 3 services poll from

C. SNS Topic fanning out to three separate SQS Queues

D. Amazone Kinesis Data Streams

Answer: C

234. A banking application must process financial transactions. Two strict requirements exist: 1) The transactions must be processed in the exact order they were received, 2) No transaction should ever be processed twice (Exactly-Once).

Which combination provides this 'Strict Consistency'?

A. Kinesis Data Streams with Partition Keys

B. SNS FIFO Topic + SQS FIFO Queue

C. SNS Standard + SQS Standard

D. Amazon Aurora Serverless

Answer: B

235. An application needs to perform three independent tasks whenever an 'Order Placed' event occurs: 1) Send a confirmation email, 2) Update the Inventory database, 3) Send data to a Machine Learning model for fraud detection. Each task has different processing speeds and failure rates. What is the most 'Simple , Resilient and Scalable' architecture?

A. RDS Read Replicas for each device

B. A single SQS Queue that al 3 services poll from

C. SNS Topic fanning out to three separate SQS Queues

D. Amazone Kinesis Data Streams

Answer: C

236. A financial company needs to scale its database's read performance because of a massive increase in reporting queries. They do not want to impact the performance of the primary write database. What should they implement?

A. RDS Read Replicas

B. Vertical Scaling (Increasing Instance Size)

C. RDS Multi-AZ Deployment

D. AWS DMS

Answer: A

237. A developer is building a serverless application that needs to store simple profile data (User ID, Name, Preferences). The database must handle millions of requests per second with consistent single-digit millisecond performance. Which service is best?

A. Amazon Redshift.

B. Amazon Aurora.

C. Amazon DynamoDB.

D. Amazon RDS PostgreSQL.

Answer: C

238. A high-frequency trading firm requires a dedicated, consistent network connection between their on-premises data center and AWS that does not traverse the public internet. Which service should they choose?

A. AWS Site-to-Site VPN.

B. Amazon CloudFront.

C. VPC Peering.

D. AWS Direct Connect

Answer: D

239. How is a Virtual Private Cloud (VPC) best defined within the context of AWS?

A. A physical data centre exclusively reserved for a single organisation's hardware.

B. A logically isolated section of the AWS Cloud where resources can be launched in a defined virtual network.

C. A public subnet that automatically exposes all EC2 instances to the internet by default.

D. A global network that connects all AWS Regions without the need for IP addressing.

Answer: B

240. Which component is mandatory for a subnet to be classified as a 'Public Subnet'?

A. A /32 CIDR block that locks all incoming traffic.

B. A route to the internet via an Internet Gateway (IGW).

C. The presence of a PostgreSQL database instance.

D. A direct connection to a private data centre via VPN.

Answer: B

241. if a CIDR block has a /28 suffix, how many 'free' bits remain for host addresses?

A. 2 bits

B. 16 bits

C. 4 bits

D. 8 bits

Answer: C

242. An enterprise is expanding from 5 VPCs to over 200 VPCs across multiple AWS accounts. They need to connect all VPCs to their on-premises data center via a single AWS Direct Connect gateway while avoiding the complexity of a 'mesh' network of VPC Peering. Which service should they implement ?

A. AWS PrivateLink.

B. AWS Transit Gateway.

C. VPC Peering with transitive routing.

D. Multiple Site-to-Site VPN tunnels.

Answer: B

AI VIET NAM – AI COURSE 2025

Cloud Practitioner Quiz - Week 2

Lớp AWS (AIO 2025)

1. Per the AWS Shared Responsibility Model, management of which of the following AWS services is the responsibility of the customer?
   A. AWS Elastic Beanstalk
   B. Amazon Elastic Compute Cloud (Amazon EC2)
   C. Amazon DynamoDB
   D. Amazon Simple Storage Service (Amazon S3)

   Answer: B

2. Which of the following will help you control the incoming traffic to an Amazon EC2 instance?
   A. Security Group
   B. Network access control list (network ACL)
   C. AWS Resource Group
   D. Route Table

   Answer: A

3. Which of the following is the responsibility of the customer when running applications using AWS Lambda?
   A. Managing the physical servers where the Lambda function runs
   B. Configuring the networking infrastructure for the Lambda service
   C. Updating the operating system and runtime environment for Lambda functions
   D. Writing and maintaining the function code and its dependencies

   Answer: D

4. A company is moving its on-premises application to AWS Cloud. The application uses in-memory caches for running custom workloads. Which Amazon Elastic Compute Cloud (Amazon EC2) instance type is the right choice for the given requirement?

A. Memory Optimized instance types

B. Storage Optimized instance types

C. Accelerated computing instance types

D. Compute Optimized instance types

Answer: A

5. A big data analytics company is moving its IT infrastructure from an on-premises data center to AWS Cloud. The company has some server-bound software licenses that it wants to use on AWS. As a Cloud Practitioner, which of the following EC2 instance types would you recommend to the company?

   A. Reserved Instance (RI)

   B. Dedicated Host

   C. Dedicated Instance

   D. On-Demand Instance

   Answer: B

6. Which of the following is a serverless AWS service?

   A. Amazon Elastic Compute Cloud (Amazon EC2)

   B. AWS Elastic Beanstalk

   C. Amazon EMR

   D. AWS Lambda

   Answer: D

7. A startup wants to provision an EC2 instance for the lowest possible cost for a long-term duration but needs to make sure that the instance would never be interrupted. As a Cloud Practitioner, which of the following options would you recommend?

   A. EC2 Spot Instance

   B. EC2 Reserved Instance (RI)

   C. EC2 Dedicated Host

   D. EC2 On-Demand Instance

   Answer: B

8. Which of the following AWS services should be used to automatically distribute incoming traffic across multiple targets?

   A. AWS Elastic Load Balancing (ELB)

   B. AWS Elastic Beanstalk

   C. Amazon OpenSearch Service

   D. AWS Auto Scaling

   Answer: A

9. Which of the following is the MOST cost-effective option to purchase an EC2 Reserved Instance (RI)?
   A. No upfront payment option with standard 1-year term
   B. All upfront payment option with the standard 1-year term
   C. Partial upfront payment option with standard 3-years term
   D. No upfront payment option with standard 3-years term

   Answer: C

10. Which AWS technology/service helps you to scale your resources to match supply with demand while still keeping your cloud solution cost-effective?
   A. AWS Systems Manager
   B. AWS CloudFormation
   C. AWS Cost Explorer
   D. AWS Auto Scaling

   Answer: D

11. Which of the following statements is INCORRECT about AWS Auto Scaling?
   A. You can automatically register new instances to a load balancer
   B. You can scale out and add more Amazon Elastic Compute Cloud (Amazon EC2) instances to match an increase in demand as well as scale in and remove Amazon Elastic Compute Cloud (Amazon EC2) instances to match a reduced demand
   C. You can automatically remove unhealthy instances
   D. You can automatically deploy AWS Shield when a DDoS attack is detected

   Answer: D

12. Which AWS service can be used to store, manage, and deploy Docker container images?
   A. Amazon Elastic Container Service (Amazon ECS)
   B. Amazon Elastic Compute Cloud (Amazon EC2)
   C. Amazon Elastic Container Registry (Amazon ECR)
   D. AWS Lambda

   Answer: C

13. Which AWS service should be used when you want to run container applications, but want to avoid the operational overhead of scaling, patching, securing, and managing servers?
   A. AWS Lambda
   B. Amazon Elastic Compute Cloud (Amazon EC2)
   C. Amazon Elastic Container Service (Amazon ECS)- EC2 launch type
   D. Amazon Elastic Container Service (Amazon ECS)- Fargate launch type

Answer: D

14. A startup company decides to move from purchasing physical servers to using AWS cloud services. Instead of paying a large upfront investment to build a data center, they only pay monthly for the computing resources they actually use.

    Which core benefit of AWS cloud computing does this scenario illustrate?

    A. Stop guessing capacity (No need to predict resource capacity in advance)

    B. Trade fixed expense for variable expense

    C. Go global in minutes

    D. Benefit from massive economies of scale

    Answer: B

15. A company wants to run its workload on Amazon EC2 instances for more than 1 year. This workload will run continuously. Which option offers a discounted hourly rate compared to the hourly rate of On-Demand Instances?

    A. AWS Graviton processor

    B. Dedicated Hosts

    C. EC2 Instance Savings Plans

    D. Amazon EC2 Auto Scaling instances

    Answer: C

16. A company wants to automatically add and remove Amazon EC2 instances. The company wants the EC2 instances to adjust to varying workloads dynamically.

    A. Amazon DynamoDB

    B. Amazon EC2 Spot Instances

    C. AWS Snow Family

    D. Amazon EC2 Auto Scaling

    Answer: D

17. Which design principle aligns with performance efficiency pillar of the AWS Well-Architected Framework?

    A. Using serverless architectures

    B. Scaling horizontally

    C. Measuring the cost of workloads

    D. Using managed services

    Answer: A

18. A company runs a web application on Amazon EC2 instances using a fixed number of On-Demand instances. During peak hours, the application experiences high traffic and slow

response times. During off-peak hours, many instances remain underutilized, increasing costs. The company wants to optimize cost while maintaining high availability and performance.

Which solution is the most appropriate?

A. Use larger EC2 instance types to handle peak traffic at all times

B. Use Auto Scaling and distribute traffic using an Elastic Load Balancing

C. Move all instances to Reserved Instances and keep a fixed number of servers

D. Deploy the application in a single Availability Zone with more powerful instances

Answer: B

19. Which of the following data transfers is completely FREE under the Amazon S3 pricing model?

A. Data transferred out from S3 to the internet

B. Data transferred in to S3 from the internet

C. Data transferred out from S3 to an EC2 instance in a different AWS Region.

D. Inter-Region data transfer between S3 buckets

Answer: B

20. Amazon S3 is designed for '11 nines' of durability. What does this mean?

A. The data is available 99.999999999

B. The probability of losing an object is 0.000000001

C. Data can be accessed from 11 different AWS Regions simultaneously.

D. The system can handle 11 trillion requests per second.

Answer: B

21. A developer needs to update a small piece of data within a 10GB file. Which storage type is LEAST efficient for this 'partial edit' task?

A. Block Storage

B. File Storage

C. Object Storage

D. Network Attached Storage (NAS)

Answer: C

22. Which AWS service enables on-premises applications to seamlessly use AWS Cloud storage by caching frequently accessed data locally while storing all data in S3?

A. AWS Storage Gateway

B. AWS Snowball

C. AWS Direct Connect

D. Amazon EFS

Answer: A

23. Which of the following Amazon S3 storage classes has NO constraint on a minimum storage duration charge for objects?
    A. Amazon S3 Standard- Infrequent Access (S3 Standard- IA)
    B. Amazon S3 One Zone- Infrequent Access (S3 One Zone-IA)
    C. Amazon S3 Glacier Flexible Retrieval
    D. Amazon S3 Standard

    Answer: D

24. A real-time stock trading platform needs a load balancer that can handle millions of TCP requests per second with ultra-low latency. Which ELB should they choose?
    A. Application Load Balancer
    B. Network Load Balancer
    C. Gateway Load Balancer
    D. Classic Load Balancer

    Answer: B

25. Which EBS volume type supports Multi-Attach, allowing it to be attached to multiple EC2 instances simultaneously?
    A. Cold HDD (sc1)
    B. General Purpose SSD (gp2/ gp3)
    C. Throughput Optimized HDD (st1)
    D. Provisioned IOPS SSD (io1/io2)

    Answer: D

26. Which feature eliminates the initial read latency when accessing a newly restored EBS volume from a snapshot?
    A. EBS Multi-Attach
    B. EBS Snapshot Archive
    C. Provisioned IOPS
    D. Fast Snapshot Restore (FSR)

    Answer: D

27. A financial services company needs to retain its data for 10 years to meet compliance norms. Which Amazon Simple Storage Service (Amazon S3) storage class is the best fit for this use case considering that the data has to be stored at a minimal cost?
    A. Amazon S3 Standard-Infrequent Access (S3 Standard-IA)
    B. Amazon S3 Glacier Deep Archive

C. Amazon S3 Glacier Flexible Retrieval

D. Amazon S3 Intelligent-Tiering

Answer: B

28. A media company uses Amazon Simple Storage Service (Amazon S3) for storing all its data. Which storage class should it consider for cost-optimal storage of the data that has random access patterns?

A. Amazon S3 Random Access (S3 Random-Access)

B. Amazon S3 Standard (S3 Standard)

C. Amazon S3 Intelligent-Tiering (S3 Intelligent-Tiering)

D. Amazon S3 Standard-Infrequent Access (S3 Standard-IA)

Answer: C

29. An e-commerce company has its on-premises data storage on an NFS file system that is accessed in parallel by multiple applications. The company is looking at moving the applications and data stores to AWS Cloud. Which storage service should the company use to move their files to AWS Cloud seamlessly if the application is hosted on Amazon Elastic Compute Cloud (Amazon EC2) instances?

A. Amazon Simple Storage Service (Amazon S3)

B. AWS Storage Gateway

C. Amazon Elastic Block Store (Amazon EBS)

D. Amazon Elastic File System (Amazon EFS)

Answer: D

30. Which of the following is the correct statement regarding the AWS Storage services?

A. Amazon Simple Storage Service (Amazon S3) is file based storage, Amazon Elastic Block Store (Amazon EBS) is block based storage and Amazon Elastic File System (Amazon EFS) is object based storage

B. Amazon Simple Storage Service (Amazon S3) is block based storage, Amazon Elastic Block Store (Amazon EBS) is object based storage and Amazon Elastic File System (Amazon EFS) is file based storage

C. Amazon Simple Storage Service (Amazon S3) is object based storage, Amazon Elastic Block Store (Amazon EBS) is block based storage and Amazon Elastic File System (Amazon EFS) is file based storage

D. Amazon Simple Storage Service (Amazon S3) is object based storage, Amazon Elastic Block Store (Amazon EBS) is file based storage and Amazon Elastic File System (Amazon EFS) is block based storage

Answer: C

31. A fleet of Amazon EC2 instances spread across different Availability Zones (AZ) needs to access, edit and share file-based data stored centrally on a system. As a Cloud Practitioner, which AWS service would you recommend for this use-case?

A. Amazon Elastic Block Store (Amazon EBS)

B. Amazon Simple Storage Service (Amazon S3)

C. EC2 Instance Store

D. Amazon Elastic File System (Amazon EFS)

Answer: D

32. A company wants a fully managed, flexible, and scalable file storage system, with low latency access, for its Windows-based applications. Which AWS service is the right choice for the company?

A. Amazon Elastic File System (Amazon EFS)

B. Amazon FSx for Windows File Server

C. Amazon Elastic Block Storage (Amazon EBS)

D. Amazon FSx for Lustre

Answer: B

33. Which Amazon Simple Storage Service (Amazon S3) storage class offers the lowest availability?

A. Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)

B. Amazon S3 Glacier Flexible Retrieval

C. Amazon S3 Intelligent-Tiering

D. Amazon S3 Standard

Answer: A

34. Which of the following is the best way to protect your data from accidental deletion on Amazon S3?

A. Amazon S3 storage classes

B. Amazon S3 Versioning

C. Amazon S3 lifecycle configuration

D. Amazon S3 Transfer Acceleration (Amazon S3TA)

Answer: B

35. What is Amazon EBS primarily used for?

A. Hosting static websites.

B. Creating a highly scalable data lake.

C. Providing Block Storage volumes for Amazon EC2 instances.

D. Archiving compliance logs for 10 years.

Answer: C

36. A research lab wants to optimize the caching capabilities for its scientific computations application running on Amazon Elastic Compute Cloud (Amazon EC2) instances. Which Amazon Elastic Compute Cloud (Amazon EC2) storage option is best suited for this use case?

    A. Instance Store

    B. Amazon Elastic Block Store (Amazon EBS)

    C. Amazon Simple Storage Service (Amazon S3)

    D. Amazon Elastic File System (Amazon EFS)

    Answer: A

37. An IT company has deployed a static website on Amazon Simple Storage Service (Amazon S3), but the website is still inaccessible. As a Cloud Practioner, which of the following solutions would you suggest to address this issue?

    A. Fix the Amazon S3 bucket policy

    B. Enable Amazon S3 versioning

    C. Disable Amazon S3 encryption

    D. Enable Amazon S3 replication

    Answer: A

38. Which of the following Amazon Simple Storage Service (Amazon S3) storage classes has NO constraint of a minimum storage duration charge for objects?

    A. Amazon S3 Standard-Infrequent Access (S3 Standard-IA)

    B. Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)

    C. Amazon S3 Glacier Flexible Retrieval

    D. Amazon S3 Standard

    Answer: D

39. Which AWS service can be used to host a static website with the LEAST effort?

    A. Amazon S3 Glacier

    B. Amazon Simple Storage Service (Amazon S3)

    C. AWS Storage Gateway

    D. Amazon Elastic File System (Amazon EFS)

    Answer: B

40. An organization would like to copy data across different Availability Zones (AZs) using Amazon EBS snapshots. Where are Amazon EBS snapshots stored in the AWS Cloud?

    A. Amazon Elastic Compute Cloud (Amazon EC2)

    B. Amazon Simple Storage Service (Amazon S3)

    C. Amazon Relational Database Service (Amazon RDS)

D. Amazon Elastic File System (Amazon EFS)

Answer: B

41. Which Amazon EC2 pricing model provides the most significant discount for fault-tolerant and flexible workloads?

    A. Spot Instances

    B. On-Demand Instances

    C. Reserved Instances

    D. Dedicated Hosts

    Answer: A

42. A media company needs to store millions of high-resolution images for their public website. These images must be accessible via a URL and the storage must scale automatically. Which service is best suited for this?

    A. Amazon EBS

    B. AWS Snowball

    C. Amazon EFS

    D. Amazon S3

    Answer: D

43. A company has a dataset with unpredictable access patterns. Sometimes the data is used daily, and other times it is not touched for months. They want to minimize costs without manually moving data. Which S3 feature helps?

    A. S3 Intelligent-Tiering

    B. S3 Lifecycle Policies

    C. S3 One Zone-IA

    D. S3 Standard

    Answer: A

44. You are creating a new EC2 instance and need to choose a volume to store the operating system (the root volume). Which storage type is used for this purpose?

    A. Amazon EFS

    B. AWS Storage Gateway

    C. Amazon EBS

    D. Amazon S3

    Answer: C

45. An analytics application generates reports every month. These reports are rarely looked at after the first week, but when they are needed, they must be accessible immediately. Which S3 class is best?

A. S3 Glacier Flexible Retrieval

B. S3 Standard

C. Amazon EFS

D. S3 Standard-IA

Answer: D

46. What happens to the data stored on an 'Instance Store' volume if the associated EC2 instance is stopped or terminated?

A. The data is moved to a new EBS volume.

B. The data is automatically backed up to S3.

C. The data is lost (Ephemeral).

D. The data is preserved and can be re-attached.

Answer: C

47. A data processing firm runs batch jobs that are stateless and can be interrupted at any time. They are looking for the absolute lowest cost for compute. Which option is best?

A. Saving Plans

B. On-Demand Instances

C. Spot Instances

D. Capacity Reservation

Answer: C

48. Which EC2 pricing model allows you to bid for unused AWS capacity?

A. On-Demand

B. Dedicated Hosts

C. Reserved Instances

D. Spot Instances

Answer: D

49. A company needs to run a specialized compliance-heavy workload that requires software licenses bound to a specific physical server's sockets and cores. Which EC2 tenancy model is the only one that supports this requirement?

A. Dedicated Instances

B. Dedicated Hosts

C. On-Demand Capacity Reservations

D. Reserved Instances

Answer: B

50. An organization is looking to migrate to AWS and wants to commit to a 3-year term. They need a discount that applies to EC2 instances regardless of the instance family, AWS Region, or even if they switch to AWS Fargate or Lambda. Which Savings Plan fits this best?

    A. EC2 Instance Savings Plans

    B. SageMaker Savings Plans

    C. Compute Savings Plans

    D. Standard Reserved Instaces

    Answer: C

51. A developer is using Spot Instances for a non-critical test environment. If AWS needs the capacity back, how much notice does the developer receive before the instance is terminated?

    A. 5 minutes

    B. No notice is provided

    C. 30 seconds

    D. 2 minutes

    Answer: D

52. A company wants to receive the maximum possible discount on their EC2 usage. They have a predictable workload for the next 3 years and are willing to pay for the entire term upfront. Which payment option provides the highest savings?

    A. Monthly Billing

    B. All Upfront

    C. No Upfront

    D. Partial Upfront

    Answer: B

53. If an On-Demand Capacity Reservation is not being used (no instances are running in it), how is the customer billed?

    A. They are billed a small "holding fee"(10

    B. They are billed at the full On-Demand rate for the reserved capacity

    C. They receive a credit for future use

    D. They are not billed at all

    Answer: B

54. Which AWS tool can provide recommendations on which Reserved Instances or Savings Plans to buy based on your historical usage?

    A. AWS Budgets

    B. AWS Cost Explorer

    C. AWS Trusted Advisor

D. Amazon Inspector

Answer: B

55. A retail company expects unpredictable traffic spikes during a flash sale. They need to automatically add servers to handle demand and remove them afterward to save costs

A. Spot Instances and Placement Groups

B. Auto Scaling Group (ASG) and Elastic Load Balancing (ELB)

C. Reserved Instances and CloudWatch

D. Dedicated Hosts

Answer: B

56. A biotech startup needs high parallel processing power for complex genomic analysis that standard CPUs process too slowly. They require specialized hardware accelerator

A. Compute Optimized

B. Memory Optimized

C. Accelerated Computing

D. Storage Optimized

Answer: C

57. A financial company must retain data for 10 years for compliance at the lowest possible cost. Data is rarely accessed, and a retrieval time of 12–48 hours is acceptable. Which Amazon S3 storage class is the best fit for this use case?

A. Amazon S3 Standard-IA

B. Amazon S3 Glacier Deep Archive

C. Amazon S3 Glacier Flexible Retrieval

D. Amazon S3 Intelligent-Tiering

Answer: B

58. In Amazon EC2, what acts as a virtual firewall at the instance level to control inbound and outbound traffic?

A. Security Group

B. Network Access Control List (NACL)

C. AWS WAF

D. Route Table

Answer: A

59. Which EC2 purchasing option is the most cost-effective for fault-tolerant, flexible workloads (such as batch processing) and can offer discounts up to 90

A. On-Demand Instances

B. Spot Instances

C. Reserved Instances

D. Dedicated Hosts

Answer: B

60. To optimize applications that require extremely low network latency and high network bandwidth between instances within the same Availability Zone (AZ), which type of Placement Group should you use?

A. Cluster Placement Group

B. Spread Placement Group

C. Partition Placement Group

D. Auto Scaling Group

Answer: A

61. A company needs to store data for a Big Data system that requires high throughput but does not require a high number of Input/Output Operations Per Second (IOPS). Which of the following Amazon EBS volume types is the most cost-effective choice?

A. Cold HDD (sc1)

B. Throughput Optimized HDD (st1)

C. Provisioned IOPS SSD (io2)

D. General Purpose SSD (gp3)

Answer: B

62. You have a large amount of old image data that is infrequently accessed (about once every few months). However, when needed, the data must be immediately available in milliseconds. Which Amazon S3 storage class is the most cost-optimized choice?

A. S3 Standard-IA

B. S3 Glacier Flexible Retrieval

C. S3 One Zone-IA

D. S3 Standard

Answer: A

63. A financial institution is migrating its core banking database to AWS. Due to strict regulatory compliance, the database must run on isolated physical hardware. The database runs 24/7, requires a consistent sub-millisecond latency, and needs up to 100,000 IOPS. The institution wants the most cost-effective purchasing option for a 3-year architecture plan. Which combination of compute and storage should the solutions architect choose?

A. EC2 Dedicated Instances with a 3-year Compute Savings Plan + Amazon EBS gp3.

B. EC2 Default Tenancy with a 3-year All Upfront Reserved Instance + Amazon EBS io1.

C. EC2 Dedicated Hosts with a 3-year Reserved Instance + Amazon EBS io2 Block Express.

D. EC2 Capacity Reservations with On-Demand pricing + Amazon EBS st1.

Answer: C

64. You are designing a global leaderboard for a popular mobile game using AWS API Gateway, AWS Lambda, and Amazon DynamoDB. During special events, read traffic becomes highly unpredictable, spiking to millions of requests per second as users constantly refresh to see the "Top 10 Players". This causes DynamoDB to return ProvisionedThroughputExceededException errors despite Auto Scaling being enabled. Which serverless-friendly solution will resolve the throttling issue and provide microsecond latency for the read-heavy workload?

    A. Switch DynamoDB to On-Demand capacity mode to instantly handle the millions of read requests.

    B. Implement Amazon DynamoDB Accelerator (DAX) to cache the frequently accessed leaderboard data.

    C. Use Amazon Kinesis Data Streams to buffer the read requests before they hit DynamoDB.

    D. Enable DynamoDB Global Tables to distribute the read requests across multiple AWS

    Answer: B

65. An AI company stores massive amounts of raw training data (petabytes) in Amazon S3. The data is heavily accessed for model training in the first 30 days. After 30 days, the data is only accessed once a month for model evaluation, but it requires millisecond retrieval times when accessed. After 1 year, the data is kept only for historical compliance and is rarely accessed (retrieval time of several hours is acceptable). Which S3 Lifecycle configuration is the most cost-effective?

    A. S3 Standard -> Transition to S3 Standard-IA at day 30 -> Transition to S3 Glacier Deep Archive at day 365.

    B. S3 Standard -> Transition to S3 One-Zone IA at day 30 ->$TransitiontoS3GlacierInstantRetrievalat$

    B. S3 Standard -> Transition to S3 Intelligent-Tiering at day 30 -> Transition to S3 Glacier Flexible Retrieval at day 365.

    C. S3 Standard -> Transition to S3 Standard-IA at day 30 -> Transition to S3 Glacier Instant Retrieval at day 365.

    Answer: A

66. A web application is running on an Auto Scaling Group (ASG) of EC2 instances behind an Application Load Balancer (ALB). The ASG spans across two Availability Zones: AZ-1 and AZ-2. Due to a recent scale-out event triggered by an AZ-specific metric, AZ-1 currently has 8 instances, while AZ-2 has only 2 instances. Users suddenly report 502 Bad Gateway errors. Upon investigation, you notice the 2 instances in AZ-2 are crashing due to 100

    A. The ASG is using a Partition Placement Group, causing traffic to isolate in AZ-2.

    B. Cross-Zone Load Balancing is currently disabled (OFF) on the load balancer.

    C. The Elastic Network Interfaces (ENIs) in AZ-1 have detached from the EC2 instances.

    D. The Application Load Balancer is misconfigured to use Layer 4 (TCP) instead of Layer 7 (HTTP).

Answer: B

67. A development team is modernizing a monolithic application by containerizing it into microservices. The containers will be deployed using Amazon ECS. The team demands a fully serverless compute environment where they do not have to manage the underlying OS or instances. However, the microservices must share a common, persistent storage volume that can be mounted simultaneously by multiple containers running across different Availability Zones. Which architecture satisfies all these requirements?

    A. ECS EC2 Launch Type with Amazon EBS Multi-Attach.

    B. ECS Fargate Launch Type with Amazon EBS gp3.

    C. ECS EC2 Launch Type with Amazon S3 mounted via a gateway.

    D. ECS Fargate Launch Type with Amazon EFS.

    Answer: D

68. A company wants to distribute traffic across multiple servers. What service should they use?

    A. EBS for storage expansion

    B. Auto Scaling Group for scaling capacity

    C. AMI for launching instances

    D. Elastic Load Balancer for traffic distribution

    Answer: D

69. Among Amazon EC2's payment options, which entity type allows users to leverage AWS's excess capacity at a cost up to 90

    A. Dedicated Hosts

    B. Spot Instances

    C. Capacity Reservations

    D. On-Demand Instances

    Answer: B

70. A startup is building an event-driven image-processing pipeline. Every time a user uploads a file, the application must resize the image, add a watermark, and store metadata. Traffic is highly unpredictable, ranging from almost zero to sudden spikes of thousands of requests per minute. The team does not want to provision or manage servers, and they are comfortable packaging any required libraries with the application. Which statement best identifies the customer's remaining responsibility if AWS Lambda is selected?

    A. Managing the physical servers that execute the function

    B. Updating the operating system and Lambda runtime

    C. Writing and maintaining the function code and its dependencies

    D. Configuring the underlying AWS networking infrastructure for the Lambda service

    Answer: C

71. What are the core differences in operating models between AWS Lambda and Amazon EC2?
    A. Lambda executes event-driven code and automatically manages the infrastructure, while EC2 requires users to manage and maintain virtual machines.
    B. Lambda is designed to run tasks continuously for extended periods, while EC2 can only run for short durations.
    C. Lambda only supports Python, while EC2 supports a variety of different operating systems.
    D. EC2 scales automatically without configuration, while Lambda requires setting up an Auto Scaling Group.

    Answer: A

72. How does Horizontal Scalability (Scaling Out) differ fundamentally from Vertical Scalability (Scaling Up)?
    A. Horizontal scaling involves adding more RAM to an existing instance, while vertical scaling adds more instances.
    B. Vertical scaling is restricted by hardware limits of a single machine, while horizontal scaling is theoretically limitless by adding more nodes.
    C. Horizontal scaling is preferred for non-distributed systems like traditional databases.
    D. Vertical scaling requires a Load Balancer, whereas horizontal scaling does not.

    Answer: B

73. What is the benefit of using a Cluster Placement Group?
    A. It spreads instances across different physical hardware to minimize correlated failures
    B. It partitions instances across different racks to support distributed data systems.
    C. It packs instances close together in a single Availability Zone to achieve low-latency, high-throughput networking.
    D. It automatically distributes traffic across multiple Availability Zones.

    Answer: C

74. A student is reviewing core AWS services related to EC2, scaling, networking, and availability. They want to verify which statements are correct.How many of the following statements are correct? (1) Elastic Load Balancer distributes incoming traffic across multiple EC2 instances; (2) Auto Scaling Group automatically increases or decreases the number of EC2 instances based on demand; (3) An AMI is used to route traffic between multiple servers; (4) Private IP addresses allow EC2 instances to communicate within a VPC; (5) Deploying resources in multiple Availability Zones improves high availability.
    A. 2
    B. 3
    C. 4
    D. 5

    Answer: C

75. A company is planning to run a batch data processing job that is fault-tolerant and can be interrupted without affecting the final output. Which Amazon EC2 purchasing option provides the most cost-effective solution for this workload?
    A. On-Demand Instances
    B. Reserved Instances
    C. Dedicated Hosts
    D. Spot Instances

    Answer: D

76. Which AWS service allows developers to run code in response to events without provisioning or managing the underlying servers?
    A. Amazon EC2
    B. AWS Lambda
    C. Amazon Elastic Kubernetes Service (EKS)
    D. Amazon Cognito

    Answer: B

77. A cloud architect needs to distribute incoming HTTP and HTTPS web traffic across multiple Amazon EC2 instances. The architecture requires advanced routing capabilities, such as routing traffic based on the URL path. Which AWS service should be used?
    A. Network Load Balancer (NLB)
    B. Amazon API Gateway
    C. Application Load Balancer (ALB)
    D. Auto Scaling Group (ASG)

    Answer: C

78. Which statement is FALSE?
    A. Amazon S3 objects can be transitioned between storage classes using Lifecycle Rules.
    B. Amazon EFS is designed as a shared file system for multiple instances.
    C. Amazon EBS is mainly used for EC2 volumes, database storage, and low-latency workloads.
    D. Amazon S3 is the best option for high-transaction relational databases because it has lower latency than block storage.

    Answer: D

79. A team stores huge backup archives and AI training datasets. They want very high durability, very large scale, and low cost. However, developers also complain that they cannot make small in-place edits to stored items. Which service is most likely being used?
    A. Amazon EBS
    B. Amazon EFS

C. Amazon S3

D. Amazon RDS

Answer: C

80. A company wants to back up an EBS volume today and later restore that backup in another Availability Zone or even another Region. Which option should they use?

A. Amazon EFS lifecycle policy

B. Amazon S3 lifecycle rule

C. Amazon EBS Snapshot

D. Amazon S3 replication only

Answer: C

81. An application runs on a single EC2 instance and uses a relational database that must support high transaction volume and low latency. Which option is the most appropriate?

A. Amazon S3, because it has high durability

B. Amazon EFS, because it can be mounted like a filesystem

C. Amazon EBS, because block storage is efficient for databases

D. Amazon Glacier, because archival tiers reduce cost

Answer: C

82. 1. Which of the following is an example of horizontal scaling in the AWS Cloud?

A. Replacing an existing EC2 instance with a larger, more powerful one.

B. Increasing the compute capacity of a single EC2 instance to address the growing demands of an application.

C. Adding more RAM capacity to an EC2 instance.

D. Adding more EC2 instances of the same size to handle an increase in traffic.

Answer: D

83. You have deployed your application on multiple Amazon EC2 instances. Your customers complain that sometimes they can't reach your application. Which AWS service allows you to monitor the performance of your EC2 instances to assist in troubleshooting these issues?

A. AWS Lambda

B. AWS Config

C. Amazon CloudWatch

D. AWS CloudTrail

Answer: C

84. Which service provides object-level storage in AWS?

A. Amazon EBS

B. Amazon Instance Store

C. Amazon EFS

D. Amazon S3

Answer: D

85. Which type of Load Balancer of Amazon ELB service can be reached via Route Table Entry?

A. Application Load Balancer

B. Network Load Balancer

C. Gateway Load Balancer

D. Classic Load Balancer

Answer: C

86. Which statement best describes Elastic Load Balancing?

A. It translates a domain name into an IP address using DNS

B. It distributes incoming application traffic across one or more Amazon EC2 instances

C. It collects metrics on connected Amazon EC2 instances

D. It automatically adjusts the number of Amazon EC2 instances to support incoming traffic

Answer: B

87. Which AWS service provides a simple and scalable shared file storage solution for use with Linux-based AWS and on-premises servers?

A. Amazon S3

B. Amazon Glacier

C. Amazon EBS

D. Amazon EFS

Answer: D

88. A company wants to host a web application on Amazon EC2 and ensure it remains operational even if an entire data center fails. Which of the following is the BEST architectural choice?

A. Deploy all EC2 instances into a Cluster Placement Group within one Region

B. Use an Auto Scaling Group to distribute instances across multiple Availability Zones (AZs)

C. Assign an Elastic IP address to a single high-capacity EC2 instance

D. Deploy the instances on a Dedicated Host within a single private subnet

Answer: B

89. A media company's video streaming platform is hosted in the AWS ap-southeast-1 Region. Users in Europe report slow load times for static content such as thumbnails and video previews. Which AWS infrastructure concept would BEST reduce this latency?

A. Deploy additional EC2 instances in a second AZ

B. Use Edge Locations to cache and serve static content closer to users

C. Migrate the application to a Region closer to Europe

D. Enable Multi-Region replication for the primary database

Answer: B

90. Which EC2 purchasing option is best suited for batch processing jobs that are fault-tolerant, can be interrupted without losing progress, and offers the HIGHEST cost savings?

A. Reserved Instances

B. On-Demand Instances

C. Spot Instances

D. Dedicated Hosts

Answer: C

91. A company needs shared file storage accessible concurrently by multiple Linux EC2 instances across 3 Availability Zones. Which storage service is the most appropriate?

A. Amazon EBS

B. EC2 Instance Store

C. Amazon EFS

D. Amazon S3

Answer: C

92. A startup uploads log files to Amazon S3 Standard. They are accessed frequently in the first month, but rarely used after that. Which S3 feature helps automatically minimize storage costs without manual intervention?

A. S3 Lifecycle Rules

B. EBS Snapshots

C. Amazon EFS replication

D. EBS Multi-Attach

Answer: A

93. A machine learning platform needs temporary, extremely high-speed storage for caching intermediate computation results during model training on an EC2 instance. The data does not need to be retained after the instance stops. Which storage option is BEST?

A. Amazon EBS gp3

B. Amazon EFS Standard

C. EC2 Instance Store

D. Amazon S3 Standard

Answer: C

94. A company needs to store compliance audit logs for 7 years to meet regulatory requirements. They are almost never accessed. The PRIMARY requirement is to optimize for the lowest cost. Which S3 storage class is most appropriate?
    A. Amazon S3 Standard-IA
    B. Amazon S3 Glacier Instant Retrieval
    C. Amazon S3 Glacier Deep Archive
    D. Amazon S3 One Zone-IA

    Answer: C

95. Which AWS service is best suited for storing a large amount of unstructured data (images, videos, documents) with high durability and low cost?
    A. Amazon EBS
    B. Amazon S3
    C. Amazon EFS
    D. EC2 Instance Store

    Answer: B

96. Which AWS storage type is best suited for running a relational database that requires high I/O performance on Amazon EC2?
    A. Amazon S3
    B. Amazon EBS
    C. Amazon EFS
    D. Amazon Glacier

    Answer: B

97. Which component acts as a virtual firewall to control inbound and outbound traffic for EC2 instances?
    A. Route Table
    B. Security Group
    C. IAM
    D. Elastic Load Balancer

    Answer: B

98. What is the primary function of 'User Data' in Amazon EC2 when configuring a new instance launch?
    A. To store personal user files in the cloud
    B. To run a bootstrap script to configure the server
    C. To define firewall rules for the instance
    D. To monitor CPU and memory usage

Answer: B

99. What is the primary function of an Amazon Machine Image (AMI)?
    A. To serve as a preconfigured template for launching EC2 instances
    B. To act as a firewall for incoming network traffic
    C. To provide persistent block storage for running instances
    D. To automatically scale the number of instances based on demand

    Answer: A

100. How can you encrypt an existing, unencrypted Amazon EBS volume?
    A. Change the setting directly in the AWS Console for that volume
    B. Take a snapshot and copy that snapshot with encryption enabled
    C. Use a script to encrypt the data while the instance is running
    D. Attach an IAM policy to force the system to automatically encrypt this volume

    Answer: B

101. What is the primary difference between the EC2 Launch Type and the Fargate Launch Type in Amazon ECS?
    A. Fargate requires the user to install the ECR Agent manually
    B. Fargate is a serverless architecture that removes the need to manage EC2 instances
    C. Only the EC2 Launch Type supports Docker containers
    D. The EC2 Launch Type is always more cost-effective for short-lived, small-scale tasks

    Answer: B

102. Which AWS service allows you to run code without provisioning or managing servers (Serverless compute)?
    A. Amazon EC2
    B. Amazon ECS
    C. Amazon EBS
    D. AWS Lambda

    Answer: D

103. A company has an application with unpredictable traffic spikes and wants a database solution where they only pay for the exact read and write requests they perform. Which Amazon DynamoDB capacity mode should they choose?
    A. Provisioned Mode
    B. On-Demand Mode
    C. Standard-IA Mode
    D. Global Tables Mode

Answer: B

104. A company hosts a video streaming platform with its main servers located in the US. However, users in Asia frequently experience high latency and slow video loading times. Which AWS infrastructure component is specifically designed to cache content closer to these users to reduce latency?

    A. Availability Zones

    B. AWS Regions

    C. Edge Locations

    D. AWS Outposts

    Answer: C

105. A startup needs to provision an EC2 instance to run a database for a long-term duration (at least 3 years). They want to achieve the lowest possible cost but must guarantee that the instance will never be interrupted by AWS. Which purchasing option should they recommend?

    A. EC2 Spot Instance

    B. EC2 On-Demand Instance

    C. EC2 Dedicated Host

    D. EC2 Reserved Instance (RI)

    Answer: D

106. A developer is designing an Amazon DynamoDB table to store user profiles and session data. What is the maximum allowed size for a single item (row) in this table?

    A. 128 KB

    B. 256 KB

    C. 400 KB

    D. 1 MB

    Answer: C

107. A company is hosting a web application in a Docker container on Amazon EC2. AWS is responsible for which of the following tasks?

    A. Scaling the web application and services developed with Docker

    B. Provisioning or scheduling containers to run on clusters and maintain their availability

    C. Performing hardware maintenance in the AWS facilities that run the AWS Cloud

    D. Managing the guest operating system, including updates and security patches

    Answer: C

108. Where can you store files in AWS?

    A. Amazon EFS

    B. Amazon SNS

C. Amazon ECS

D. Amazon EMR

Answer: A

109. Which EC2 storage option provides temporary block storage located on the physical host machine and loses its data if the instance is stopped or terminated?

A. Amazon Elastic Block Store (Amazon EBS)

B. Amazon Elastic File System (Amazon EFS)

C. EC2 Instance Store

D. Amazon S3

Answer: C

110. Using Amazon EC2 falls under which of the following cloud computing models?

A. Iaas SaaS

B. IaaS

C. SaaS

D. PaaS

Answer: B

111. Amazon Glacier is an Amazon S3 storage class that is suitable for storing which type of data?

A. Active archives

B. Dynamic websites' assets

C. Active databases

D. Cached data

Answer: A

112. Which EC2 placement group strategy divides instances into logical segments, ensuring that each segment runs on separate underlying hardware racks to support large distributed systems like Hadoop or Cassandra?

A. Cluster Placement Group

B. Spread Placement Group

C. Partition Placement Group

D. Dedicated Hosts

Answer: C

113. Which is not a type of Storage in AWS services?

A. Block Storage

B. File Storage

C. Disk Storage

D. Object Storage

Answer: C

114. Which feature allows you to fully initialize an Amazon EBS volume created from a snapshot to eliminate initial read latency?

A. Snapshot Archive

B. Recycle Bin for Snapshots

C. Fast Snapshot Restore (FSR)

D. Cross-Region Snapshot Copy

Answer: C

115. When deploying containers using Amazon Elastic Container Service (Amazon ECS), which launch type allows you to run tasks on AWS-managed infrastructure without provisioning or managing the underlying EC2 instances?

A. EC2 Launch Type

B. Fargate Launch Type

C. External Launch Type

D. Kubernetes Launch Type

Answer: B

116. What must be enabled as a prerequisite before you can set up Amazon DynamoDB Global Tables for cross-region data replication?

A. DynamoDB Accelerator (DAX)

B. Amazon Kinesis Data Streams

C. On-Demand Capacity Mode

D. DynamoDB Streams

Answer: D

117. Which of the following procedures will help reduce your Amazon S3 costs?

A. Use the Import/Export feature to move old files automatically to Amazon Glacier

B. Use the right combination of storage classes based on different use cases

C. Pick the right Availability Zone for your S3 bucket

D. Move all the data stored in S3 standard to EBS

Answer: B

118. If an Application Load Balancer is configured to route incoming traffic based on the domain name, such as routing one.example.com to Service A and other.example.com to Service B, which routing method is being utilized?

A. Host-based routing

B. Request-based routing

C. Path-based routing

D. Header-based routing

Answer: A

119. An application runs on multiple Amazon EC2 instances that access a shared file system simultaneously. Which AWS storage service should be used?

A. Amazon EBS

B. Amazon EFS

C. Amazon S3

D. AWS Artifact

Answer: B

120. Which Amazon API Gateway endpoint type is designed for worldwide users by routing requests through CloudFront Edge Locations to reduce latency?

A. Regional

B. Edge-Optimized

C. Private

D. Global

Answer: B

121. Which of the following is a key limitation of object storage systems like Amazon S3?

A. They cannot store massive datasets cost-effectively.

B. They provide very limited metadata capabilities.

C. They require a rigid, hierarchical folder structure to operate.

D. They do not support partial edits; the entire object must be overwritten to make a change.

Answer: D

122. A data science team is setting up a shared ML dataset for distributed cluster training. Which storage option provides the moderate speed and high scalability required for this specific workload?

A. Amazon Elastic File System (Amazon EFS)

B. Amazon S3 Glacier

C. Relational Database running on Amazon EBS

D. Single Disk Storage

Answer: A

123. How does block storage manage and locate data on the backend?

A. By nesting files within a hierarchy of directories and folders using file paths.

B. By breaking data into fixed-size blocks, each with a unique address stored in a mapping table.

C. By using a flat address space with globally unique identifiers and customizable metadata.

D. By organizing data into relational tables with primary and foreign keys.

Answer: B

124. A company needs to process large-scale log files where the lowest storage cost is the primary priority, and the data is accessed infrequently. Which Amazon EBS volume type is most appropriate?

A. Provisioned IOPS SSD (io1)

B. General Purpose SSD (gp3)

C. Cold HDD (sc1)

D. Throughput Optimized HDD (st1)

Answer: C

125. When configuring an Amazon S3 Lifecycle rule to transition objects to a cheaper storage class, what is the recommended minimum object size to transition so that the transition costs do not outweigh the storage savings?

A. 16 KiB

B. 128 KiB

C. 1 MiB

D. 1 GiB

Answer: B

126. Which component is a pre-configured template (including operating system, software, etc.) used to launch an EC2 instance?

A. EBS Snapshot

B. Amazon Machine Image (AMI)

C. User Data

D. Security Group

Answer: B

127. You want to execute automated commands (such as installing applications) as soon as an EC2 instance is launched for the first time. Which feature should you use?

A. Instance Metadata

B. Elastic IP

C. User Data

D. Lifecycle Rules

Answer: C

128. Which EC2 instance type is designed for workloads that require heavy data processing in memory (RAM), such as distributed databases?

    A. Compute Optimized

    B. Memory Optimized

    C. Storage Optimized

    D. General Purpose

    Answer: B

129. You have deployed your application on multiple Amazon EC2 instances. Your customers complain that sometimes they can't reach your application. Which AWS service allows you to monitor the performance of your EC2 instances to assist in troubleshooting these issues?

    A. AWS Lambda.

    B. AWS Config.

    C. Amazon CloudWatch.

    D. AWS CloudTrail.

    Answer: C

130. A company is expecting a short-term spike in internet traffic for its application. During the traffic increase, the application cannot be interrupted. The company also needs to minimize cost and maximize flexibility. Which Amazon EC2 instance type should the company use to meet these requirements?

    A. On-Demand Instances

    B. Spot Instances

    C. Reserved Instances

    D. Dedicated Hosts

    Answer: A

131. Which service stores objects, provides real-time access to those objects, and offers versioning and lifecycle capabilities?

    A. Amazon Glacier.

    B. AWS Storage Gateway.

    C. Amazon S3.

    D. Amazon EBS.

    Answer: C

132. A company runs a data analytics platform that processes large amounts of data every night. The workload is fault-tolerant and can handle interruptions. The company wants to minimize costs as much as possible.

    A. On-Demand Instances

B. Reserved Instances

C. Savings Plans

D. Spot Instances

Answer: D

133. Which of the following statement about Amazon EC2 Security Groups is TRUE?

A. A security group can only be associated with one EC2 instance

B. Security group rules require a reboot of the instance to take effect

C. An EC2 instance can be associated with multiple security groups

D. Security groups only control inbound traffic

Answer: C

134. A company is running an application on an Amazon EC2 instance in Region A. The company wants to deploy the same application in Region B with identical configuration, including OS, installed software, and settings, for disaster recovery purposes.

What is the MOST appropriate way to achieve this?

A. Create a snapshot of the EC2 instance and directly launch it in Region B

B. Create an AMI from the EC2 instance, copy the AMI to Region B, and launch a new instance from it

C. Export the EC2 instance as a file and manually import it into Region B

D. Use Auto Scaling to automatically replicate the instance to another region

Answer: B

135. A machine learning team is training models on large datasets stored in Amazon S3. The training jobs run on EC2 instances that repeatedly read the same subset of data during each training epoch. Which storage strategy will most effectively reduce network bottlenecks and improve training performance without changing the data lake architecture?

A. Use EBS or instance store as a local cache of frequently accessed S3 objects

B. Use Amazon S3 with S3 Select for every training batch read

C. Move all data from S3 to dedicated EBS volumes before each training job

D. Migrate all S3 data to EFS and mount it on training instances

Answer: A

136. A distributed application running across hundreds of EC2 instances accesses the same EFS file system. You observe increased latency at scale. Which configuration change is most appropriate to improve scalability?

A. Switch the EFS performance mode from General Purpose to Max I/O

B. Add more EFS mount targets in the same Availability Zone

C. Increase the size of the EFS file system by writing dummy data

D. Convert the EFS file system to One Zone storage class

Answer: A

137. You run a big data analytics workload on EC2 that processes large, sequentially accessed files. The workload is cost-sensitive and can tolerate moderate latency but needs high throughput. Which Amazon EBS volume type is most cost-effective while still meeting these needs?

A. Cold HDD (sc1)

B. Throughput Optimized HDD (st1)

C. Provisioned IOPS SSD (io2)

D. General Purpose SSD (gp3)

Answer: B

138. You have deployed a web application on multiple Amazon EC2 instances behind an Application Load Balancer. Traffic fluctuates significantly throughout the day. You want to: Automatically scale based on demand; Reduce cost during low traffic periods; Ensure high availability without interruption. Which solution is MOST optimal?

A. Use an Auto Scaling Group with Scheduled Scaling based on fixed times

B. Use an Auto Scaling Group with Dynamic Scaling (target tracking) based on CPU utilization

C. Use a fixed number of On-Demand instances and manually scale when needed

D. Use only Spot Instances without any On-Demand instances

Answer: B

139. You store log files in Amazon S3: 0–30 days -> frequently accessed; 30–90 days -> infrequently accessed; 90 days -> rarely accessed but must be retained long-term. You want to minimize cost as much as possible while still allowing retrieval when needed. Which strategy is the best fit?

A. Keep all data in S3 Standard

B. Lifecycle: Standard → Standard-IA → Glacier Flexible Retrieval

C. Lifecycle: Standard → One Zone-IA → Glacier Deep Archive

D. Lifecycle: Standard → Intelligent-Tiering only (no further transitions)

Answer: B

140. You are running a mission-critical relational database on Amazon EC2. The workload has the following characteristics: Requires consistently high IOPS; Low latency is critical; Cannot tolerate performance variability; Data durability is very important. Which Amazon EBS volume type is the MOST appropriate?

A. gp3 (General Purpose SSD)

B. io2 (Provisioned IOPS SSD)

C. st1 (Throughput Optimized HDD)

D. sc1 (Cold HDD)

Answer: B

141. A company is designing a storage architecture for its AI platform. They need to handle: (1) raw data ingestion at petabyte scale with low cost, (2) feature serving with low latency and consistency between training and serving, and (3) experiment metadata tracking with easy querying and lineage tracking. Based on the Storage Layer in AI Systems, which combination of storage layers and their associated tools correctly matches all three requirements?

A. Raw Data Storage (Data Warehouse/Data Lake) → Feature Storage (Feast) → Metadata Storage (MLflow)

B. Raw Data Storage (Pinecone) → Feature Storage (FAISS) → Metadata Storage (Feast)

C. Raw Data Storage (Feast) → Feature Storage (MLflow) → Metadata Storage (Pinecone)

D. Raw Data Storage (MLflow) → Feature Storage (Pinecone) → Metadata Storage (FAISS)

Answer: A

142. A machine learning team needs to store a shared NLP dataset that will be used for distributed training across multiple compute instances in different Availability Zones. The dataset needs moderate read speed and high scalability. Based on the storage impact analysis for ML training, which AWS storage service and reasoning is most appropriate?

A. Amazon S3 — because it offers the lowest latency for ML I/O operations

B. Amazon EBS — because it supports multi-attach across Availability Zones for all volume types

C. AWS Glacier — because it provides the most cost-effective solution for large datasets

D. Amazon EFS — because it offers moderate speed, high scalability, and is suitable for distributed training with shared access across AZs

Answer: D

143. Regarding Amazon S3 Storage Classes, what happens when you configure Lifecycle Rules to transition objects automatically?

A. Objects can only move from Glacier Deep Archive to Standard

B. Objects can transition from Standard → Standard IA → Glacier tiers, and this movement can be automated using Lifecycle Rules

C. Lifecycle Rules only apply to objects smaller than 128 KB

D. Transition between storage classes requires manual intervention and cannot be automated

Answer: B

144. In the AI system storage pipeline, what are the key requirements for Embedding Storage?

A. Version control, reproducibility, and deployment tracking

B. Massive scalability (TB → PB), low cost, and high durability

C. Fast similarity search (ANN), semantic retrieval, and low latency

D. Query experiments easily, track lineage, and debug models

Answer: C

145. Which storage type is most suitable for running relational databases (SQL) and Virtual Machine (VM) disks?

A. Object Storage — because of its rich metadata capabilities

B. File Storage — because of its hierarchical directory structure

C. Block Storage — because of its extremely fast performance and support for high-transaction workloads

D. Archival Storage — because of its low cost

Answer: C

146. A customer showed interest in a company product and wanted to quickly review its application. To do it, the boss wants to launch an instance as fast as it can and optimizing the cost shows we can easily stop it after the customer's review. Which purchasing options below are preferred?

A. Spot instance

B. Reserved Instance

C. On-demand Instance

D. Saving Plans

Answer: C

147. Which of the following that an ALB can't route traffic to?

A. S3 Buckets

B. EC2 Instances

C. Lambda Functions

D. IP Addresses

Answer: A

148. Which service provides temporary block storage that is physically attached to the host computer?

A. Instance store

B. EBS

C. S3 standard

D. Amazon EFS

Answer: A

149. What attaching an EBS volume to an EC2 instance, where must the volume be located?

A. In any AWS region

B. Should be at your physical service where you company placed

C. In the same AZ as the EC2 instance

D. In the same VPC as the EC2 instance

Answer: C

150. What is the keys difference between EBS and EFS in terms of Availability Zones?

A. EFS is only available in a single AZ

B. Neither service supports multiple AZs

C. EBS is locked to a single AZ, while EFS data is stored redundantly across multiple AZs.

D. EBS can work across multiple AZs by default

Answer: C

151. A company is building a distributed ML training system on multiple EC2 instances. The training nodes must access the same dataset at the same time. The workload does not require ultra-low latency like a relational database, but it does require shared access and scalability. Which storage option is the best fit?

A. Amazon EBS

B. Amazon EFS

C. Amazon S3

D. Instance store

Answer: B

152. How does EFS scale its storage capacity?

A. Manually increase the size in the AWS console

B. It scales automatically as you add or remove files

C. It scales based on the number of EC2 instances attached

D. You must use Auto Scaling Groups to scale EFS

Answer: B

153. Which ELB is best suited for handling millions of requests per second while maintaining ultra-low latency?

A. Application Load Balancer

B. Network Load Balancer

C. Classic Load Balancer

D. Amazon Load Balancer

Answer: B

154. A company needs storage for a content platform where data is stored as files in folders, accessed by file path, and shared by several application servers. Which storage type and AWS service best match this requirement?

A. Block storage — Amazon EBS

B. File storage — Amazon EFS

C. Object storage — Amazon S3

D. Archival storage — Amazon Glacier

Answer: B

155. Which statement is TRUE about object storage?

A. It is best for high-transaction relational databases because it supports partial in-place edits efficiently.

B. It has lower latency than block storage and is the default choice for SQL databases.

C. It is cost-effective for massive datasets, supports rich metadata, and is suitable for backups and AI/ML training sets.

D. It is mainly designed for EC2 root volumes and ultra-low-latency workloads.

Answer: C

156. A solutions architect is choosing between S3, EBS, and EFS for a workload. The workload is single-instance, random I/O heavy, latency-sensitive, and behaves like a transactional system. Which is the best choice?

A. Amazon S3

B. Amazon EFS

C. Amazon EBS

D. Amazon S3 Glacier Instant Retrieval

Answer: C

157. Which service allows multiple EC2 instances to access the same file system simultaneously across different AZs ?

A. EBS

B. EFS

C. ELB

D. ENI

Answer: B

158. Elastic block storage (EBS), Elastic network interfaces (ENI), Elastic load balancer (ELB), Elastic file system (EFS). Which service here, its setup default can only work with a single EC2 instance?

A. EBS

B. ENI

C. ELB

D. EFS

Answer: A

159. A user needs to automatically discover, classify, and protect sensitive data stored in Amazon S3. Which AWS service can meet these requirements?
    A. Amazon Inspector
    B. Amazon Macie
    C. Amazon GuardDuty
    D. AWS Secrets Manager

    Answer: B

160. What kind of database is Amazon DynamoDB?
    A. Document database
    B. Relational database
    C. NoSQL database
    D. Graph database

    Answer: C

161. Which Amazon S3 storage class has the lowest cost?
    A. S3 Intelligent-Tiering
    B. S3 Standard
    C. S3 Glacier Deep Archive
    D. S3 One Zone-IA

    Answer: C

162. What is AWS EFS?
    A. AWS Elastic File System
    B. AWS Efficient File System
    C. AWS Enterprise File System
    D. AWS External File Storage

    Answer: A

163. Economies of scale continually reduce AWS Cloud pricing.
    A. False
    B. True
    C. Only for enterprise customers
    D. Only for storage services

    Answer: B

164. Which database is ideal for large organizations and enterprises?
    A. PostgreSQL

B. MySQL

C. Amazon Aurora

D. SQLite

Answer: C

165. A company needs to run a large-scale data analysis workload that is fault-tolerant and can handle interruptions. Which Amazon EC2 purchasing option provides the lowest cost (up to 90

A. On-Demand Instances

B. Reserved Instances

C. Spot Instances

D. Dedicated Hosts

Answer: B

166. Which of the following is a key characteristic of AWS Lambda?

A. You must provision and manage the underlying virtual servers.

B. It is a serverless compute service that runs code in response to events and scales automatically.

C. It is designed for long-running applications that must stay active 24/7.

D. You are billed based on the number of instances you have running per hour.

Answer: B

167. What is the primary advantage of using AWS Fargate for containerized applications?

A. It provides a physical server fully dedicated to a single customer.

B. It allows you to choose and manage the specific EC2 instances that run your containers.

C. It is a serverless compute engine for containers that removes the need to provision or manage servers.

D. It acts as a static public IPv4 address for your container environment.

Answer: C

168. A developer wants to automatically install software packages and update the operating system the first time an Amazon EC2 instance starts. Which feature should they use?

A. Amazon Machine Image (AMI)

B. EC2 User Data

C. Security Groups

D. Elastic Block Store (EBS) Snapshots

Answer: B

169. A developer accidentally terminated an EC2 instance. They noticed that the root volume (which contained the operating system) was automatically deleted, but an additional data volume attached to the same instance was kept. What is the default behavior of Amazon EBS regarding the DeleteOnTermination attribute?

A. Both root and additional volumes are deleted by default.

B. Both root and additional volumes are kept by default.

C. The root volume is deleted by default, but additional volumes are kept by default.

D. The root volume is kept by default, but additional volumes are deleted by default.

Answer: C

170. You are building a serverless web application. You have written your backend logic using AWS Lambda. You now need to expose these Lambda functions as RESTful APIs so your frontend application can call them over HTTPS. Which AWS service should you use to create and manage these APIs?

A. Amazon API Gateway

B. Elastic Load Balancer (ELB)

C. AWS Step Functions

D. Amazon Route 53

Answer: A

171. A media production company has petabytes of raw 4K video footage. Once a video project is finished, the raw footage is archived. The company wants the absolute lowest storage cost possible. If an old project needs to be reopened, the video editors are completely fine waiting up to 12 hours for the files to be restored. Which Amazon S3 storage class should they use?

A. S3 Standard-IA

B. S3 Glacier Instant Retrieval

C. S3 Glacier Flexible Retrieval

D. S3 Glacier Deep Archive

Answer: D

172. You are configuring an S3 Lifecycle Rule to automate storage costs. Your goal is to keep objects in S3 Standard for the first 30 days. After 30 days, the objects are infrequently accessed but need immediate retrieval. After 365 days, the objects are kept only for compliance and can take a day to retrieve. Which transition sequence is valid and most cost-effective?

A. S3 Standard S3 Glacier Deep Archive S3 Standard-IA

B. S3 Standard S3 Intelligent-Tiering S3 One Zone-IA

C. S3 Standard S3 Standard-IA S3 Glacier Deep Archive

D. S3 Standard S3 Glacier Flexible Retrieval S3 Glacier Instant Retrieval

Answer: C

173. The identification process of an online financial services company requires that new users must complete an online interview with their security team. The completed recorded interviews are only required in the event of a legal issue or a regulatory compliance breach. What is the most cost-effective service to store the recorded videos?

A. S3 Intelligent-Tiering.

B. AWS Marketplace.

C. Amazon S3 Glacier Deep Archive.

D. Amazon EBS.

Answer: C

174. A company wants to focus on business activities instead of managing compute and capacity. Which AWS service can be used to automatically add or remove Amazon EC2 instances based on demand?

A. Elastic Load Balancer.

B. Amazon EC2 Auto Scaling.

C. Amazon Route 53.

D. Amazon CloudFront.

Answer: B

175. If an instance store reboots, does the data persist?

A. Yes, always

B. No

C. Only for Linux instances

D. Only if backed up

Answer: B

176. A company needs to physically migrate 50 Petabytes of data from their local data center to AWS because transferring it over their internet connection would take several years. Which service is designed for this?

A. AWS Storage Gateway

B. AWS Snowball Edge

C. AWS Snowmobile

D. AWS Direct Connect

Answer: C

177. Which services are parts of the AWS serverless platform?

A. Amazon EC2, Amazon S3, Amazon Athena.

B. Amazon Kinesis, Amazon SQS, Amazon EMR.

C. AWS Step Functions, Amazon DynamoDB, Amazon SNS.

D. Amazon Athena, Amazon Cognito, Amazon EC2.

Answer: C

178. Which of the following are characteristics of Amazon S3?

A. A global file system

B. A cloud-based object storage service

C. A local file store

D. A network file system

Answer: B

179. A startup wants to store secondary backup copies of data that can be easily recreated if lost at the absolute lowest cost possible, without needing high resilience. Which storage class is best?

A. Amazon S3 Standard

B. Amazon S3 One Zone-IA

C. Amazon S3 Standard-IA

D. Amazon S3 Intelligent-Tiering

Answer: B

180. A global corporation has a predictable baseline of compute usage across multiple AWS Regions. They use a mix of EC2 instances, AWS Lambda, and AWS Fargate. Which purchasing model provides the highest level of flexibility to cover all these services?

A. Standard Reserved Instances

B. EC2 Instance Savings Plans

C. Compute Savings Plans

D. Regional Reserved Instances

Answer: C

181. A solutions architect is designing a system for a tightly coupled scientific simulation that requires extremely low-latency communication between nodes using a Cluster Placement Group. What is a true limitation of this strategy?

A. It cannot span across multiple Availability Zones

B. It is limited to a maximum of 7 instances per Region

C. It requires the use of Dedicated Hosts for all instances.

D. It only supports T-series instance types

Answer: A

182. AWS Elastic Block Store Snapshot is

A. Differential data backup

B. Full data backup

C. Incremental data backup

D. Manual data copy

Answer: C

183. What is the maximum amount of RAM you can allocate to an AWS Lambda function?
    A. 10 GB
    B. 3 GB
    C. 1 GB
    D. 32 GB
    Answer: A

184. Which Amazon EFS storage class offers the lowest "first byte read latency"?
    A. EFS Standard
    B. EFS Infrequent Access (IA)
    C. EFS Archive
    D. They all have the same latency.
    Answer: A

185. Which AWS storage service is designed to transfer petabytes of data in and out of the cloud?
    A. AWS Snowmobile
    B. AWS Redshift
    C. Amazon S3
    D. AWS DataSync
    Answer: A

186. Which tool lets you visualize and manage your AWS costs?
    A. AWS Cost Explorer
    B. AWS Budgets
    C. AWS Price Calculator
    D. AWS Billing Dashboard
    Answer: A

187. Which service lets you run code without managing servers
    A. AWS ECS
    B. AWS Lambda
    C. Amazon EC2 Auto Scaling
    D. AWS Fargate
    Answer: B

188. Which of the following factors directly affects the training speed of a machine learning model (ML Training Speed)?
    A. Monthly storage cost

B. Maximum capacity of a single disk

C. User API interface

D. Storage system I/O performance

Answer: D

189. Which AWS service provides a history of application data changes with immutability?

A. AWS Quantum Ledger Database

B. AWS Neptune

C. AWS DocumentDB

D. AWS ElastiCache

Answer: A

190. Why is AWS Glacier considered NOT suitable for directly training machine learning models?

A. Extremely high latency

B. Storage cost is too expensive

C. Does not support log data storage

D. Low data durability

Answer: A

191. Which AWS storage service is recommended for distributed training thanks to its high scalability and moderate speed?

A. Amazon EBS

B. Amazon EFS

C. Amazon S3 Glacier

D. Local Storage Device

Answer: B

192. In the scenario of enabling the Hibernate feature for an EC2 instance, which of the following technical requirements is mandatory for the root volume?

A. The root volume must have a size at least twice the instance's RAM.

B. The root volume must be an Instance Store volume to ensure RAM write speed.

C. The root volume must be an encrypted Amazon EBS volume.

D. The root volume must have the "Delete on Termination"attribute set to false

Answer: C

193. A solutions architect needs to run a cluster of 7 EC2 instances such that each instance is placed on a separate physical hardware rack within the same Availability Zone to avoid simultaneous failures. Which type of Placement Group best meets this requirement?

A. Dedicated Placement Group

B. Cluster Placement Group

C. Partition Placement Group

D. Spread Placement Group

Answer: D

194. A company runs a web application on Amazon EC2. Traffic spikes during peak hours and drops at night. The company wants a solution that automatically adjusts the number of instances based on actual load while minimizing operational effort. Which solution is the most appropriate?

A. Use Scheduled Scaling

B. Use Target Tracking Scaling

C. Manually increase instance size

D. Use Reserved Instances

Answer: B

195. What is the key difference in hardware control capabilities between Dedicated Hosts and Dedicated Instances?

A. Dedicated Instances allow users to choose the exact physical rack location in the data center.

B. Both provide the same level of control over the physical server's BIOS configuration.

C. Dedicated Instances always remain on the same physical hardware even after a Stop/Start operation.

D. Dedicated Hosts provide visibility and control over instance placement on physical servers to meet software licensing requirements (BYOL).

Answer: D

196. An application requires high availability and fault tolerance. If one instance fails, traffic must automatically be routed to another instance, and the system must self-recover.

Which solution is the most appropriate?

A. Use EC2 only

B. EC2 + Auto Scaling

C. EC2 + Load Balancer

D. EC2 + Load Balancer + Auto Scaling

Answer: D

197. What is Amazon DynamoDB Accelerator (DAX) capable of?

A. Improves visualization of graphs

B. Improves write performance

C. Improves read performance of NoSQL data

D. Improves relational queries

Answer: C

198. A company wants to run short-term workloads at the lowest possible cost and is willing to accept that instances may be terminated at any time.

Which solution is the most appropriate?

A. On-Demand

B. Reserved

C. Spot Instances

D. Dedicated Hosts

Answer: C

199. What problem is the Amazon EBS “Snapshot Archive” feature specifically designed to solve?

A. Increase the speed of creating new volumes from snapshots through Fast Snapshot Restore.

B. Protect important snapshots from accidental deletion for up to one year.

C. Automatically create incremental backups based on real-time data changes.

D. Optimize costs for infrequently accessed backups with an expected restore time of 24 to 72 hours.

Answer: D

200. A company runs a steady workload on EC2 over a long period of time. They want to minimize costs as much as possible without changing the architecture.

Which solution is the most appropriate?

A. On-Demand Instances

B. Spot Instances

C. Reserved Instances

D. Auto Scaling

Answer: C

201. A developer is using an AWS Lambda function to process large video files uploaded to S3. The function consistently takes 18 minutes to complete the processing. To ensure the function finishes successfully, which action should be taken?

A. Increase the Memory (RAM) allocation to the maximum (10GB) to provide more CPU power.

B. Change the Timeout setting in the Lambda configuration to 20 minutes

C. Migrate the workload to Amazon EC2 or AWS Fargate.

D. Enable Provisioned Concurrency to keep the function "warm"for longer processing.

Answer: C

202. A company wants to quickly deploy multiple instances with the same configuration and pre-installed software, while minimizing setup time.

Which solution is the best?

A. Configure each instance via SSH

B. Use snapshots each time

C. Create an AMI and use it in a Launch Template

D. Manually copy files

Answer: C

203. You are using an Amazon EBS Volume (General Purpose SSD) attached to an EC2 instance in Availability Zone (AZ) us-east-1a. There is a major power outage that takes down the entire us-east-1a zone. How can you access your data in another zone (us-east-1b)?

A. EBS is a Regional service, so you can simply attach the volume to a new instance in us-east-1b.

B. You cannot; EBS volumes are tied to a specific AZ. You must restore the data from a Snapshot.

C. AWS automatically replicates EBS volumes across all AZs in a Region for high availability.

D. Use S3 Transfer Acceleration to move the EBS data to the new zone.

Answer: B

204. How much time does the Auto Scaling Group need to take the next activates (default) ?

A. 3 seconds

B. 300 miliseconds

C. 300 seconds

D. 3000 seconds

Answer: C

205. What is an Amazon Machine Image (AMI) primarily used for?

A. To automatically scale EC2 resources

B. To provide a preconfigured template to launch EC2 instances

C. To distribute network traffic across instances

D. To provide network-attached storage for EC2

Answer: B

206. A company is using a third-party service to back up 10 TB of data to a tape library. The on-premises backup server is running out of space. The company wants to use AWS services for the backups without changing its existing backup workflows. Which AWS service should the company use to meet these requirements?

A. Amazon Elastic Block Store (Amazon EBS)

B. AWS Storage Gateway

C. Amazon Elastic Container Service (Amazon ECS)

D. AWS Lambda

Answer: B

207. A company runs a web application on multiple EC2 instances behind an Application Load Balancer (ALB). The instances need to share a common directory containing user-uploaded files with low latency.

Which solution is the most appropriate?

A. Store the files in Amazon S3 and mount it directly to the EC2 instances

B. Use an EBS volume and attach it to all instances

C. Use Amazon EFS and mount it to all instances

D. Store the files in the instance store

Answer: C

208. Which feature allows you to run a startup script automatically when an EC2 instance first boots?

A. Auto Scaling Groups

B. AWS Step Functions

C. EC2 User Data

D. Elastic Network Interfaces

Answer: C

209. A company wants a fully managed Windows file server for its Windows-based applications. Which AWS service will meet this requirement?

A. Amazon FSx

B. Amazon Elastic Kubernetes Service (Amazon EKS)

C. Amazon Elastic Container Service (Amazon ECS)

D. Amazon EMR

Answer: A

210. A company has an uninterruptible application that runs on Amazon EC2 instances. The application constantly processes a backlog of files in an Amazon Simple Queue Service (Amazon SQS) queue. This usage is expected to continue to grow for years. What is the MOST cost-effective EC2 instance purchasing model to meet these requirements?

A. Spot Instances

B. On-Demand Instances

C. Savings Plans

D. Dedicated Hosts

Answer: C

211. A company needs to store millions of log files. The files will be accessed frequently during the first 30 days, after which they will rarely be accessed but must be retained for a long period.

Which solution is the most cost-effective?

A. Amazon S3 Standard with a lifecycle policy transitioning to Glacier

B. Amazon S3 Standard

C. Amazon S3 Intelligent-Tiering

D. Amazon EFS Infrequent Access

Answer: A

212. A company plans to deploy containers on AWS. The company wants full control of the compute resources that host the containers. Which AWS service will meet these requirements?

A. Amazon Elastic Kubernetes Service (Amazon EKS)

B. AWS Fargate

C. Amazon EC2

D. Amazon Elastic Container Service (Amazon ECS)

Answer: C

213. Which EC2 purchasing option allows you to use spare AWS capacity at up to a 90

A. Spot Instances

B. Eco Plans

C. Savings Instances

D. Savings Plans

Answer: A

214. Which Amazon EC2 pricing model is the MOST cost efficient for an uninterruptible workload that runs once a year for 24 hours?

A. On-Demand Instances

B. Reserved Instances

C. Spot Instances

D. Dedicated Instances

Answer: A

215. A system needs to store static files (images, CSS, JavaScript) for a public website, with high scalability and CDN integration.

Which solution is the best?

A. EBS

B. EFS

C. S3 + CloudFront

D. Instance Store

Answer: C

216. What is the maximum number of instances per Availability Zone allowed in a Spread Placement Group?

A. 3

B. 5

C. 7

D. Based on developer's settings

Answer: C

217. A company needs to run a batch processing job. This task can be abruptly interrupted and resumed later without causing errors or affecting the final result. Which Amazon EC2 pricing model will provide the lowest cost for the company in this scenario?

A. On-Demand Instances

B. Spot Instances

C. Reserved Instances

D. Dedicated Hosts

Answer: B

218. What is the default termination behavior for the root EBS volume when an EC2 instance is terminated?

A. It is kept and must be manually deleted.

B. It is automatically backed up to a snapshot, then deleted.

C. It is automatically deleted.

D. It is moved to the Recycle Bin.

Answer: C

219. An e-commerce website frequently experiences unpredictable spikes in traffic. Which of the following architectures combines two AWS services to automatically increase the number of servers during high load and evenly distribute incoming traffic across them?

A. AWS Lambda and Amazon Route 53

B. Amazon CloudFront and Amazon S3

C. Amazon EC2 Auto Scaling and Elastic Load Balancing (ELB)

D. AWS Elastic Beanstalk and Amazon RDS

Answer: C

220. An engineer notices that Amazon EFS has higher latency than Amazon EBS when running database workloads.

    What is the main reason?

    A. Amazon EFS does not use SSDs

    B. Amazon EFS is a network file system

    C. Amazon EFS is object storage

    D. Amazon EFS does not support caching

    Answer: B

221. Which EBS volume type provides the lowest-cost storage option and is designed specifically for infrequently accessed data?

    A. Provisioned IOPS SSD (io1/io2)

    B. Throughput Optimized HDD (st1)

    C. General Purpose SSD (gp2/gp3)

    D. Cold HDD (sc1)

    Answer: D

222. A developer wants to automatically run a code script to compress an image immediately every time a user uploads a new image to Amazon S3. They only want to pay for the exact compute time the code consumes and do not want to provision, configure, or manage any servers. Which compute service is best suited for this requirement?

    A. Amazon EC2

    B. Amazon ECS

    C. AWS Elastic Beanstalk

    D. AWS Lambda

    Answer: D

223. A developer wants to increase the CPU power for an AWS Lambda function. Which configuration change is required to achieve this?

    A. Switch the Lambda execution environment to a dedicated EC2 instance.

    B. Enable 'Turbo Mode' in the Lambda configuration.

    C. Increase the Memory (RAM) allocated to the function.

    D. Update the CPU allocation setting in the Lambda Console.

    Answer: C

224. In order to use the EC2 Hibernate feature, which requirement must be met for the root volume?

    A. The root volume must be stored on an Instance Store.

    B. The root volume must be an encrypted Amazon EBS volume.

C. The root volume must be detached before hibernating.

D. The root volume must be backed by an EFS file system.

Answer: B

225. Your company is planning to migrate its microservices architecture, currently packaged as Docker containers, to AWS. The technical team wants to use a fully managed container orchestration service that natively supports the open-source Kubernetes engine. Which service should they choose?

A. Amazon Elastic Container Service (Amazon ECS)

B. Amazon Elastic Kubernetes Service (Amazon EKS)

C. AWS Fargate

D. AWS Elastic Beanstalk

Answer: B

226. Which Amazon EC2 purchasing option is best suited for a 24/7 database workload with a predictable traffic pattern for the next 3 years?

A. On-Demand Instances

B. Dedicated Hosts

C. Spot Instances

D. Reserved Instances (RI)

Answer: D

227. A company is designing a shared storage solution for a containerized application running on multiple Amazon EC2 instances across different Availability Zones. The application requires a POSIX-compliant file system with automatic scaling and high availability.

Which solution is the most appropriate?

A. Use Amazon EBS volumes and attach them to all EC2 instances across Availability Zones

B. Use Amazon EFS and mount it on all EC2 instances

C. Store data in Amazon S3 and access it using the S3 API

D. Use instance store volumes for shared storage

Answer: B

228. A company is looking for a storage solution that can store a vast amount of unstructured data, such as images and videos. Each item must be stored with a unique identifier and descriptive metadata. Which storage type should they use?

A. Block Storage

B. File Storage

C. Object Storage

D. Single Disk Storage

Answer: C

229. A company needs to store large volumes of AI training datasets (images and text files) that are accessed frequently during model training. The solution must scale to petabytes, minimize cost, and require no strict schema. Which AWS storage service best fits this requirement?

A. Amazon EBS (gp3 volume)

B. Amazon EFS Standard

C. Amazon S3 Standard

D. Amazon RDS on EBS

Answer: C

230. Compared to a Single Disk Storage, which of the following is a key characteristic of a Distributed Storage System (Max Scale)?

A. Lowest Durability and High Speed

B. Very High Durability and Very High Throughput

C. Low Cost and Limited Throughput

D. Low Latency and High Speed

Answer: B

231. Which OSI layer does the Application Load Balancer (ALB) operate at in AWS Structure ?

A. Layer 3

B. Layer 4

C. Layer 7

D. Layer 2

Answer: C

232. A developer is building a real-time AI recommendation system. They need a storage layer that ensures the data used for training is the same as the data used for serving (consistency) and requires low latency for instant predictions. Which storage layer should they use?

A. Raw Data Storage

B. Feature Storage

C. Metadata Storage

D. Model Storage

Answer: B

233. A data engineering team uploads log files to Amazon S3. The logs are actively analyzed for the first 30 days, rarely accessed from day 31–90, and must be archived cheaply from day 91 onward. Which feature of Amazon S3 automates this storage class transition?

A. S3 Replication Rules

B. S3 Lifecycle Rules

C. S3 Bucket Versioning

D. S3 Access Points

Answer: B

234. A developer is running a temporary test environment where Low Cost and Low Latency (High Speed) are the top priorities. Data loss is not a concern for this specific task. Which storage type should they choose?

A. Storage Cluster

B. Distributed Storage System

C. Single Disk Storage

D. Object Storage

Answer: C

235. A big data analytics platform requires Very High Throughput to process massive datasets and Very High Durability to ensure no data is ever lost. They can tolerate higher latency (slower response time). Which storage is most suitable?

A. Single Disk Storage

B. Distributed Storage System (Max Scale)

C. Block Storage

D. Local Storage

Answer: B

236. An AI startup stores 500 TB of raw training data in an S3 bucket. The data is accessed frequently for the first 30 days, but after that, the access pattern becomes unpredictable—some files are never touched again, while others are needed randomly for re-training. Which S3 storage strategy offers the best balance between cost-optimization and performance without manual intervention?

A. S3 Standard-IA with a lifecycle policy to move data after 30 days.

B. S3 One Zone-IA.

C. S3 Glacier Deep Archive.

D. S3 Intelligent-Tiering.

Answer: D

237. Which of the following storage options provides High Durability and High Throughput while maintaining Normal Latency (Medium Speed)?

A. Single Disk Storage

B. Distributed Storage System

C. Storage Cluster

D. Tape Drive Storage

Answer: C

238. A startup is building a distributed ML training cluster where multiple EC2 instances across different Availability Zones must simultaneously read from a shared dataset. Which AWS storage service is most appropriate?

A. Amazon EBS io2

B. Amazon S3 Glacier

C. Amazon EBS gp3

D. Amazon EFS

Answer: D

239. A student is reviewing AWS storage services. He learns that one service is designed to store data as objects in a flat structure, supports REST API access, and is highly scalable and durable for storing large datasets such as logs, backups, and AI training data. Which AWS service is being described?

A. Amazon S3 provides object storage with high scalability and durability.

B. Amazon EBS provides block storage attached to EC2 for low latency workloads.

C. Amazon EFS provides shared file storage for multiple EC2 instances.

D. Amazon EC2 provides virtual servers for running applications.

Answer: A

240. Your image data uploaded to S3 is frequently accessed during the first 30 days, then access gradually declines, and after 90 days, it is rarely used. You want to automate the transition of this data from S3 Standard to cheaper storage classes. Which feature should you use?

A. S3 Replication Rules

B. S3 Lifecycle Rules

C. EBS Snapshots

D. EFS Infrequent Access

Answer: B

241. In the structure of an object in object storage, which component contains descriptive information such as Content-Type and other custom attributes?

A. Unique ID

B. Metadata

C. Data

D. Storage Pool

Answer: B

242. In the architecture of an Object Storage system (like Amazon S3), what is the primary characteristic of its addressing system compared to traditional file systems?

A. It uses a nested directory tree with file paths.

B. It uses a flat address space where each object has a unique identifier.

C. It organizes data into fixed-size 4KB blocks with physical addresses.

D. It relies on a Master Boot Record (MBR) to locate data across clusters.

Answer: B

243. In the context of AI systems, what is a key requirement for 'Raw Data Storage'?

A. Extremely low latency for real-time inference.

B. Massive scalability from TB -> PB and low cost.

C. Strict schema enforcement and relational consistency.

D. Frequent partial edits to individual data blocks.

Answer: B

244. A startup is building an AI data lake to store raw datasets, processed outputs, model artifacts, and logs. The team wants virtually unlimited scalability, high durability, low-cost storage, and API-based access for cloud-native applications. Which AWS storage service is the best fit?

A. Amazon EBS

B. Amazon EFS

C. Amazon S3

D. EC2 Instance Store

Answer: C

245. A company stores compliance audit logs in Amazon S3. The logs are accessed only once or twice per year for regulatory review. The company wants to minimize storage costs while maintaining data durability. Which S3 storage class is MOST cost-effective for this use case?

A. S3 Glacier Deep Archive

B. S3 Standard

C. S3 Standard-IA

D. S3 Intelligent-Tiering

Answer: A

246. A developer configures a lifecycle rule on an Amazon S3 bucket to transition objects to Amazon S3 Glacier after 60 days. What will happen when objects reach 60 days?

A. Objects are automatically deleted and permanently removed from the bucket storage.

B. Objects are automatically moved to Glacier and become cheaper but slower to access.

C. Objects are copied to Glacier while the original objects remain unchanged in S3.

D. Objects are compressed and stored in the same storage class with reduced cost.

Answer: B

247. A machine learning team runs training jobs on multiple Amazon EC2 instances across different Availability Zones, and all instances must read and update the same shared dataset using a standard file system interface without managing storage capacity manually. Which AWS storage service is the most appropriate?

   A. Amazon S3

   B. Amazon EFS

   C. Amazon EBS

   D. Amazon S3 Glacier Deep Archive

   Answer: B

248. A company applies a lifecycle rule to transition objects to Amazon S3 Glacier after 90 days and to delete them after 365 days. What is the expected outcome?

   A. Objects remain in S3 Standard and only metadata is moved to Glacier after 90 days.

   B. Objects are archived in Glacier and cannot be deleted after the lifecycle policy is applied.

   C. Objects are moved to Glacier after 90 days and automatically deleted after 365 days

   D. Objects are duplicated into Glacier and the original objects remain until deleted manually

   Answer: C

249. A company is designing a data pipeline where raw CSV logs are ingested daily. After 30 days, these logs are rarely accessed but must be retained for 3 years for regulatory compliance. If an audit occurs, the logs must be retrievable immediately without a restore step. What is the most cost-effective Amazon S3 Lifecycle configuration?

   A. Keep the objects in S3 Standard for 3 years, then expire them.

   B. Transition the objects to S3 Standard-IA after 30 days, then transition them to S3 Glacier Flexible Retrieval after 3 years.

   C. Transition the objects to S3 Standard-IA after 30 days, then expire them after 1,095 days (3 years).

   D. Transition the objects to S3 Glacier Instant Retrieval after 30 days, then expire them after 3 years.

   Answer: D

250. A developer needs block storage for an EC2 instance running a relational database requiring consistent sub-millisecond latency and more than 80,000 IOPS. Which Amazon EBS volume type should be selected?

   A. st1 (Throughput Optimized HDD)

   B. gp2 (General Purpose SSD)

   C. sc1 (Cold HDD)

   D. io2 Block Express (Provisioned IOPS SSD)

   Answer: D

251. Why is Amazon Glacier considered unsuitable for active Machine Learning (ML) training tasks?
    A. It has extremely high latency that prevents timely data access.
    B. It is too expensive for large datasets.
    C. It lacks the durability required for training datasets.
    D. It cannot store unstructured data like images or video.

    Answer: A

252. What is an AMI (Amazon Machine Image) used for?
    A. Monitoring EC2 performance
    B. Storing backups of databases
    C. Launching EC2 instances with predefined configurations
    D. Encrypting EC2 data

    Answer: C

253. What is the main purpose of EC2 User Data?
    A. Store user credentials
    B. Run scripts during instance startup
    C. Monitor CPU usage
    D. Assign IP addresses

    Answer: B

254. Which EC2 instance type is best suited for memory-intensive workloads like large databases?
    A. Memory Optimized
    B. Compute Optimized
    C. General Purpose
    D. Storage Optimized

    Answer: A

255. Which statement about Elastic IP (EIP) is correct?
    A. It changes every time the instance restarts
    B. It is a private IP address
    C. It can only be used within a VPC
    D. It is a static public IPv4 address

    Answer: D

256. What is a key characteristic of Amazon EBS?
    A. Data is lost when the instance stops

B. It is shared across multiple instances by default

C. It provides persistent block storage

D. It only works with serverless services

Answer: C

257. Which EC2 pricing option provides the lowest cost but can be interrupted?

A. On-Demand Instances

B. Dedicated Hosts

C. Reserved Instances

D. Spot Instances

Answer: D

258. You successfully push a new Docker image tagged v2.0 to Amazon ECR, but your website still runs the old v1.0 code. You verify the v2.0 image is physically present in the ECR repository.

What should you do to run your new v2.0 Docker Image to your website?

A. Restart Run the Docker Image with v2.0 tag in Amazon ECR.

B. Update the Amazon ECS Task Definition to point to the v2.0 tag.

C. Increase the storage quota for the Amazon ECR registry.

D. Wait for Amazon ECR to automatically deploy the image to the servers.

Answer: B

259. A research intern needs to process a short video dataset that is 'interrupt-tolerant'. Which purchasing model would minimize costs for this non-critical task?

A. Reserved Instances

B. Spot Instances

C. On-Demand Instances

D. Dedicated Hosts

Answer: B

260. What does horizontal scaling (scale out) mean?

A. Adding more instances to handle load

B. Increasing CPU and RAM of one instance

C. Reducing storage size

D. Moving to a different region

Answer: A

261. Which statement best describes AWS Lambda?

A. Requires manual server provisioning

B. Runs continuously like EC2

C. Executes code in response to events

D. Only supports Java

Answer: C

262. Which of the following workload types is least suited for Reserved Instances?

A. Long-term workloads

B. Batch jobs

C. Permanent workloads

D. Non-changing workloads

Answer: B

263. A European bank with a stable customer base needs a database server running 24/7. They need a specific EC2 configuration, want to save costs while ensuring high availability. Which EC2 pricing model is most appropriate?

A. On-Demand Instances

B. Reserved Instances

C. Savings Plans

D. Spot Instances

Answer: B

264. Which of the following is not a valid EC2 instance type?

A. Data Lake Optimized

B. General Purpose

C. Compute Optimized

D. Memory Optimized

Answer: A

265. A development team has multiple Linux-based EC2 instances across different Availability Zones that need concurrent read/write access to a shared, POSIX-compliant file system. The solution must be highly available, automatically scale, and require no management of underlying servers. Which storage service is the most appropriate?

A. Amazon Elastic Block Store (Amazon EBS)

B. Amazon Elastic File System (Amazon EFS)

C. Amazon FSx for Windows File Server

D. Amazon S3

Answer: B

266. What is statement is WRONG about EBS and EFS file systems?

A. EFS works across multiple AZs.

B. EBS can be mounted by only one EC2 instance.

C. Both are only compatible with Linux-based AMIs.

D. EFS can be mounted by multiple EC2 instances simultaneously.

Answer: C

267. A company plans to host its database on one or more EC2 instances to store persistent transaction data. Which storage option is not suitable for this use case?

A. Elastic File System

B. Instance Store

C. Elastic Block Store

D. nan

Answer: B

268. Which AWS service provides virtual servers in the cloud?

A. Amazon S3

B. Amazon EC2

C. AWS Lambda

D. Amazon RDS

Answer: B

269. A company is building an application on AWS. The application stores: - Website images and video files that need to be accessed over the internet, - A database running on one EC2 instance that requires low-latency block storage, - Shared application files that must be accessed by multiple EC2 instances at the same time.

Which combination of AWS storage services is the most appropriate?

A. Use Amazon EBS for website images, Amazon S3 for the database, and Amazon EFS for shared files

B. Use Amazon S3 for website images and video files, Amazon EBS for the database, and Amazon EFS for shared files

C. Use Amazon EFS for website images and video files, Amazon S3 for the database, and Amazon EBS for shared files

D. Use Amazon S3 for website images and video files, Amazon EFS for the database, and Amazon EBS for shared files

Answer: B

270. Your ML classification model is now in production. It serves a global user base with a stable, predictable baseline of traffic 24/7. Your primary goal is to minimize long-term operating costs (OPEX) over the next three years.

Which combination of AWS features should you choose?

A. EC2 Spot Instances using the T-family (General Purpose burstable).

B. EC2 Reserved Instances using the Inf2-family (Compute Optimized)

C. AWS Lambda triggered by API Gateway.

D. EC2 Reserved Instances using the G-family (Accelerated Computing)

Answer: B

271. A company stores backup files that are accessed infrequently (about once every 3–6 months) but must be available in milliseconds when needed for compliance audits. The data is mission-critical and should remain highly durable across multiple Availability Zones. The team wants to minimize storage costs without sacrificing immediate access or high availability. Which Amazon S3 storage class is the most appropriate choice for these backups?

A. S3 Standard

B. S3 Intelligent-Tiering

C. S3 Standard-Infrequent Access (S3 Standard-IA)

D. S3 Glacier Instant Retrieval

Answer: C

272. A company with no remote offices wants to host an internal website that is used only during business hours. Which EC2 purchasing option is most suitable?

A. On-Demand

B. Reserved Instances

C. Spot Instances

D. Savings Plans

Answer: A

273. Which EC2 Placement Group type is best for applications that require very low latency between instances?

A. Spread

B. Cluster

C. Partition

D. Regional

Answer: B

274. A batch processing application is highly fault-tolerant and can be paused and restarted later without affecting the results. Which Amazon EC2 instance purchasing option provides the most cost savings (up to 90

A. A. Spot Instances

B. B. On-Demand Instances

C. C. Dedicated Hosts

D. D. Reserved Instances

Answer: A

275. What is the default behavior of an EBS root volume when an EC2 instance is terminated?

A. It is moved to another instance

B. It is deleted automatically

C. It is archived to Amazon Glacier

D. It remains and must be deleted manually

Answer: B

276. A web system is deployed using an Application Load Balancer and Auto Scaling Group across multiple Availability Zones. What is the primary purpose of this architecture?

A. Reduce operational cost

B. Increase availability and prevent downtime

C. Improve EC2 CPU performance

D. Reduce storage usage

Answer: B

277. A financial services company has a workload that runs continuously 24/7 for the next 3 years with a fixed instance type and region. The company wants to minimize cost while ensuring the instance is never interrupted. Which purchasing option provides the GREATEST discount under these conditions?

A. EC2 Spot Instance

B. EC2 Reserved Instance with All Upfront payment on a 3-year term

C. EC2 On-Demand Instance

D. EC2 Reserved Instance with No Upfront payment on a 1-year term

Answer: B

278. In the lab, why does the Web Server Security Group only allow traffic from the ALB, instead of allowing public access (0.0.0.0/0)?

A. To improve system performance

B. To reduce costs

C. To enhance security by preventing direct internet access

D. Because ALB does not support public traffic

Answer: C

279. You want to automatically install a Web Server as soon as the EC2 instance is finished launching for the first time. Which feature should you use?

A. EC2 Metadata

B. Amazon Machine Image (AMI)

C. EC2 User Data

D. Elastic Network Interface (ENI)

Answer: C

280. If EC2 CPU utilization exceeds 50

A. EC2 instances will be terminated

B. Auto Scaling Group will launch additional instances

C. The Load Balancer will stop working

D. Security Groups will automatically change

Answer: B

281. What is the main difference between a Public IP and an Elastic IP on AWS?

A. An Elastic IP can be remapped from one instance to another, whereas a Public IP cannot.

B. A Public IP can only be used within a VPC, while an Elastic IP is used on the Internet.

C. A Public IP is fixed, while an Elastic IP changes every time the instance reboots.

D. There is no functional difference.

Answer: A

282. What is the greatest benefit of using 'Hibernate' compared to a normal instance 'Stop'?

A. Data in the Instance Store will be preserved.

B. Saves on EBS storage costs.

C. No need to encrypt the Root volume.

D. Applications can resume from their previous running state (in-memory state).

Answer: D

283. A financial services company needs to retain its data for 10 years to meet compliance norms. Which Amazon Simple Storage Service (Amazon S3) storage class is the best fit for this use case considering that the data has to be stored at a minimal cost?

A. Amazon S3 Standard-Infrequent Access (S3 Standard-IA)

B. Amazon S3 Glacier Flexible Retrieval

C. Amazon S3 Intelligent-Tiering

D. Amazon S3 Glacier Deep Archive

Answer: D

284. An e-commerce company has its on-premises data storage on an NFS file system that is accessed in parallel by multiple applications. The company is looking at moving the applications and data stores to AWS Cloud. Which storage service should the company use to move their files to AWS Cloud seamlessly if the application is hosted on Amazon Elastic Compute Cloud (Amazon EC2) instances?

A. Amazon Simple Storage Service (Amazon S3)

B. Amazon Elastic Block Store (Amazon EBS)

C. Amazon Elastic File System (Amazon EFS)

D. AWS Storage Gateway

Answer: C

285. When using EBS Multi-Attach (io1/io2 family only), how many EC2 instance(s) can it support?

A. 4

B. 8

C. 16

D. All of the above

Answer: D

286. Which of the following Amazon Simple Storage Service (Amazon S3) storage classes has NO constraint of a minimum storage duration charge for objects?

A. Amazon S3 Standard-Infrequent Access (S3 Standard-IA)

B. Amazon S3 Glacier Flexible Retrieval

C. Amazon S3 Standard

D. Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)

Answer: C

287. A compliance team requires that a copy of all S3 objects in a critical bucket be maintained in another AWS Region to protect against regional disasters. The copy must occur automatically when new objects are created. Which feature should you configure?

A. S3 Lifecycle transition to Glacier Deep Archive in the same Region

B. Amazon EFS asynchronous replication between Regions

C. S3 Cross-Region Replication (CRR) on the bucket

D. EBS snapshot copy from the primary Region to the secondary Region

Answer: C

288. You manage an S3 bucket storing log files that grow quickly. Older logs (over 90 days) are rarely accessed, but must remain immediately retrievable when needed. How should you configure storage classes using lifecycle policies?

A. Store all logs in S3 Glacier Deep Archive to minimize cost

B. Immediately transition logs to S3 One Zone-IA upon upload

C. Delete logs after 90 days using lifecycle expiration

D. Transition objects to S3 Standard-IA after 90 days and keep them there

Answer: D

289. A company wants to build a system that allows users to upload files, processes the files immediately after upload, and does not require managing servers. Which architecture is the MOST appropriate?

    A. Amazon EC2 with Amazon EBS

    B. Amazon S3 with AWS Lambda trigger

    C. Amazon EFS with Amazon EC2

    D. Amazon RDS with AWS Lambda

    Answer: B

290. An application experiences spiky traffic during peak hours, has almost no traffic at night, and needs to optimize cost. Which compute solution is MOST appropriate?

    A. Amazon EC2 On-Demand

    B. Amazon EC2 Reserved Instances

    C. AWS Lambda

    D. Dedicated Hosts

    Answer: C

291. An application needs storage that can be attached directly to an EC2 instance like a disk. Which service should be used?

    A. Amazon S3

    B. Amazon EFS

    C. Amazon EBS

    D. Amazon S3 Glacier

    Answer: C

292. A global web application needs to store static content and reduce latency for users worldwide. Which solution is MOST appropriate?

    A. Amazon EC2 with Amazon EBS

    B. Amazon S3 with Amazon CloudFront

    C. Amazon EFS with Application Load Balancer

    D. Amazon RDS with Amazon Route 53

    Answer: B

293. A system needs to store data long-term at low cost with infrequent access. Which solution is MOST appropriate?

    A. Amazon S3 Standard

    B. Amazon S3 Glacier

    C. Amazon EBS

    D. Amazon EFS

Answer: B

294. What is Amazon EC2?
    A. A database service for storing structured data
    B. A scalable virtual server service in the cloud (IaaS)
    C. A tool for designing websites only
    D. A physical data center owned by users

    Answer: B

295. What is an AMI (Amazon Machine Image)?
    A. A backup file stored in S3
    B. A preconfigured template used to launch EC2 instances
    C. A networking protocol
    D. A monitoring tool only

    Answer: B

296. What does EC2 User Data allow you to do?
    A. Store files permanently
    B. Automatically run scripts when the instance starts
    C. Change instance type after launch
    D. Delete EC2 instances automatically

    Answer: B

297. What is the main purpose of Security Groups in EC2?
    A. Store application logs
    B. Control network traffic to and from EC2 instances
    C. Increase CPU performance
    D. Backup EC2 instances

    Answer: B

298. What is an Elastic IP (EIP)?
    A. A dynamic IP that changes frequently
    B. A static public IPv4 address provided by AWS
    C. A private IP inside VPC
    D. A DNS service

    Answer: B

299. Which AWS service allows you to run code in response to events without provisioning or managing any servers, and charges you only for the compute time you consume?

A. AWS Elastic Beanstalk

B. Amazon ECS

C. Amazon EC2

D. AWS Lambda

Answer: D

300. Which AWS service provides resizable compute capacity in the cloud, allowing you to launch virtual servers?

A. Amazon VPC

B. Amazon S3

C. Amazon RDS

D. Amazon EC2

Answer: D

301. A company has a steady-state, predictable workload that will run continuously for the next 3 years. Which Amazon EC2 pricing model provides the most cost-effective solution?

A. Reserved Instances

B. Spot Instances

C. Dedicated Hosts

D. On-Demand Instances

Answer: A

302. According to the architectural concepts of storage systems, which type of storage stores data in a flat address space, utilizes rich metadata, and is accessed primarily via REST APIs

A. File Storage

B. Object Storage

C. Block Storage

D. Single-Disk Storage

Answer: B

303. You are designing a system for big data processing and data warehousing that requires high sequential throughput. You need a low-cost EBS volume type, and it will NOT be used as a boot volume. Which EBS volume type is the most appropriate

A. Cold HDD(s1)

B. General purpose SSD(gp3)

C. Object Storage

D. Throughput Optimized SSD(st1)

Answer: D

304. Which kind of storage is the most relevant for normal latency and high durability
   A. Distributed Storage System
   B. Spot Instances
   C. Storage Cluster
   D. Single-Disk Storage

   Answer: C

305. Which of the following best describes an Availability Zone (AZ) in the AWS Global Infrastructure?
   A. A single geographic area containing multiple AWS Regions.
   B. One or more discrete data centers with redundant power, cooling, and networking.
   C. A global network of edge locations used for content delivery.
   D. A virtual server environment used to host web applications.

   Answer: B

306. A company wants to reduce latency for users by caching content (such as images or videos) closer to their geographic location. Which AWS component should they use?
   A. AWS Regions.
   B. Availability Zones.
   C. Edge Locations.
   D. Backup Regions.

   Answer: C

307. According to the Cloud Service Models, which model provides the most control to Network Architects and IT Administrators over the underlying infrastructure?
   A. Software as a Service (SaaS).
   B. Platform as a Service (PaaS).
   C. Infrastructure as a Service (IaaS).
   D. Function as a Service (FaaS).

   Answer: C

308. Which service is a serverless, pay-as-you-go compute service that scales automatically based on incoming requests?
   A. Amazon EC2
   B. AWS Lambda
   C. Amazon Lightsail
   D. AWS Outposts

   Answer: B

309. Which S3 feature allows you to automatically transition objects to a cheaper storage class after a certain period?
    A. S3 Versioning
    B. S3 Replication
    C. S3 Lifecycle Policy
    D. S3 Transfer Acceleration

    Answer: C

310. Which of the following is a managed, elastic file system that supports the NFS protocol and allows multiple EC2 instances to access data simultaneously?
    A. Amazon EBS.
    B. Amazon S3.
    C. Amazon EFS.
    D. Amazon S3 Glacier.

    Answer: C

311. Which AWS service provides a managed registry for storing, managing, and deploying Docker container images?
    A. Amazon ECS
    B. Amazon EKS
    C. Amazon ECR
    D. AWS App Runner

    Answer: C

312. Which storage type is characterized by a "flat address space"where data is stored as objects with unique IDs and rich metadata?
    A. Block Storage.
    B. File Storage.
    C. Object Storage.
    D. Local Instance Store.

    Answer: C

313. What can be used to create a point-in-time backup of an Amazon EBS volume and can be copied across different AWS Regions?
    A. EBS Multi-attach.
    B. EBS Snapshots.
    C. S3 Lifecycle Rules.
    D. EFS Mount Targets.

Answer: B

314. Which storage option is best for low-latency database workloads?
    A. S3
    B. Glacier
    C. EBS SSD
    D. EFS Infrequent Access

    Answer: C

315. For long-term backup with infrequent access, which service should you use?
    A. S3 Standard
    B. EFS
    C. Glacier
    D. EBS

    Answer: C

316. According to the AWS Shared Responsibility Model, which of the following is AWS responsible for?
    A. Managing guest OS on EC2
    B. Securing physical data centers
    C. Configuring Security Groups
    D. Encrypting application data

    Answer: B

317. How many EC2 instances can an EBS volume be attached to at the same time?
    A. Multiple instances
    B. One instance (except Multi-Attach)
    C. nan
    D. Unlimited

    Answer: B

318. In AWS infrastructure terminology, the ability of a system to continue functioning correctly even when some of its components fail is called:
    A. Scalability.
    B. Consistency.
    C. Fault Tolerance.
    D. Decoupling.

    Answer: C

319. Which AWS S3 feature allows a user to define rules to move objects to cheaper storage classes or delete them after a specific number of days?

A. Bucket Versioning.

B. Object Tags.

C. Lifecycle Policies.

D. Cross-Region Replication.

Answer: C

320. Which AWS service allows you to deploy applications without managing infrastructure but still gives access to underlying EC2 instances?

A. AWS Lambda

B. AWS Elastic Beanstalk

C. Amazon S3

D. Amazon DynamoDB

Answer: B

321. A company needs to store archive data that is accessed less than once a year but must be preserved for several years at the lowest possible cost. Which S3 storage class is best?

A. S3 Standard.

B. S3 Intelligent-Tiering.

C. S3 Glacier Instant Retrieval.

D. S3 Glacier Deep Archive.

Answer: D

322. Which EC2 pricing model is best for fault-tolerant workloads with the lowest cost?

A. On-Demand.

B. Reserved.

C. Spot Instances

D. Dedicated Hosts

Answer: C

323. What is a key limitation of Amazon EBS compared to Amazon EFS regarding instance attachment?

A. EBS is object-based

B. EFS only attaches to one instance

C. EBS usually attaches to one instance at a time

D. EFS is much slower

Answer: C

324. What is the main benefit of using AWS Lambda?
   A. Full control of OS.
   B. No need to manage servers.
   C. Unlimited compute power.
   D. Fixed pricing.
   Answer: B

325. What happens to the data in an Instance Store if the associated EC2 instance is terminated?
   A. Data is saved to S3
   B. Data is persistent
   C. Data is lost
   D. Data is moved to EBS
   Answer: C

326. Which AWS service is used for storing objects with high durability?
   A. Amazon EC2
   B. Amazon S3
   C. Amazon RDS
   D. AWS Lambda
   Answer: B

327. Which of the following services can automatically scale EC2 instances based on demand?
   A. AWS Lambda
   B. AWS Auto Scaling
   C. Amazon CloudWatch
   D. AWS IAM
   Answer: B

328. Where are Amazon EBS snapshots stored to ensure high durability and availability?
   A. Inside the EC2 instance
   B. On an Instance Store
   C. In Amazon S3
   D. On a Tape Gateway
   Answer: C

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AI VIET NAM – AI COURSE 2025

Cloud Practitioner Quiz - Week 1

Lớp AWS (AIO 2025)

1. Which of the following is an advantage that users experience when they move on-premises workloads to the AWS Cloud?
   A. Elimination of expenses for running and maintaining data centers.
   B. Price discounts that are identical to discounts from hardware providers
   C. Distribution of all operational controls to AWS.
   D. Elimination of operational expenses.

   Answer: A

2. Which AWS benefit is demonstrated by on-demand technology services that enable companies to replace upfront fixed expenses with variable expenses?
   A. High availability
   B. Economies of scale
   C. Pay-as-you-go pricing
   D. Global reach

   Answer: C

3. A company wants to know more about the benefits offered by cloud computing. The company wants to understand the operational advantage of agility. How does AWS provide agility for users?
   A. The ability to ensure high availability by deploying workloads to multiple regions.
   B. A pay-as-you-go model for many services and resources.
   C. The ability to transfer infrastructure management to the AWS Cloud.
   D. The ability to provision and deprovision resources quickly with minimal effort.

   Answer: D

4. Which AWS Cloud benefit is shown by an architecture's ability to withstand failures with minimal downtime?

A. Agility

B. Elasticity

C. Scalability

D. High availability

Answer: D

5. Which AWS Cloud deployment model uses AWS Outposts as part of the application deployment infrastructure?

A. On-premises

B. Serverless

C. Cloud-native

D. Hybrid

Answer: D

6. Which of the following is an advantage that the AWS Cloud provides to users?

A. Users eliminate the need to guess about infrastructure capacity requirements.

B. Users decrease their variable costs by maintaining sole ownership of IT hardware.

C. Users maintain control of underlying IT infrastructure hardware.

D. Users maintain control of operating systems for managed services.

Answer: A

7. A company wants the ability to automatically acquire resources as needed and release the resources when they are no longer needed. Which cloud concept describes this functionality?

A. Availability

B. Elasticity

C. Durability

D. Reliability

Answer: B

8. Which benefit of cloud computing gives a company the ability to deploy applications to users all over the world through a network of AWS Regions, Availability Zones, and edge locations?

A. Economy of scale

B. Global reach

C. Agility

D. High availability

Answer: B

9. A user is moving a workload from a local data center to an architecture that is distributed between the local data center and the AWS Cloud. Which type of migration is this?

A. On-premises to cloud native

B. Hybrid to cloud native

C. On-premises to hybrid

D. Cloud native to hybrid

Answer: C

10. A company needs to deploy applications in the AWS Cloud as quickly as possible. The company also needs to minimize the complexity that is related to the management of AWS resources. Which AWS service should the company use to meet these requirements?

    A. AWS Config

    B. AWS Elastic Beanstalk

    C. Amazon EC2

    D. Amazon Personalize

    Answer: B

11. A global media company stores high-resolution video files in an Amazon S3 bucket located in the us-east-1 (N. Virginia) Region. While users in the United States experience fast download speeds, users in Singapore and Tokyo report significant delays and "slow content delivery"when accessing the same files. Which architectural change would provide the best user experience and performance for the global audience while minimizing the load on the origin S3 bucket?

    A. Create a secondary S3 bucket in the ap-southeast-1 (Singapore) Region and manually copy all video files to it.

    B. Deploy Amazon CloudFront and configure the N. Virginia S3 bucket as the origin.

    C. Upgrade the S3 bucket to a Multi-AZ configuration to increase data replication speed.

    D. Instruct global users to use a VPN to connect directly to the N. Virginia network path.

    Answer: B

12. When a company moves from an on-premises data center to the AWS Cloud, which financial shift typically occurs in their accounting model?

    A. Moving from variable operational expense (OPEX) to upfront capital expense (CAPEX).

    B. Moving from upfront capital expense (CAPEX) to variable operational expense (OPEX)

    C. Elimination of all capital and operational expenditures.

    D. Moving from fixed variable costs to fixed capital investments.

    Answer: B

13. A startup needs to get a new product to market quickly and requires the ability to provision compute and storage resources in just a few minutes. Which cloud characteristic best meets this need?

    A. Reliability.

    B. High Availability.

C. Agility.

D. Elasticity.

Answer: C

14. Adeveloper wants to deploy a web application without worrying about the underlying infrastructure, such as the operating system or server hardware. Which cloud computing model should they use?

    A. Infrastructure as a Service (IaaS).

    B. Software as a Service (SaaS).

    C. Networking as a Service (NaaS).

    D. Platform as a Service (PaaS).

    Answer: D

15. Which AWS Cloud benefit allows an organization to automatically match the supply of resources with changing workload demands?

    A. Elasticity.

    B. Security.

    C. Global Reach.

    D. Fault Tolerance.

    Answer: A

16. How do customers directly benefit from Amazon's "Massive Economies of Scale"?

    A. They receive free hardware upgrades every three years.

    B. They receive periodic price reductions as a result of Amazon's operational efficiencies.

    C. They gain the ability to scale up and down as needed.

    D. They can use any third-party software license for free.

    Answer: B

17. Underthe Shared Responsibility Model, which task remains the customer's responsibility when using Infrastructure as a Service (IaaS) like Amazon EC2, but becomes the responsibility of AWS when using a managed Platform as a Service (PaaS) or serverless service?

    A. Data encryption.

    B. Physical security of data centers.

    C. Configuration of security groups.

    D. Patching the guest operating system.

    Answer: D

18. A company hosts a legacy application in its corporate data center but uses Amazon EC2 instances for its modern web servers. Which term correctly describes this infrastructure model?

    A. Cloud-native architecture.

B. Partner network model.

C. Hybrid architecture.

D. Monolithic architecture.

Answer: C

19. To prevent a "Single Point of Failure"in a monolithic application, a Cloud Practitioner is advised to re-architect the application into components that run independently. This is an example of which design principle?

    A. Vertical scaling.

    B. Tightly coupled design.

    C. Manual monitoring.

    D. Loose coupling.

    Answer: D

20. Which specific advantage of cloud computing addresses the traditional IT challenge of over provisioning or under-provisioning infrastructure based on "peak load"estimates?

    A. Stop guessing about capacity.

    B. High availability.

    C. Increased speed and agility.

    D. Physical isolation of workloads.

    Answer: A

21. Which statement best describes a Multi-AZ architecture?

    A. Deploying resources across multiple Availability Zones within a single cloud region

    B. Deploying application services across multiple cloud regions around the world

    C. Deploying multiple servers inside one data center to increase computing capacity

    D. Deploying resources at edge locations to reduce latency for global users

    Answer: A

22. How are Availability Zones (AZs) connected to each other within a Region?

    A. Via public internet connections.

    B. Through physical transport of storage media.

    C. Using high-bandwidth, low-latency private networking.

    D. They are not connected to ensure isolation.

    Answer: C

23. A web application runs on multiple servers inside a cloud environment. The company wants to distribute incoming HTTP and HTTPS requests across several backend servers to improve reliability and scalability. This is supported by Application Load Balancer. What is the main function of ALB?

A. Providing virtual machine instances for running backend application code.

B. Resolving domain names into IP addresses for client network connections.

C. Storing and caching static website files for faster delivery to global users.

D. Distributing incoming web requests across multiple application servers.

Answer: D

24. A fintech platform has three requirements: (1) It must tolerate the failure of a single data center facility with synchronous database protection. (2) It must remain recoverable if an entire geographic Region becomes unavailable. (3) It must deliver static web content with low latency to users worldwide. Which architecture best aligns with AWS Global Infrastructure best practices?

    A. Deploy the application and database across multiple Availability Zones in one Region, and use CloudFront for static content.

    B. Deploy the application in one Region, place backups in Edge Locations, and use Local Zones for disaster recovery.

    C. Deploy the application across multiple Availability Zones in a primary Region, replicate to a secondary AWS Region for disaster recovery, and use CloudFront for global static content delivery.

    D. Deploy the database in two Local Zones and use Direct Connect for synchronous cross-city replication.

    Answer: C

25. Which statement best defines cloud computing?

    A. Running applications only on personal computers without connecting to the internet.

    B. Installing physical servers locally and maintaining hardware inside an organization building.

    C. Storing data permanently on removable storage devices such as USB drives.

    D. Delivering computing resources through the internet with on-demand access and pay-as-you-go pricing.

    Answer: D

26. Which AWS service provides resizable compute capacity in the cloud and gives you full administrative control over the operating system?

    A. AWS Lambda

    B. Amazon EC2

    C. AWS Fargate

    D. Amazon S3

    Answer: B

27. What is the primary characteristic of an Availability Zone (AZ) regarding fault isolation?

    A. It is a geographic area isolated by international boundaries for data sovereignty.

B. It consists of multiple Regions connected by low-latency networks.

C. It has independent power, cooling, and networking to ensure full fault isolation.

D. It provides edge caching to reduce latency for global users.

Answer: C

28. A global e-commerce website receives visitors from many countries. The company wants to direct users to the correct application endpoint using their domain name and also speed up delivery of images, videos, and static content for users around the world. Which combination of services should the company use to achieve these goals?

    A. Use Amazon Route 53 for domain name resolution and Amazon CloudFront for global content delivery.

    B. Use Amazon CloudFront for domain name resolution and Amazon Route 53 for content caching.

    C. Use Amazon Route 53 for caching static files and Amazon CloudFront for managing domain records

    D. Use Amazon CloudFront for routing DNS traffic and Amazon Route 53 for video streaming services

    Answer: A

29. A company runs a reduced-capacity but fully functional copy of its application stack in a secondary Region. During a disaster, it plans to scale out that environment to handle production traffic. Which strategy is this?

    A. Backup and Restore

    B. Pilot Light

    C. Warm Standby

    D. Active/Active

    Answer: C

30. Which cloud computing model provides virtualized computing resources such as virtual machines, storage, and networking over the internet?

    A. Software as a Service (SaaS)

    B. Platform as a Service (PaaS)

    C. Infrastructure as a Service (IaaS)

    D. Function as a Service (FaaS)

    Answer: C

31. A small e-commerce company hosts its website on one physical server located in its office. One day, the server suddenly stops working because of a hardware problem. As a result, the entire website becomes unavailable and customers cannot access the online store. The IT team realizes that the system design caused the entire service to stop when a single component failed. Which concept best describes the problem in this situation?

A. Single Point of Failure where one component failure stops the entire system

B. Resource Pooling where multiple users share the same infrastructure resources

C. Rapid Elasticity where computing resources scale automatically with demand

D. Broad Network Access where services are available through internet networks

Answer: A

32. A company wants to deploy its application in multiple geographic regions so users around the world can access it with lower latency. Which AWS Cloud benefit supports this requirement?

A. High availability

B. Global reach

C. Elasticity

D. Pay-as-you-go pricing

Answer: B

33. A startup has continuously changing workloads, needs rapid deployment, does not want to invest in a data center upfront, and wants to pay only for the resources it actually uses. According to the slide content, which of the following best explains why cloud is more suitable than on-premises?

A. Cloud provides full control over physical infrastructure and completely eliminates internet dependency

B. Cloud changes upfront fixed capital costs into variable costs based on usage and supports flexible resource scaling

C. Cloud requires the company to manage all networking, storage, and virtualization by itself

D. Cloud is only suitable for large enterprises with stable workloads over many years

Answer: B

34. A small startup wants to build a web application for online food delivery. The team wants to focus mainly on developing application features rather than managing servers, operating systems, or runtime environments. They still want to write and deploy their own code, but they prefer the cloud provider to handle most of the infrastructure and platform maintenance. Which service model is the most appropriate solution for this situation?

A. On-Premises infrastructure where the company manages servers, software, and networks

B. Infrastructure as a Service (IaaS) providing virtual servers while users manage operating systems

C. Platform as a Service (PaaS) providing runtime environments while users deploy applications

D. Software as a Service (SaaS) providing complete applications managed entirely by providers

Answer: C

35. A startup is launching a video streaming app and wants to ensure that users in Tokyo and New York experience the same fast loading times for thumbnail images. Which service should they implement?

A. Amazon EC2 Auto Scaling

B. Amazon CloudFront using Edge Locations

C. Multi-AZ deployment in the Singapore Region

D. AWS Direct Connect

Answer: B

36. Which component of the AWS Global Infrastructure consists of one or more discrete data centers, each with redundant power, networking, and connectivity, housed in separate facilities?

A. AWS Regions.

B. Edge Locations.

C. Availability Zones.

D. Private Networking.

Answer: C

37. To achieve high availability for a web application, compute resources should be provisioned across a minimum of how many Availability Zones?

A. One.

B. Two.

C. Three.

D. Four.

Answer: B

38. Which statement accurately describes the relationship between Regions and Availability Zones?

A. Regions contain multiple Availability Zones.

B. Data centers contain multiple Regions.

C. Availability Zones contain multiple Edge Locations.

D. Edge Locations are housed within a single Availability Zone.

Answer: A

39. Which AWS service is specifically designed as a global service to provide DNS management across the entire global infrastructure?

A. Amazon EC2.

B. Amazon S3.

C. Amazon Route 53.

D. Amazon DynamoDB.

Answer: C

40. A company currently operates in one AWS Region and wants to start supporting a second Region. What is the correct procedure to begin using this new Region?

A. Contact an AWS Account Manager to sign a separate regional contract.

B. Move an existing Availability Zone from the current Region to the new one.

C. Download a Region-specific version of the Management Console.

D. Begin deploying resources in the second Region directly via the Management Console.

Answer: D

41. A company requires synchronous replication of its database to ensure zero data loss if a single data center facility fails. Which feature of the Global Infrastructure makes this possible?

A. High-speed fiber connections between different AWS Regions.

B. Low-latency, redundant networking links between Availability Zones within a Region.

C. Automated caching of database writes at global Edge Locations.

D. The use of AWS Direct Connect to link multiple data centers.

Answer: B

42. An organization needs to protect its application against a service disruption that could affect an entire geographic area, such as a major natural disaster. What is the recommended architectural action?

A. Deploy the application across all Availability Zones within one Region.

B. Use a hybrid cloud model with on-premises resources in the same city.

C. Deploy the application across multiple AWS Regions.

D. Replicate the data across multiple Edge Locations globally.

Answer: C

43. Which factor is most critical when a customer must choose an AWS Region to ensure compliance with data sovereignty and legal requirements?

A. Proximity to the customer's corporate office for on-site audits.

B. The specific geographic location where data will be stored and processed.

C. The number of Edge Locations available in that specific country.

D. Whether the application is presented in the local language.

Answer: B

44. Which component of the AWS Global Infrastructure is used by Amazon CloudFront to ensure low-latency delivery of content to end-users worldwide?

A. Regional Data Centers.

B. Virtual Private Gateways.

C. Availability Zones.

D. Edge Locations.

Answer: D

45. In the AWS Shared Responsibility Model, which of the following is the sole responsibility of AWS regarding Global Infrastructure?
    A. Configuring Network ACLs to block malicious IP addresses.
    B. Managing security groups for EC2 instances.
    C. Management and physical security of Edge Locations.
    D. Patching the guest operating system of resources in an AZ.

    Answer: C

46. Which of the following can be components of a VPC in the AWS Cloud?
    A. Amazon API Gateway
    B. Amazon S3 buckets and objects
    C. AWS Storage Gateway
    D. Internet gateway

    Answer: D

47. A company is migrating its application from an on-premises infrastructure to the cloud to improve high availability and reduce infrastructure management overhead. The company decides to run its application on Amazon EC2.

    According to the Shared Responsibility Model of Amazon Web Services, which of the following responsibilities belongs to the customer?
    A. Maintaining the physical servers in AWS data centers
    B. Managing and patching the operating system running on the EC2 instance
    C. Ensuring the availability of the underlying AWS networking infrastructure
    D. Maintaining the cooling systems and power supply in AWS data centers

    Answer: B

48. Which of the 'Six Advantages of Cloud Computing' specifically addresses the ability of a company to avoid over-provisioning or under-provisioning based on workload estimates?
    A. Trade fixed expense for variable expense
    B. Benefit from massive economies of scale
    C. Stop guessing capacity
    D. Go global in minutes

    Answer: C

49. Over Night is a startup that provides event management and online ticket distribution services. The company wants to avoid large upfront investments in hardware, quickly launch its platform, and automatically scale resources to handle sudden increases in demand. Which solution would BEST meet these requirements?
    A. Build an on-premises data center and purchase enough servers to handle the highest expected traffic

B. Use a co-location data center and manage all physical servers and networking infrastructure internally.

C. Deploy the application in the AWS Cloud using scalable services that automatically adjust capacity based on demand.

D. Purchase several physical servers and host the application in the company's office.

Answer: C

50. A healthcare company is working with sensitive patient records which is currently stored in an on-premise way. The company intends to extend globally and expecting high demand due to the computational resource of upcoming AI usage for medical image analysis. Which architecture should the company choose?

A. Build more on-premise data center over the globe for expansion and security.

B. Move everything to cloud for easy expand and scale as well as lower latency due to powerful pre-installed infrastructure.

C. Keep everything on-premise for full control and security.

D. Build a hybrid model where keeping patient record on-premise while computational resource utilizing cloud infrastructure.

Answer: D

51. DeepScan runs an AI X-ray analysis app on Amazon EC2. Despite using Security Groups, EBS encryption, and a hardened Ubuntu OS, a data leak occurs due to an outdated OpenSSL library running on the EC2 instance.

A. AWS: They must ensure all software/libraries on their infrastructure are patched.

B. AWS: They are responsible for all encryption-related vulnerabilities (like OpenSSL).

C. Customer (DeepScan): They are responsible for managing the guest OS, applications, and installed libraries.

D. Shared: AWS is responsible for the software flaw, and the customer for failing to detect it.

Answer: C

52. Which of the following is a core design principle of the "Security"pillar of the Well-Architected Framework?

A. Stop guessing capacity

B. Implement a strong identity foundation

C. Scale horizontally to increase workload availability

D. Perform operations as code

Answer: B

53. What type of cloud computing service is AWS Lambda (Serverless)?

A. Infrastructure as a Service (IaaS)

B. Platform as a Service (PaaS)

C. Software as a Service (SaaS)

D. Function as a Service (FaaS)

Answer: D

54. Which AWS Cloud characteristic allows customers to scale resources up or down automatically based on demand?

A. Resource pooling

B. Rapid elasticity

C. Broad network access

D. Measured service

Answer: B

55. Which AWS service is considered a global service?

A. Amazon EC2

B. Amazon RDS

C. AWS IAM

D. Amazon S3

Answer: C

56. What is the primary purpose of AWS Edge Locations?

A. Run virtual machines

B. Store relational databases

C. Cache content closer to users to reduce latency

D. Host application servers

Answer: C

57. Which cloud service model allows customers to manage operating systems and applications, while AWS manages the underlying infrastructure?

A. SaaS

B. PaaS

C. IaaS

D. On-premises

Answer: C

58. Which AWS infrastructure component provides geographic isolation

A. Availability Zone

B. Region

C. Edge location

D. Data center

Answer: B

59. What best describes the concept of scalability?

A. The ability for a system to withstand a certain amount of failure and still remain funcitonal

B. The ability for a system to grow in size, capacity, and/or scope

C. The ability for a system to grow and shrink based on demand

D. The ability for a system be accessible when you attempt to acesss it

Answer: B

60. What is the goal of High Availability architecture?

A. Minimize downtime

B. Reduce cost

C. Increase storage capacity

D. Simplify deployment

Answer: A

61. A company wants to deploy an application for users in Vietnam and Southeast Asia. Which AWS Region would most likely provide the lowest latency?

A. us-east-1 (N. Virginia)

B. ap-south-1 (Mumbai)

C. eu-west-1 (Ireland)

D. ap-southeast-1 (Singapore)

Answer: D

62. A company wants to improve application availability within a single region. Which architecture should they implement?

A. Deploy instances in multiple AZs

B. Deploy instances in a single AZ

C. Deploy in a single EC2 instance

D. Deploy using only edge locations

Answer: A

63. An application stores frequently accessed static images globally. Which AWS service can cache these images near users?

A. RDS

B. CloudFront

C. EC2

D. EBS

Answer: B

64. A company wants to automatically redirect traffic if one Availability Zone fails. Which AWS component should they use?

A. Application Load Balancer

B. Amazon S3

C. AWS IAM

D. Amazon Glacier

Answer: A

65. Which AWS deployment strategy protects against data center failures within a region?

A. Multi-Region architecture

B. Multi-AZ architecture

C. Edge computing

D. Hybrid cloud

Answer: B

66. A company wants to build an architecture that continues running even if an entire AWS Region becomes unavailable. What should they implement?

A. Multi-AZ deployment

B. Auto Scaling

C. Multi-Region architecture

D. Edge caching

Answer: C

67. An e-commerce website deploys EC2 instances in two Availability Zones behind an Application Load Balancer. What AWS architectural principle does this represent?

A. Edge caching

B. Hybrid deployment

C. Resource pooling

D. Fault isolation and high availability

Answer: D

68. Which architecture provides the highest resilience against large-scale disasters but increases complexity and cost?

A. Single AZ deployment

B. Multi-AZ deployment

C. Multi-Region deployment

D. Edge caching

Answer: C

69. Which term describes the cost advantage that AWS has over a company running its own private cloud?

A. Total Cost of Ownership (TCO)

B. Economies of Scale

C. Operating Expenses

D. Capital Expenses

Answer: B

70. A startup is launching a new app. They expect a massive spike in traffic during a 2-hour marketing event, followed by very low traffic. Which cloud characteristic allows them to handle the spike without paying for idle servers afterward?

A. Reliability

B. Durability

C. Elasticity

D. Sustainability

Answer: C

71. A global e-learning platform stores videos in S3 in the Tokyo Region. Students from Europe complain about slow loading times. Which solution BEST improves performance for these users while keeping data in the same Region?

A. Deploy in a single AZ with Auto Scaling only

B. Deploy across multiple AZs behind an Application Load Balancer

C. Deploy in two different Regions without load balancing

D. Use CloudFront Edge Locations to host the system directly

Answer: B

72. A global e-learning platform stores videos in S3 in the Tokyo Region. Students from Europe complain about slow loading times. Which solution BEST improves performance for these users while keeping data in the same Region?

A. Move S3 buckets to a European Region only

B. Enable Multi-AZ for the S3 bucket

C. Use CloudFront with Edge Locations to cache video content closer to users

D. Create a second S3 bucket in Europe and manually sync data daily

Answer: C

73. A startup wants to develop a new web application. They do not have a system administration team to maintain the operating system (OS) or perform software patching, but they must retain full control over their application code and data. Based on the shared responsibility model, which cloud computing model best fits this requirement?
    A. Software as a Service (SaaS)
    B. On-premise
    C. Infrastructure as a Service (IaaS)
    D. Platform as a Service (PaaS)

    Answer: D

74. Which of the following is NOT one of the five main characteristics of cloud computing?
    A. On-demand self-service
    B. Rapid elasticity
    C. Manual provisioning
    D. Measured service

    Answer: C

75. Which AWS service is a typical example of Infrastructure as a Service (IaaS)?
    A. Amazon WorkMail
    B. AWS Elastic Beanstalk
    C. Amazon WorkDocs
    D. Amazon EC2

    Answer: D

76. Which of the following services is an example of Software as a Service (SaaS) on AWS?
    A. Amazon S3
    B. AWS Elastic Beanstalk
    C. Amazon EC2
    D. Amazon WorkDocs

    Answer: D

77. Which AWS Cloud benefit is characterized by the ability to swap upfront fixed capital expenses (CAPEX) for variable operational expenses (OPEX)?
    A. Massive economies of scale
    B. Trade fixed expense for variable expense
    C. High availability
    D. Stop guessing capacity

    Answer: B

78. Which of the following is an example of an AWS "Global Service"that is not tied to a specific geographic Region?
    A. Amazon EC2
    B. Amazon RDS
    C. AWS IAM (Identity and Access Management)
    D. Amazon VPC

    Answer: C

79. Which cloud concept describes the ability to automatically acquire resources as needed and release them when they are no longer required to match demand?
    A. Reliability
    B. High Availability
    C. Elasticity
    D. Durability

    Answer: C

80. Which of the following components is the customer responsible for managing in the Platform as a Service (PaaS) model?
    A. Operating Systems (OS)
    B. Applications and Data
    C. Networking and Storage
    D. Virtualization

    Answer: B

81. A company wants full control over its infrastructure and hosts its servers inside its own building. Which deployment model is being used?
    A. Public Cloud
    B. Hybrid Cloud
    C. Edge Computing
    D. Private Cloud (On-premises)

    Answer: D

82. What is an AWS Region?
    A. A single data center located in a major city.
    B. A physical geographic area that contains multiple Availability Zones.
    C. A collection of Edge Locations used for content caching.
    D. A virtual network boundary for your AWS resources.

    Answer: B

83. What is the primary purpose of an AWS Edge Location?
    A. To host primary compute workloads like EC2 instances.
    B. To provide long-term archival storage for data.
    C. To deliver content closer to end-users to reduce latency.
    D. To provide synchronous data replication between Regions.

    Answer: C

84. Which Disaster Recovery (DR) strategy involves keeping a minimal, scaled-down version of a functional environment always running in a secondary Region?
    A. Backup and Restore
    B. Pilot Light
    C. Warm Standby
    D. Multi-Site (Active-Active)

    Answer: C

85. A European fintech company must comply with strict GDPR data residency requirements that mandate customer data must be stored within a specific country's legal jurisdiction. What is the most critical factor for them when setting up their AWS environment?
    A. Selecting the Region with the lowest pricing for S3 storage.
    B. Ensuring they have the maximum number of AZs available in their chosen continent.
    C. Selecting the specific AWS Region located within the required geographic/legal boundary.
    D. Using IAM policies to restrict global access to their data.

    Answer: C

86. Why is synchronous data replication typically limited to a Multi-AZ architecture rather than a Multi-Region architecture?
    A. Because Edge Locations do not support synchronous protocols.
    B. Because AWS does not allow data transfer between different Regions.
    C. Because the physical distance between Regions introduces latency that makes synchronous sync difficult.
    D. Because Regional services like Amazon RDS cannot operate in multiple AZs.

    Answer: C

87. Under the Shared Responsibility Model, which task is a customer's responsibility when using Infrastructure as a Service (IaaS) but becomes AWS's responsibility when using a managed Platform as a Service (PaaS)?
    A. Physical security of the data center
    B. Patching the guest operating system
    C. Maintaining the virtualization layer

D. Disposal of physical storage disks

Answer: B

88. A company wants to ensure its application remains operational even if an entire data center facility within a Region fails. Which architecture should they implement?

A. Single-AZ deployment with higher-capacity servers.

B. Multi-AZ deployment within the same Region.

C. Deploying the application in a single Edge Location.

D. Moving the entire workload to an on-premises Private Cloud

Answer: B

89. A company wants its application to remain operational if a data center in a Region fails. Which architecture best meets this requirement?

A. Deploy across multiple Regions

B. Deploy across multiple AZs

C. Deploy in one AZ with Auto Scaling

D. Use Edge Locations only

Answer: B

90. Which AWS infrastructure component provides the lowest latency access to users globally?

A. Regions

B. AZs

C. Edge Locations

D. VPC

Answer: C

91. What is the main difference between an Availability Zone (AZ) and a Region?

A. AZ spans multiple Regions

B. Region contains multiple AZs

C. AZ is global

D. Region is virtual only

Answer: B

92. Which scenario is the BEST use case for Multi-Region deployment?

A. Improve availability within a Region

B. Reduce AZ latency

C. Disaster recovery and global users

D. Reduce storage cost

Answer: C

93. A developer wants to deploy a web application without worrying about managing the underlying infrastructure, such as the operating system or server hardware. Which cloud computing model should they use?
    A. Infrastructure as a Service (IaaS)
    B. Software as a Service (SaaS)
    C. Platform as a Service (PaaS)
    D. On-premises

    Answer: C

94. Which AWS service is considered global?
    A. Amazon EC2
    B. Amazon RDS
    C. AWS IAM
    D. Amazon VPC

    Answer: C

95. Which AWS infrastructure component is primarily used to cache content closer to end-users to reduce latency and improve performance?
    A. Regions
    B. Availability Zones
    C. Edge Locations
    D. Virtual Private Clouds (VPC)

    Answer: C

96. Which of the following AWS services is considered a Global service?
    A. Amazon EC2
    B. Amazon RDS
    C. AWS IAM
    D. Amazon VPC

    Answer: C

97. A company is over-provisioning its on-premises resources to handle peak traffic, leading to high waste during off-peak hours. Which AWS Cloud advantage directly addresses this specific inefficiency?
    A. Go global in minute
    B. Stop guessing capacity
    C. Massive economies

D. Increase speed and agility

Answer: B

98. A retail company uses AWS to improve its business operations. To stay ahead of competitors, they launched a new Mobile Payment feature in just two days by instantly spinning up new development environments. During a holiday sale, their web server fleet automatically doubled in size to handle a massive traffic spike, then shrank back to its original size once the sale ended.

    Which cloud concepts are described in order ?

    A. A. 1. Elasticity; 2. Agility

    B. B. 1. Agility; 2. Elasticity

    C. C. 1. Scalability; 2. High Availability

    D. D. 1. Reliability; 2. Scalability

    Answer: B

99. A IT company currently operates with a fully Cloud-Native architecture on AWS. They are planning to transition to a Hybrid architecture by establishing an on-premises data center to handle their highly stable internal workloads.

    What is the Benefits the company gain from this transition ?

    A. A. Elimination of the need for an internal team to manage and maintain physical infrastructure.

    B. B. Establish Shared Responsibility model where AWS manages the physical security of the new on-premises data center.

    C. C. Don't need to guess about capacity by automatically scaling resources to match fluctuating traffic.

    D. D. Total physical control and customization over hardware and underlying infrastructure for specific compliance needs.

    Answer: D

100. A startup have a small software engineering team to write the application code and manage user data, but they do not want to spend time managing or patching the underlying operating systems, runtime environments, or server hardware. Which cloud computing model should they choose ?

    A. A. Infrastructure as a Service (IaaS), because it provides the team with full control over the operating system and runtime environment.

    B. B. Software as a Service (SaaS), because the cloud provider will manage everything including the application and data, eliminating the need for their engineering team.

    C. C. Platform as a Service (PaaS), because the provider manages the underlying infrastructure, operating system, and runtime.

    D. D. On-Premises, because it allows their software engineering team to have complete control over the physical data center.

    Answer: C

101. Which of the following is a key characteristic of cloud computing that allows users to automatically increase or decrease computing resources based on demand?
    A. A. Resource pooling
    B. B. Rapid elasticity
    C. C. Broad network access
    D. D. Measured service

    Answer: B

102. A company already has an on-premises data center but wants to integrate some workloads with AWS services while keeping sensitive data locally. Which deployment model should the company use?
    A. A. Public Cloud
    B. B. Private Cloud
    C. C. Hybrid Cloud
    D. D. Community Cloud

    Answer: C

103. A startup launches a new online course platform. During enrollment periods, the number of users increases dramatically, but outside the enrollment season traffic is very low. The company wants infrastructure that can automatically scale up during high demand and scale down when demand decreases Which cloud characteristic BEST addresses this requirement?
    A. A. Broad Network Access
    B. B. Rapid Elasticity
    C. C. Resource Pooling
    D. D. Measured Service

    Answer: B

104. A company wants to avoid large upfront costs for servers and data centers and prefers paying only for the resources they use. Which cloud advantage supports this business goal?
    A. A. Global reach
    B. B. Trade fixed expense for variable expense
    C. C. Resource pooling
    D. D. Broad network access

    Answer: B

105. Currently, your company (a FinTech company working with stock market data) has its servers (EC2) and databases (RDS) hosted in Tokyo. To ensure that your application does not go down if the entire Tokyo region experiences a natural disaster (such as an earthquake), while also reducing network latency for customers in London (Europe), which architecture should you choose?

A. A. Single AZ

B. B. Multi-AZ

C. C. Multi-Region

D. D. Edge-only

Answer: C

106. Which of the following is NOT a characteristic of an AWS Region?

A. A. It consists of multiple Availability Zones (AZs) located within the same geographic area.

B. B. The resources and infrastructure within a Region are fully isolated from other Regions.

C. C. It shares the same physical networking infrastructure with nearby Regions for risk redundancy.

D. D. The Availability Zones within a Region are connected through low-latency networks.

Answer: C

107. When building an Electronic Emergency Medical Records system on the AWS platform, the system stores extremely sensitive personal patient information. At the same time, because it is used in an emergency clinic, doctors require that patient medical records be retrieved almost instantly in order to make life-saving decisions. The project is also allocated a limited monthly budget for ongoing operations. Based on the key factors used when selecting an AWS Region, which of the following options represents the most appropriate priority order (from most important to least important) for the system architecture?

A. A. Cost Optimization → Service Availability → Latency Optimization → Compliance

B. B. Latency Optimization → Compliance → Cost Optimization → Service Availability

C. C. Compliance → Latency Optimization → Service Availability → Cost Optimization

D. D. Service Availability → Compliance → Latency Optimization → Cost Optimization

Answer: C

108. A company deploys its application across multiple Availability Zones (AZs) behind an Application Load Balancer (ALB). What does this primarily provide?

A. A. Lower storage cost

B. B. Global caching

C. C. High availability

D. D. Cross-region disaster recovery

Answer: C

109. Which statement best describes an Availability Zone?

A. A single data center globally shared

B. B. Multiple isolated data centers in one Region

C. C. A globally distributed caching system

D. D. A logical network boundary

Answer: B

110. Which architecture pattern provides the highest level of resilience?

A. A. Single AZ

B. B. Multi-AZ

C. C. Multi-Region

D. D. Edge-only

Answer: C

111. A company has a Linux server located in its office. Employees log in to the server using SSH to run code and process data. The server is manually administered by the IT department. What is the most accurate description of this system?

A. Private cloud

B. Public cloud

C. Traditional on-premise server

D. Hybrid cloud

Answer: C

112. An application is designed to run on EC2 instances across 2 different Availability Zones (Multi-AZ). However, both instances read/write data to a single EBS volume. Does this design truly provide High Availability (HA) for the system?

A. A. Yes, because the compute power is distributed across two independent AZs

B. B. No because the distance between AZs will increase data read/write latency

C. C. Yes, because AWS automatically replicates EBS volumes to all other AZs in a Region

D. D. No, because an EBS volume is a zonal resource, creating a 'Single Point of Failure'

Answer: D

113. Previously, an enterprise had to spend a large amount of money to purchase servers and networking equipment before deploying a system. When moving to the cloud, what type of cost does this expense become?

A. Capital expenditure (CAPEX)

B. Operational expenditure (OPEX)

C. Depreciation cost

D. Hardware ownership cost

Answer: B

114. Which tasks are AWS responsibilities according to the AWS Shared Responsibility Model?

A. Patching networking devices

B. Defining user password policies

C. Configuring security groups

D. Patching an EC2 instance operating system

Answer: A

115. Who is responsible for decommissioning underlying storage devices that reach the end of their useful life used to host data on AWS?

A. Customer

B. AWS

C. Account creator

D. Auditing team

Answer: B

116. Which of the following is a customer responsibility, according to the AWS Shared Responsibility Model?

A. Identity access management

B. Hard drive disposal

C. Data center hardware security

D. Availability zone security

Answer: A

117. A company runs a e-commerce web application on AWS. During a large sales event, the traffic increases by 10 times and the system automatically launches additional servers to handle the load. After the event ends, the extra servers are automatically terminated.

Which AWS characteristic is best represented in this scenario?

A. Resource pooling

B. Rapid elasticity

C. Measured service

D. Broad network access

Answer: B

118. A company is moving a legacy application to AWS. In the past, they had to purchase $50,000 worth of hardware every 3 years to ensure they had enough "headroom" for growth, yet 40$

A. Go global in minutes

B. Benefit from massive economies of scale

C. Stop guessing capacity

D. Increase speed and agility

Answer: C

119. A company is comparing the costs of running their own data center versus moving to AWS. They realize that because AWS serves hundreds of thousands of customers, AWS can purchase hardware at much lower costs than the company ever could. This leads to lower prices for the company. Which cloud advantage is this?

    A. Trade capital expense for variable expense

    B. Stop guessing capacity

    C. Increase speed and agility

    D. Benefit from massive economies of scale

    Answer: D

120. A development team wants to deploy a web application. They do not want to manage the operating system or the runtime, but they want to keep full control over the application's environment configuration and code versions. Which service model fits best?

    A. SaaS

    B. PaaS

    C. On-premise

    D. IaaS

    Answer: B

121. Which AWS Cloud feature enables users to have the ability to pay based on current needs, rather than projected needs?

    A. AWS Budgets

    B. Pay-as-you-go pricing

    C. Volume discounts

    D. Saving Plans

    Answer: B

122. Which AWS service acts as the global DNS service that directs users to different AWS resources?

    A. Amazon Cloudfront

    B. Amazon Route 53

    C. Elastic Load Balacing

    D. AWS Global Accelerator

    Answer: B

123. Which AWS infrastructure component is primarily designed to cache content close to users to reduce latency?

    A. Availability Zone

    B. Region

    C. Edge Location

D. Data Center

Answer: C

124. A company wants to design a highly available architecture within a single AWS region.

They deploy application servers across three Availability Zones.

However, all instances access a single shared file system hosted on an EC2 instance in one AZ.

Which risk still exists in this architecture?

A. Load balancing cannot work across AZs

B. A failure in the AZ hosting the file system can cause application failure

C. Route 53 will redirect traffic to another region automatically

D. Edge locations will cache the file system data

Answer: B

125. An engineering team claims that deploying infrastructure across multiple Availability Zones guarantees disaster recovery.

Which statement best challenges this assumption?

A. AZs share the same physical power grid

B. Route 53 does not support multi-AZ routing

C. Edge locations must be used for redundancy

D. Region-level failures are still possible

Answer: D

126. A company wants to reduce latency for global users while also ensuring regional disaster recovery.

Which architecture best satisfies both goals?

A. Deploy the application in one region with many AZs

B. Deploy in multiple regions and use latency-based routing with health checks

C. Use a single region and multiple edge locations

D. Use weighted routing across Availability Zones

Answer: B

127. A financial corporation requires its core system to have a disaster recovery (DR) capability with a near-zero Recovery Time Objective (RTO/Zero Downtime). The system must remain uninterrupted even if an extreme weather event takes down all data centers in a vast geographic area (Full Region Outage). They are willing to accept the highest level of cost and complexity. Which architectural design is the most accurate solution?

A. Deploy a Multi-AZ architecture within a single Region using Synchronous Replication.

B. Use a Pilot Light strategy across multiple Regions to scale up when needed.

C. Deploy Edge Locations with Amazon CloudFront and Route 53 to accelerate global content delivery.

D. Implement a Multi-Region architecture using an Active-Active model.

Answer: D

128. Which of the following global network services is primarily used by AWS at Edge Locations to cache static content closer to end-users, thereby reducing latency and offloading requests from the origin servers?

A. Amazon RDS

B. Amazon EC2

C. Amazon CloudFront

D. Amazon EBS

Answer: C

129. A retail business frequently hosts seasonal Flash Sale events causing sudden traffic spikes, but their systems remain completely idle for most of the remaining days of the year. From a financial perspective, which characteristic of Cloud Computing offers the greatest benefit for this business to migrate from an On-Premises infrastructure to AWS?

A. The ability to completely eliminate all Operational Expenses (OPEX).

B. The ability to leverage AWS Outposts to install physical servers directly in their offices.

C. The ability to increase infrastructure complexity to enhance customer data security.

D. The ability to trade fixed Capital Expenses (CAPEX) for Variable Expenses, combined with Elasticity.

Answer: D

130. What is the CORE difference between Elasticity and Scalability in Cloud Computing?

A. Elasticity handles long-term, sustainable growth; Scalability handles short-term, instantaneous fluctuations

B. Elasticity handles short-term, instantaneous fluctuations; Scalability handles long-term, sustainable growth.

C. Elasticity only applies to Auto Scaling Groups; Scalability applies to all services.

D. There is no difference; the two concepts are the same.

Answer: B

131. You are acting as a Solution Architect for an online retail corporation. During a briefing, the Customer Service department reported that the website frequently experiences high latency during the checkout process, leading to a high cart abandonment rate. Additionally, the Chief Financial Officer (CFO) requires the new solution to be cost-optimized and operationally simple.

The business has defined the following technical requirements:

1. Performance: Internal network latency must be extremely low (< 10ms) for real-time transaction processing. 2. Availability: If a data center experiences a hardware failure, the system must automatically fail over immediately without impacting the customer experience. 3. Scope: Currently, the customer base is primarily concentrated within a single country.

Which architecture would you propose as the MOST optimal solution?

A. Deploy a Multi-Region (Global Resilience) architecture to ensure the system remains operational even during a regional disaster.

B. Deploy a Single-Region, Multi-AZ architecture using an Active/Active configuration and Synchronous Replication.

C. Use a Hybrid Cloud model by placing payment servers at the corporate headquarters to reduce cloud operational costs.

D. Deploy the application across multiple Edge Locations to ensure fast access for customers from any geographic location.

Answer: B

132. Company X faces two simultaneous scenarios:

Scenario A: The number of users increases by 40Scenario B: Every Friday, traffic triples in two hours; the system automatically adds servers at 8 PM and removes them at 10 PM. Match the correct concepts:

A. A = Elasticity; B = Scalability

B. A = Scalability; B = Elasticity

C. Both A and B are Elasticity

D. Both A and B are Scalability

Answer: B

133. A company deploys its Amazon EC2 instances across 2 AZs within a Region. A load balancer automatically routes traffic to healthy instances. The database is replicated across the same two AZs. Which statement is the most correct one?

A. The architecture is Fault Tolerance because it can still operate when one AZ fails.

B. The architecture is not Highly Available, because it runs only in a single Region.

C. The architecture provides Global Resilience, because multiple Availability Zones are used.

D. The architecture provides High Availability, but not Fault Tolerance, because the system may experience brief downtime during failover.

Answer: D

134. A company needs to ensure its application can survive a catastrophic event that takes down an entire AWS Region. Which design strategy provides the highest level of resilience and the lowest Recovery Time Objective?

A. Deploying the application across multiple Availability Zones with an Application Load Balancer.

B. Implementing a Multi-Region Active-Active strategy with Route 53 for global traffic routing.

C. Setting up a Pilot Light environment in a different Availability Zone within the same Region.

D. Using CloudFront to cache content at Edge Locations to prevent regional service interruption

Answer: B

135. A company based in Vietnam wants to deploy a web application serving primarily Southeast Asian users. They need to ensure the application remains operational if a data center fails but have a limited budget and do not require regional disaster recovery. Which architecture should they choose: Multi-AZ or Multi-Region?

    A. Multi-Region, because deployment across multiple regions is necessary to better serve Southeast Asian users and ensure the fastest access speeds.

    B. Multi-AZ, because Multi-AZ is completely free while Multi-Region requires additional fees, so Multi-AZ is the most cost-effective choice.

    C. Multi-Region, because Multi-AZ is not secure enough; only Multi-Region can guarantee application uptime in case of failure.

    D. Multi-AZ, because this architecture protects against single-center data center failures, has low latency (<2ms between AZs), moderate cost, and low complexity, suitable for limited requirements and budgets.

    Answer: D

136. Which layer of the computing stack is the responsibility of the customer in Software as a Service (SaaS)

    A. Applications

    B. Data

    C. Operating System

    D. nan

    Answer: D

137. "Which of the following is NOT a characteristic of Amazon Elastic Compute Cloud (Amazon EC2)? C. Amazon EC2 can launch as many or as few virtual servers as needed. D. Amazon EC2 offers scalable computing."

    A. Amazon EC2 is considered a Serverless Web Service

    B. Amazon EC2 eliminates the need to invest in hardware upfront

    C. Amazon EC2 can launch as many or as few virtual servers as needed

    D. Amazon EC2 offers scalable computing

    Answer: A

138. Which classification is correct for Amazon Elastic Compute Cloud (Amazon EC2)?

    A. On Premise

    B. SaaS

    C. PaaS

    D. IaaS

    Answer: D

139. An e-commerce company increases the number of servers in its architecture to handle a steady increase in customer traffic over the years. Which concept does this represent?

    A. Elasticity

    B. Agility

    C. Fault tolerance

    D. Scalability

    Answer: D

140. Which design principles are enabled by the AWS Cloud to improve the operation of workloads?

    A. Minimize platform design

    B. Minimum viable product

    C. Customized hardware

    D. Remove single points of failure

    Answer: D

141. New department (in Da Nang) can access to a company's system set up in Ha Noi right after connecting the internet. Which answer described the feature advantage of cloud through this situation ?

    A. Global reach

    B. Agility

    C. Elastiscity

    D. Both A and B

    Answer: D

142. A development team can now provision hundreds of virtual servers in minutes, compared to the weeks it took to procure and deploy physical hardware. Which AWS Cloud advantage does this describe?

    A. Benefit from massive economies of scale

    B. Increase speed and agility

    C. Stop guessing capacity

    D. Go global in minutes

    Answer: B

143. A company uses AWS Cloud to help develop their workload from Viet Nam to another team in Uruguay. One of an advantage cloud computing have in this example is

    A. Increase speed and agility

    B. Benefit from massive economies of scale

    C. Go global in minute

    D. Trade fixed expense for variable expense

Answer: C

144. Small start up have just established which wants to use the newest AI technologies for business, so they decided to choose clound model for their system. Which answers below here are not the advatages of cloud model?
    A. Don't have to spend money on maintaining and running server
    B. Business concentration
    C. Fully control and customise your Software environment
    D. Change from CAPEX (Capital Expenditure) to OPEX (Operation expenditure)

Answer: C

145. In the On-Premises model, who is responsible for managing all layers from Networking to Applications?
    A. The cloud service provider
    B. A third-party outsourcing vendor
    C. The organization that owns the system
    D. AWS manages everything

Answer: C

146. Amazon EC2 is an example of which cloud service model?
    A. Software as a Service (SaaS)
    B. Platform as a Service (PaaS)
    C. Infrastructure as a Service (IaaS)
    D. Hybrid Cloud

Answer: C

147. A company is using Amazon EC2 to run their web applications. According to the Shared Responsibility Model, which of the following tasks is the customer's responsibility?
    A. Maintaining the physical network infrastructure of the data center.
    B. Replacing failed physical hard drives in the servers.
    C. Managing the virtualization software (hypervisor).
    D. Updating patches for the Guest Operating System (Guest OS) installed on the EC2 instances.

Answer: D

148. Which of the following is NOT one of the Five Characteristics of AWS?
    A. On-Demand Self-Service
    B. Rapid Elasticity
    C. Measured Service

D. Automatic Code Deployment

Answer: D

149. When a large financial organization requires full control over its infrastructure for strict security reasons but still wants to use virtualization technology, which deployment model should they choose?

A. SaaS (Software as a Service)

B. Hybrid Cloud

C. Public Cloud

D. Private Cloud

Answer: D

150. You ONLY want to manage Applications and Data. Which type of Cloud Computing model should you use?

A. On-premises

B. PaaS

C. IaaS

D. SaaS

Answer: B

151. A company is planning to run a global marketing application in the AWS Cloud. The application will feature videos that can be viewed by users. The company must ensure that all users can view these videos with low latency. Which AWS service should the company use to meet this requirement?

A. AWS Auto Scaling

B. Amazon Kinesis Video Streams

C. Elastic Load Balancing

D. Amazon CloudFront

Answer: D

152. Which of the following is NOT one of the Five Characteristics of Cloud Computing?

A. Rapid elasticity and scalability

B. Multi-tenancy and resource pooling

C. Dedicated Support Agent to help you deploy applications

D. On-demand service

Answer: C

153. What is the key difference between IaaS and PaaS in terms of user responsibility?

A. IaaS requires the user to manage from OS to Applications; PaaS only requires managing Applications and Data.

B. IaaS and PaaS have exactly the same scope of responsibility.

C. PaaS requires the user to manage more layers than IaaS.

D. IaaS only requires the user to manage Applications, while PaaS requires managing Networking as well.

Answer: A

154. Which pillar of the AWS Well-Architected Framework refers to the ability of a system to recover from infrastructure or service disruptions and dynamically acquire computing resources to meet demand?

A. Security

B. Reliability

C. Performance efficiency

D. Cost optimization

Answer: B

155. What is a Single Point of Failure (SPOF), and what is the recommended solution?

A. A system with too many redundant servers; the solution is to reduce the number of servers.

B. A single component whose failure causes the entire system to stop functioning; the solution is Loose Coupling with multiple servers.

C. A software bug that occurs only once; the solution is to update the software.

D. A situation where operational costs exceed the budget; the solution is to migrate to the cloud.

Answer: B

156. Which of the following is NOT an advantage of Cloud Computing?

A. Trade capital expense (CAPEX) for operational expense (OPEX)

B. Train your employee less

C. Go global in minutes

D. Stop spending money running and maintaining data centers

Answer: B

157. A company is planning to replace its physical on-premises compute servers with AWS serverless compute services. The company wants to be able to take advantage of advanced technologies quickly after the migration. Which pillar of the AWS Well-Architected Framework does this plan represent?

A. Security

B. Performance efficiency

C. Operational excellence

D. Reliability

Answer: B

158. Which of the following statements correctly compares Cloud and On-Premises?
    A. On-Premises has variable costs based on usage, while Cloud has fixed upfront costs.
    B. On-Premises is easier to scale than Cloud.
    C. Cloud offers faster implementation time and variable costs based on usage.
    D. On-Premises is preferred by small and mid-sized businesses.

Answer: C

159. Which of the following services has a global scope?
    A. EC2
    B. IAM
    C. Lambda
    D. Rekognition

Answer: B

160. While AWS has the capabilities to meet many different hosting needs and models, which cloud deployment model best describes AWS as a whole?
    A. Private
    B. Public
    C. Hybrid
    D. Community

Answer: B

161. Which deployment model is characterized by the deployment of resources on-premises, using virtualization and resource management tools?
    A. Public Cloud
    B. Private Cloud
    C. Hybrid Cloud
    D. Infrastructure as a Service (IaaS)

Answer: B

162. Your company wants to save money by no longer running applications themselves or paying software developers for custom application code. Which cloud service category would you be looking at offerings for?
    A. DaaS
    B. IaaS
    C. SaaS
    D. PaaS

Answer: C

163. A pharmaceutical research company operates a private data center that is approaching capacity. One of its main workloads is a genomics analysis platform used by internal research teams. The platform processes large datasets and interacts with several systems on the company's campus network. In recent years, the company has also begun participating in multi-institution studies that require temporary increases in compute capacity and secure data access for approved external collaborators. The company wants to support growth without unnecessarily redesigning all existing systems at once. Which option is MOST appropriate?

    A. Migrate the entire genomics platform and all related services to a public cloud provider.

    B. Use a hybrid architecture that keeps tightly coupled local components in the private environment while extending compute and selected access services to the cloud

    C. Expand the private data center by purchasing additional servers, storage, and networking equipment for all future growth.

    D. Rebuild the platform entirely around managed cloud-native services to eliminate dependence on the current environment.

    Answer: B

164. Which statement is true in relation to data stored within an AWS Region?

    A. Data is always replicated to another region

    B. Data is not replicated outside of a region unless you configure it

    C. Data is always automatically replicated to at least one other availability zone

    D. Data is automatically archived after 90 days

    Answer: B

165. AWS Elastic Beanstalk belongs to which cloud service model, and which AWS services does it automatically provision?

    A. IaaS — it only provides Amazon S3

    B. SaaS — it provides Amazon WorkDocs and WorkMail.

    C. PaaS — it takes Application Code as input and automatically provisions EC2 Instances, Auto Scaling Group, Elastic Load Balancer, Amazon RDS, and Amazon S3.

    D. Hybrid — it connects on-premises infrastructure to the cloud through AWS Outposts.

    Answer: C

166. What benefits of AWS Cloud allow an e-commerce website to automatically add servers to handle the surge in traffic on Black Friday, and then automatically reclaim resources when the sale ends?

    A. Massive economies of scale

    B. Rapid Elasticity

    C. Global Reach

    D. High Availability

Answer: B

167. A startup has a small team, no existing data center, seasonal traffic fluctuations, and needs rapid time-to-market. Which deployment model is most suitable?
    A. On-Premises — because startups need full control over infrastructure for data security.
    B. Cloud — because the startup meets key conditions for cloud adoption: small team, unpredictable workload, need for rapid deployment, no existing data center, and no strong ops team.
    C. Hybrid — because startups should combine both on-premises and cloud to optimize costs.
    D. On-Premises — because long-term cloud costs will always be higher than building a data center.

Answer: B

168. A company often faces the risk of resource waste due to over-provisioning or system crashes due to resource shortages during peak hours (under-provision) when operating an on-premises data center. What core benefits of AWS Cloud will directly help them overcome this limitation?
    A. Benefit from massive economies of scale
    B. Increase speed and agility
    C. Stop guessing capacity
    D. Go global in minutes

Answer: C

169. A large company has multiple departments. Each department has its own AWS account. Each department has purchased Amazon EC2 Reserved Instances. Some departments do not use all the Reserved Instances that they purchased, and other departments need more Reserved Instances than they purchased. The company needs to manage the AWS accounts for all the departments so that the departments can share the Reserved Instances. Which AWS service or tool should the company use to meet these requirements?
    A. AWS Systems Manager
    B. Cost Explorer
    C. AWS Trusted Advisor
    D. AWS Organizations

Answer: D

170. A large enterprise operates an internal ERP system serving 3,000 users with stable workloads, and also needs a course registration portal with seasonal traffic spikes during enrollment periods. Which deployment strategy is most appropriate?
    A. Migrate both systems to the cloud to simplify management.
    B. Keep both systems on-premises since the enterprise already has an existing infrastructure team.
    C. Keep the ERP system on-premises and deploy the registration portal on the cloud.

D. Migrate the ERP system to the cloud and keep the registration portal on-premises.

Answer: C

171. A user is moving a workload from a local data center to an architecture that is distributed between the local data center and the AWS Cloud. Which type of migration is this?

A. On-premises to cloud native

B. Hybrid to cloud native

C. On-premises to hybrid

D. Cloud native to hybrid

Answer: C

172. The director of a business decides to use Amazon WorkMail for internal email. They don't have their own software engineering team. With this service, AWS manages everything from infrastructure and operating systems to software and data (All). What cloud service model is this business using?

A. SaaS

B. IaaS

C. PaaS

D. On-Premises

Answer: A

173. Which component of the AWS global infrastructure is made up of one or more discrete data centers that have redundant power, networking, and connectivity?

A. AWS Region

B. Availability Zone

C. Edge location

D. AWS Outposts

Answer: B

174. What is AWS EC2

A. AWS EC2 is a serverless compute service

B. AWS EC2 is a virtual server in the AWS Cloud

C. AWS EC2 is a data center

D. AWS EC2 is a resource monitoring service

Answer: B

175. Which cloud service model provides the highest level of control over the operating system and installed applications?

A. SaaS (Software as a Service)

B. PaaS (Platform as a Service)

C. IaaS (Infrastructure as a Service)

D. FaaS (Function as a Service)

Answer: C

176. A software company is building a customer analytics platform. Its engineers want full control over the operating system, installed packages, runtime versions, background services, and patching schedule because they use several custom dependencies and need to tune the environment for performance. However, they do not want to manage physical servers, storage hardware, or network infrastructure. Which service model is the MOST appropriate?

A. Platform as a Service (PaaS), because the company mainly wants to deploy application code without managing the underlying environment

B. Infrastructure as a Service (IaaS), because the company wants control over the OS and software stack while the provider manages the hardware and networking

C. Software as a Service (SaaS), because the provider should manage the full application stack and infrastructure

D. On-premises deployment, because control over runtime and packages requires the company to manage all infrastructure itself

Answer: B

177. Which of these statements about Availability Zones is not true? (Select the best answer)

A. Availability Zones are designed for fault isolation

B. Availability Zones are made up of one or more data centers

C. Availability zones are connected to each other using high-speed private links

D. A data center can be used for more than one availability zone

Answer: A

178. A user is comparing purchase options for an application that runs on Amazon EC2 and Amazon RDS. The application cannot sustain any interruption. The application experiences a predictable amount of usage, including some seasonal spikes that last only a few weeks at a time. It is not possible to modify the application. Which purchase option meets these requirements MOST cost-effectively?

A. Review the AWS Marketplace and buy Partial Upfront Reserved Instances to cover the predicted and seasonal load.

B. Buy Reserved Instances for the predicted amount of usage throughout the year. Allow any seasonal usage to run on Spot Instances.

C. Buy Reserved Instances for the predicted amount of usage throughout the year. Allow any seasonal usage to run at an On-Demand rate.

D. Buy Reserved Instances to cover all potential usage that results from the seasonal usage.

Answer: C

179. Which statement best describes the difference between CAPEX and OPEX in cloud computing?

A. CAPEX refers to paying only for resources used, while OPEX requires upfront investment.

B. CAPEX requires large upfront investment in infrastructure, while OPEX allows paying for resources as they are consumed.

C. CAPEX and OPEX are identical cost models used in cloud computing.

D. OPEX requires purchasing physical servers before deployment.

Answer: B

180. Which tasks are the customer's responsibility, according to the AWS shared responsibility model?

A. Establish the global infrastructure.

B. Perform client-side data encryption.

C. Secure data centers.

D. Patch the operating system in a PaaS environment.

Answer: B

181. Which of the following AWS services is categorized as a "Global Service,"meaning its operations are not restricted to a specific geographical Region?

A. Amazon EC2 (Elastic Compute Cloud)

B. Amazon RDS (Relational Database Service)

C. AWS IAM (Identity and Access Management)

D. Amazon VPC (Virtual Private Cloud)

Answer: C

182. What are Edge Locations?

A. Physical hard drives

B. AWS Regions

C. Data centers that store petabytes of data

D. Data centers that deliver data fast to the users

Answer: D

183. A startup deploys its web application on a single Amazon EC2 instance in one Availability Zone. After gaining more users, the application sometimes becomes unavailable when the instance fails. The team wants to improve availability without significantly increasing operational complexity.

Which solution is the most appropriate?

A. Deploy multiple EC2 instances across different Availability Zones behind an Application Load Balancer.

B. Increase the size of the current EC2 instance to a larger instance type

C. Move the application to a different AWS Region

D. Store application files in Amazon S3 only

Answer: A

184. A company is comparing the costs of their on-premises data center and the AWS Cloud. During a Total Cost of Ownership (TCO) analysis, which expense would be considered a "Fixed Cost"(CAPEX) in the on-premises environment?
    A. Monthly electricity bills for server cooling.
    B. Pay-as-you-go charges for Amazon EC2 instances.
    C. Upfront purchase of physical server chassis and networking cables.
    D. Salaries for a cloud-managed service provider's support team.

Answer: C

185. What is the main purpose of an Availability Zone (AZ) in the AWS global infrastructure?
    A. To provide a global endpoint for all AWS services
    B. To isolate infrastructure failures within a region and improve high availability
    C. To connect AWS to on-premises data centers
    D. To provide edge caching for content delivery

Answer: B

186. What is true about Regions?
    A. They are the physical locations of your customers
    B. All regions are located in one specific geographic area
    C. A region is a physical location that has multiple availability zones
    D. A physical location is composed of multiple regions and availability zones

Answer: C

187. According to the classification of responsibility, if a company uses a Software as a Service (SaaS) application like Amazon WorkDocs, who is responsible for managing the data and the application's runtime environment?
    A. The customer is responsible for both data and runtime.
    B. AWS is responsible for both data and runtime.
    C. The customer manages the data, while AWS manages the runtime and infrastructure.
    D. An external third-party auditor manages both.

Answer: C

188. Which of the following is an advantage that users experience when they move on-premises workloads to the AWS Cloud?
    A. Elimination of expenses for running and maintaining data centers
    B. Price discounts that are identical to discounts from hardware providers
    C. Distribution of all operational controls to AWS

D. Elimination of operational expenses

Answer: A

189. A large company is interested in avoiding long-term contracts and moving from fixed costs to variable costs. What is the value proposition of AWS for this company?

A. Economies of scale

B. Pay-as-you-go pricing

C. Volume pricing discounts

D. Automated cost optimization

Answer: B

190. A startup deploys its application to multiple AWS Regions so users from different continents can access the service with low latency. Which AWS Cloud advantage does this demonstrate?

A. Global reach

B. Elasticity

C. Reliability

D. Resource pooling

Answer: A

191. Which of the following identifiers is an example of an AWS Region?

A. Edge-Location-Tokyo

B. aws-global-1

C. us-east-1a

D. us-east-1

Answer: D

192. A software engineer needs to set up a new testing environment over the weekend. They log into the AWS Management Console and provision three virtual servers within minutes, completely bypassing the need to contact an IT administrator or wait for human approval. Which characteristic of cloud computing does this represent?

A. Rapid Elasticity

B. On-Demand Self-Service

C. Measured Service

D. Resource Pooling

Answer: B

193. Which component does Amazon CloudFront use to cache content closer to end-users to reduce latency?

A. AWS Outposts

B. Availability Zones

C. Edge Locations

D. Regional Edge Caches

Answer: C

194. How are Availability Zones (AZs) within the same Region connected?

A. High-bandwidth satellite network

B. Private internal network with low latency (< 2 ms)

C. Only when backup is required

D. Through the public Internet

Answer: B

195. When choosing an AWS Region, what is the most important factor for Data Sovereignty?

A. Latency to end-users

B. Service availability

C. Compliance and Data Residency

D. Cost of services

Answer: C

196. Which service is used to efficiently route users to AWS resources globally?

A. Amazon Route 53

B. Amazon VPC

C. Amazon EC2

D. AWS Lambda

Answer: A

197. A company is looking for the most cost-effective disaster recovery strategy. They are willing to accept a recovery time of several hours and a potential loss of data from their last daily backup. Which strategy should they choose?

A. Active-Active

B. Warm Standby

C. Pilot Light

D. Backup Restore

Answer: D

198. Which component allows you to place compute and storage resources closer to users in large urban areas that do not have a full AWS Region?

A. AWS Local Zones

B. AWS Regions

C. Availability Zones

D. AWS Direct Connect

Answer: A

199. A Japanese company hosts their applications on Amazon EC2 instances in the Tokyo Region. The company has opened new branches in the United States, and the US users are complaining of high latency. What can the company do to reduce latency for the users in the US while minimizing costs?

A. Applying the Amazon Connect latency-based routing policy.

B. Registering a new US domain name to serve the users in the US.

C. Building a new data center in the US and implementing a hybrid model.

D. Deploying new Amazon EC2 instances in a Region located in the US.

Answer: D

200. An organization needs to implement a Disaster Recovery strategy where they maintain a scaled-down but fully functional version of their application environment in a different AWS Region. Which DR strategy are they using?

A. Backup and Restore

B. Pilot Light

C. Warm Standby

D. Multi-Site Active-Active

Answer: C

201. A healthcare portal must deliver medical images quickly to patients around the world. Which AWS infrastructure component helps deliver content closer to users to reduce latency?

A. Availability Zone

B. Region

C. Edge Location

D. VPC

Answer: C

202. Which of the following businesses is better suited for using On-premise infrastructure rather than Cloud computing?

A. A. A video streaming business with rapid, seasonal growth in viewership.

B. B. A startup that lacks the capital to invest in its own data center.

C. C. A commercial bank with a large, stable customer base that must strictly comply with data security regulations.

D. D. An e-commerce company that frequently runs high-traffic flash sales

Answer: C

203. Which component of the AWS global infrastructure is made up of one or more discrete data centers that have redundant power, networking, and connectivity?

A. AWS Region

B. Availability Zone

C. Edge location

D. AWS Outposts

Answer: B

204. The ability to horizontally scale Amazon EC2 instances based on demand is an example of which concept?

A. Economy of scale

B. High Availability

C. Disaster recovery

D. Elasticity

Answer: D

205. What does the "On-Demand Self-Service"characteristic of Cloud Computing mean?

A. Customers must call a support agent to request server provisioning.

B. Resources are self-provisioned according to demand, without requiring a human intermediary.

C. Providers automatically delete user data when it is not in use.

D. Users must assemble physical servers at their data center.

Answer: B

206. A company operates in a geographic area with high political instability. Which AWS deployment strategy should they choose to ensure maximum business continuity and resilience

A. A. Deploy the application across all Availability Zones within one Region

B. B. Deploy the application on a single On-premise server with a backup on AWS S3.

C. C. Deploy the application across multiple AWS Regions

D. D. Use a single Edge Location to host the entire application infrastructure

Answer: C

207. A hospital wants to run EC2 instances and RDS databases close to their AUS patients. Which statement is correct?

A. EC2 is a Global Service and can run anywhere without specifying a Region

B. RDS is a Regional Service and must be deployed in a specific Region

C. CloudFront requires deployment in a Region

D. IAM must be deployed in every Region

Answer: B

208. How do customers directly benefit from Amazon's "Massive Economies of Scale"?

A. They receive free hardware upgrades every three years.

B. They can use any third-party software license for free.

C. They gain the ability to scale up and down as needed.

D. They receive periodic price reductions as a result of Amazon's operational efficiencies.

Answer: D

209. When should a business prioritize choosing Platform as a Service (PaaS)?

A. A. When the business needs to rent raw virtual servers to install operating systems and configure networks from scratch.

B. B. When the business wants to provide complete software to end-users without any source code customization.

C. C. When the business wants to focus entirely on developing, running, and managing applications without worrying about maintaining the underlying infrastructure (OS, storage, networking).

D. D. When the business requires total control over physical hardware and data center cooling systems

Answer: C

210. One of the most important AWS best-practices to follow is the cloud architecture principle of elasticity. How does this principle improve your architecture's design?

A. By automatically scaling your on-premises resources based on changes in demand.

B. By automatically scaling your AWS resources using an Elastic Load Balancer.

C. By reducing interdependencies between application components wherever possible.

D. By automatically provisioning the required AWS resources based on changes in demand

Answer: D

211. In a High Availability architecture within an AWS Region, which of the following is the core physical characteristic that allows Availability Zones (AZs) to effectively perform Synchronous Data Replication?

A. Each AZ is located in a different country to ensure the highest level of geographic independence.

B. AZs are interconnected via a dedicated, high-bandwidth intra-region fiber network with ultra-low latency (under 1ms).

C. Each AZ shares the same power supply and cooling system to optimize operational costs.

D. Amazon Route 53 asynchronously coordinates traffic between AZs over the public Internet.

Answer: B

212. How is a Single Point of Failure (SPOF) defined in system design?

A. It is the only network connection point that makes the system run faster.

B. It is a single component, node, or process in a system that if it fails, the entire system stops working or crashes.

C. It is a feature that helps balance the load between servers.

D. It is a backup server that automatically turns on when an error occurs.

Answer: B

213. You ONLY want to manage Applications and Data. Which type of Cloud Computing model should you use?

A. On-Premises

B. Infrastructure as a Service (IaaS)

C. Software as a Service (SaaS)

D. Platform as a Service (PaaS)

Answer: D

214. Which cloud service model does Amazon EC2 belong to?

A. SaaS (Software as a Service).

B. PaaS (Platform as a Service).

C. IaaS (Infrastructure as a Service).

D. FaaS (Function as a Service).

Answer: C

215. What does On-Premises infrastructure mean?

A. Hosting IT systems on a public cloud provider

B. Hosting IT systems within an organization's own physical location

C. Renting software through the internet

D. Using third-party infrastructure managed by providers

Answer: B

216. Which component of the AWS Global Infrastructure consists of one or more discrete data centers, each with redundant power, networking, and connectivity, housed in separate facilities?

A. AWS Regions.

B. Edge Locations.

C. Availability Zones.

D. Private Networking.

Answer: C

217. Which Global Infrastructure identity is composed of one or more discrete data centers with redundant power, networking, and connectivity, and are used to deploy infrastructure?

A. Edge Locations

B. Availibility Zones

C. Regions

D. Local Zones

Answer: B

218. Which of the following is NOT one of the five key characteristics of cloud computing?
    A. On-Demand Self-Service
    B. Rapid Elasticity
    C. Resource Pooling
    D. Dedicated Physical Ownership

Answer: D

219. When comparing On-Premises and Cloud, which statement is TRUE regarding control and flexibility?
    A. On-Premises is more flexible than Cloud but offers less control.
    B. Cloud provides full hardware control to the user but has very low flexibility.
    C. On-Premises provides full user control but is less flexible, whereas Cloud has limited user control but is highly flexible.
    D. Both models offer the exact same level of control and flexibility.

Answer: C

220. A company is considering migrating its on-premises data centre to Amazon Web Services to reduce costs and improve scalability. Which of the following is a primary benefit of adopting the AWS Cloud, as outlined in the AWS Well-Architected Framework?
    A. Increased hardware maintenance responsibilities
    B. Fixed pricing for all compute resources
    C. High availability and elasticity
    D. Mandatory coding for all deployments

Answer: C

221. In the Software as a Service (SaaS) model, what is the user responsible for?
    A. Managing infrastructure and operating systems
    B. Managing runtime environment and applications
    C. Only using the software provided
    D. Managing hardware and networking

Answer: C

222. Which of the following is NOT one of the Five Characteristics of Cloud Computing?
    A. Rapid elasticity and scalability
    B. Multi-tenancy and resource pooling
    C. Dedicated Support Agent to help you deploy applications

D. On-demand self service

Answer: C

223. Which deployment model combines cloud resources with existing on-premises infrastructure?

A. Public Cloud

B. Private Cloud

C. Hybrid Cloud

D. SaaS

Answer: C

224. Which cloud characteristic allows resources to automatically scale up or down quickly when demand changes?

A. Resource Pooling

B. Measured Service

C. Rapid Elasticity

D. Broad Network Access

Answer: C

225. AWS are able to continue to reduce their pricing due to:

A. Pay-as-you go pricing

B. The AWS global infrastructure

C. Economies of scale

D. Reserved instance pricing

Answer: C

226. When performing a Total Cost of Ownership (TCO) analysis, which of the following expenses is typically associated with on-premises environments but is reduced or eliminated in the AWS Cloud?

A. Software development.

B. Power consumption and cooling for physical servers.

C. Project management.

D. Market research.

Answer: B

227. In Disaster Recovery (DR) strategies, which of the following characteristics distinguishes the "Pilot Light"model from the "Warm Standby"model?

A. Core resources (such as databases) are always running, but other components are only created after a disaster occurs.

B. Data is only periodically saved as backups, and no resources are running.

C. The entire system is fully replicated and runs in parallel in an Active-Active state.

D. A reduced version of the entire system is always running and ready to handle low traffic.

Answer: A

228. A financial organization wants to keep part of its system in an internal data center while using the cloud to expand resources when needed.

Which deployment model best fits this requirement?

A. Public Cloud

B. Private Cloud

C. Hybrid Cloud

D. Multi-Cloud

Answer: C

229. Which factor is most critical when a customer must choose an AWS Region to ensure compliance with data sovereignty and legal requirements?(such as GDPR in Europe)

A. Proximity to the customer's corporate office for on-site audits.

B. The specific geographic location where data will be stored and processed.

C. The number of Edge Locations available in that specific country.

D. Whether the application is presented in the local language.

Answer: B

230. If your business has a highly stable workload, runs continuously for 5 years with high server utilization, and already has an infrastructure operations team, what is the most cost-effective strategy?

A. Migrate everything to the Public Cloud immediately.

B. Continue using or investing in On-premises because CAPEX will be lower than OPEX in this long-term scenario.

C. Use the SaaS model for the entire system.

D. Hire more engineers to build a complex internal Cloud.

Answer: B

231. A company wants to achieve Disaster Recovery (DR) with "Zero Downtime". Based on the DR Strategy Ladder, which deployment pattern is this?

A. Backup and Restore

B. Pilot Light

C. Warm Standby

D. Multi-Site (Active-Active)

Answer: D

232. A company deploys a web application in a single region but uses two Availability Zones. What is the main purpose of this architecture?
    A. Reduce data storage costs
    B. Increase system fault tolerance if a data center fails
    C. Serve users across multiple continents faster
    D. Ensure data complies with national regulations

    Answer: B

233. In the physical infrastructure of AWS, how is each Availability Zone (AZ) designed to ensure fault isolation?
    A. It has independent power supplies, cooling systems, and networking infrastructure.
    B. They share one large data center, but it is divided into different rooms.
    C. Each AZ is located in a different country to ensure legal safety.
    D. They are connected to other AZs through the public internet to ensure independence.

    Answer: A

234. Which improves HA in microservices architecture?
    A. Monolithic app
    B. Loose coupling with ELB and independent AZ deployments
    C. Tight dependencies
    D. Single DB

    Answer: B

235. A fintech company must ensure that customer data is stored and processed within the territory of Europe to comply with legal regulations. Which of the following is the most important consideration when designing the AWS architecture?
    A. Use multiple Availability Zones
    B. Choose a region that meets legal requirements
    C. Use Edge Locations close to customers
    D. Increase the number of EC2 instances

    Answer: B

236. A university needs to build a course registration portal. The system has very low traffic most months but experiences massive spikes during the enrollment season. Which solution is most appropriate?
    A. Buy many physical servers and place them at the university just to handle the enrollment season.
    B. Deploy the system on the Cloud to easily scale resources according to the seasonal campaign and launch quickly.

C. Do nothing and accept that the system will crash when many students access it.

D. Buy pre-packaged software (SaaS) that does not support expansion.

Answer: B

237. A global website wants to reduce latency when users around the world load images and JavaScript files. Which AWS infrastructure component best supports this goal?

A. Availability Zones

B. Edge Locations

C. Virtual Private Cloud (VPC)

D. Amazon EC2

Answer: B

238. What is the main difference between Multi-AZ and Multi-Region deployments?

A. Multi-AZ is for caching global content, while Multi-Region is for compute workloads

B. Multi-AZ prevents data center failure impact (High Availability), while Multi-Region provides disaster recovery against regional failures

C. Multi-AZ uses asynchronous replication, while Multi-Region uses synchronous replication

D. Multi-AZ requires user traffic routing via DNS, while Multi-Region relies on simple Load Balancers

Answer: B

239. A company wants to keep sensitive customer data in its own infrastructure while using cloud resources for scalable computing workloads. Which deployment model is the best fit?

A. Public Cloud

B. Private Cloud

C. Hybrid Cloud

D. Multi-tenant Cloud

Answer: C

240. An e-commerce business experiences a massive spike in traffic during a holiday sale. The AWS Cloud automatically adds more servers to handle the traffic, and then removes them when the sale ends. Which cloud characteristic is this?

A. Agility

B. Global Reach

C. Elasticity

D. Reliability

Answer: C

241. An organization is moving from Infrastructure as a Service (IaaS) to Platform as a Service (PaaS) to host their web application. Which of the following tasks will the organization NO LONGER be responsible for after this migration?

A. A. Managing user access and identity (IAM).

B. B. Configuring the application code and business logic.

C. C. Patching and maintaining the underlying Operating System (OS).

D. D. Ensuring the security of the data stored within the application.

Answer: C

242. Your media company does not have a software engineering team but needs an online document collaboration tool and an email system for employees. Which model should you choose?

A. Buy On-premises servers and configure the email system manually.

B. Rent IaaS (like Amazon EC2) and configure the web server yourself.

C. Use the SaaS model (like Amazon WorkDocs, Amazon WorkMail) because you only subscribe to the software without needing an engineering team.

D. Rent PaaS to write your own source code for the email application.

Answer: C

243. Which of the following is NOT one of the six main advantages of cloud computing described by AWS?

A. Go global in minutes

B. Trade variable expense for fixed expense

C. Stop spending money running and maintaining data centers

D. Benefit from massive economies of scale

Answer: B

244. Under the Shared Responsibility Model, when an organization migrates to a Software as a Service (SaaS) solution, which of the following remains a primary responsibility of the customer?

A. A. Physical security of the data center.

B. B. Updating and patching the operating system.

C. C. Managing data access permissions and user identities.

D. D. Maintaining the underlying network infrastructure.

Answer: C

245. A company wants to ensure its database remains available even if a primary data center suffers a total power failure. Which AWS feature or service configuration best addresses this requirement with minimal manual intervention?

A. Creating a Read Replica in the same Availability Zone.

B. Configuring an Amazon RDS Multi-AZ deployment.

C. Manually taking snapshots every hour and storing them in S3.

D. Using an Application Load Balancer to distribute traffic to a single EC2 instance.

Answer: B

246. Which of the following does NOT belong to the AWS Cloud Computing models?
    A. Platform as a Service (PaaS)
    B. Infrastructure as a Service (IaaS)
    C. Software as a Service (SaaS)
    D. Networking as a Service (NaaS)

    Answer: D

247. A global media company wants to reduce "latency"for its users in South America who are accessing video content stored in a "Region"located in Northern Virginia, USA. Which component of the Cloud Global Infrastructure should they use to cache content closer to their users?
    A. A. Availability Zones (AZs)
    B. B. Edge Locations
    C. C. Regional Edge Caches
    D. D. Local Zones

    Answer: B

248. Which design principle improves High Availability (HA) in cloud architecture?
    A. Deploy all resources in a single Availability Zone
    B. Remove redundancy to reduce cost
    C. Use redundancy across multiple Availability Zones
    D. Store all data locally on a single server

    Answer: C

249. A company wants to reduce the physical compute footprint that developers use to run code. Which service would meet that need by enabling serverless architectures?
    A. Amazon Elastic Compute Cloud (Amazon EC2)
    B. AWS Lambda
    C. Amazon DynamoDB
    D. AWS CodeCommit

    Answer: B

250. Which of the following AWS services is classified as a Global Service rather than a Regional Service?
    A. Amazon EC2 (Virtual Servers)
    B. Amazon S3 (Object Storage)
    C. Amazon RDS (Relational Database Service)
    D. AWS IAM (Identity and Access Management)

    Answer: D

251. Which feature of the AWS Cloud will support an international company's requirement for low latency to all of its customers?
    A. Fault tolerance
    B. Global reach
    C. Pay-as-you-go pricing
    D. High availability

    Answer: B

252. Which statement best describes Global cloud services?
    A. They operate within a single Availability Zone
    B. They operate within a single Region only
    C. They are available and managed across multiple regions globally
    D. They only work with edge locations

    Answer: C

253. A company is designing a globally distributed application and needs to identify which AWS services are considered Global Services rather than Regional Services. Which of the following are Global Services?
    A. Amazon EC2
    B. Amazon RDS
    C. Amazon Route 53
    D. Amazon S3

    Answer: C

254. What is cloud computing?
    A. Storing files only on a USB
    B. Delivering IT resources over the internet with pay-as-you-go pricing
    C. Installing software only on a local computer
    D. Building your own physical server room

    Answer: B

255. Which of the following is an AWS Cloud architecture design principle?
    A. Implement single points of failure
    B. Implement loose coupling
    C. Implement monolithic design
    D. Implement vertical scaling

    Answer: B

256. Which of the following is a characteristic of cloud computing?

A. Fixed hardware capacity

B. Rapid elasticity (resources can scale quickly)

C. Only works without internet

D. Requires buying physical servers

Answer: B

257. In the Disaster Recovery Strategy Ladder, which pattern has the highest cost but achieves zero/near-zero downtime and high readiness?

A. Backup Restore

B. Warm Standby

C. Active-Active

D. Pilot Light

Answer: C

258. A user is planning to launch three EC2 instances behind a single Elastic Load Balancer. The deployment should be highly available

A. Launch the instances across multiple Availability Zones in a single AWS Region

B. Launch the instances as EC2 Spot Instances in the same AWS Region and the same Availability Zone

C. Launch the instances in multiple AWS Regions, and use Elastic IP addresses.

D. Launch the instances as EC2 Reserved Instances in the same AWS Region, but in different Availability Zones

Answer: A

259. Which of the following is an advantage that users experience when they move on-premises workloads to the AWS Cloud?

A. Elimination of expenses for running and maintaining data centers

B. Price discounts that are identical to discounts from hardware providers

C. Distribution of all operational controls to AWS

D. Elimination of operational expenses

Answer: A

260. Which of the following is a core characteristic of AWS Cloud Computing?

A. A. On-Demand Self-Service

B. B. Full Physical Hardware Control

C. C. Fixed Monthly Pricing

D. D. Static Infrastructure Capacity

Answer: A

261. AWS Managed Data Analytics services (like Amazon EMR or Athena) are generally categorized under which cloud delivery model?

    A. A. IaaS

    B. B. PaaS

    C. C. SaaS

    D. D. On - premises

    Answer: B

262. A company wants to set up a highly available workload in AWS with a disaster recovery plan that will allow the company to recover in case of a regional service interruption. Which configuration will meet these requirements?

    A. Run on two Availability Zones in one AWS Region, using the additional Availability Zones in the AWS Region for the disaster recovery site.

    B. Run on two Availability Zones in one AWS Region, using another AWS Region for the disaster recovery site.

    C. Run on two Availability Zones in one AWS Region, using a local AWS Region for the disaster recovery site.

    D. Run across two AWS Regions, using a third AWS Region for the disaster recovery site.

    Answer: B

263. A startup needs to keep sensitive data on-site due to strict local regulations but wants to use AWS for its web applications. Which deployment model should they use?

    A. A. Hybrid Cloud

    B. B. Public Cloud

    C. C. Private Cloud

    D. D. None of the above

    Answer: A

264. You are launching a game globally and need an AWS architecture that meets three requirements: 1. Database high availability (HA) against a single data center failure. 2. Low-latency delivery of static content for global users. 3. Centralized identity management for admins across all regions. Which architecture correctly meets all requirements?

    A. Multi-AZ database; Edge Locations to cache static content; Global service (like IAM) for identity management.

    B. Multi-Region database; Availability Zones (AZ) to distribute static content; Regional service for identity management.

    C. Single-AZ database; Edge Locations for backend logic; Global service for the database.

    D. Multi-AZ database; Edge Locations to cache static content; Regional service (like IAM) for identity management.

    Answer: A

265. A company no longer needs to purchase and maintain physical servers after migrating its workloads to AWS. Which cloud benefit does this demonstrate?
    A. Stop spending money running and maintaining data centers
    B. Increase speed and agility
    C. Global reach
    D. Elasticity

    Answer: A

266. A company expands its application to customers in Europe and Asia by deploying resources in multiple AWS Regions. Which cloud benefit does this demonstrate?
    A. Elasticity
    B. Global reach
    C. Massive economies of scale
    D. Pay-as-you-go pricing

    Answer: B

267. Each AWS Region is composed of multiple Availability Zones. Which of the following best describes what an Availability Zone is?
    A. A distinct data center with redundant power and networking within an AWS Region.
    B. An independent data center located on a different continent than the Region.
    C. A fault-isolated location within an AWS Region that has independent power, cooling, and physical security.
    D. A logical grouping of compute resources that can be accessed globally.

    Answer: C

268. Which AWS Cloud benefit allows organizations to convert upfront infrastructure costs into variable expenses?
    A. Trade CAPEX for OPEX
    B. High availability
    C. Elasticity
    D. Global reach

    Answer: A

269. AWS provides disaster recovery capability by allowing customers to deploy infrastructure into multiple
    A. Availability Zones
    B. Regions
    C. Edge Locations
    D. VPCs

Answer: B

270. A company needs to deploy an application with minimal latency for users across Europe, Asia, and North America. Which AWS feature should be used?
    A. Multi-AZ deployment across all Regions
    B. Multiple AWS Regions with CloudFront distribution
    C. Edge Locations only
    D. A single Region with Elastic IP addresses

Answer: B

271. A company has deployed an application in the Northern California (us-west-1) region. After examining traffic, management notices that about 30
    A. Increase the compute capacity in us-west-1
    B. Deploy application resources to an AWS Region closer to Asia and use Route 53 for geolocation routing
    C. Use CloudFront to cache all content in Edge Locations
    D. Replicate data across multiple Availability Zones in us-west-1

Answer: B

272. In a Platform as a Service (PaaS) model, which of the following layers is the customer responsible for managing?
    A. Operating System and Virtualization
    B. Applications and Data
    C. Runtime and Middleware
    D. Servers and Storage

Answer: B

273. What is the main difference between Edge Locations and AWS Regions?
    A. Edge Locations are used for compute resources; Regions are for storage only
    B. Edge Locations cache content and serve it with low latency; Regions contain full AWS services
    C. Edge Locations are physically larger than Regions
    D. There is no difference; they are the same thing

Answer: B

274. A development team can launch new infrastructure resources within minutes using the AWS Management Console or APIs. Which cloud advantage does this represent?
    A. Massive economies of scale
    B. Increased speed and agility

C. Data durability

D. Global infrastructure

Answer: B

275. CloudFront is primarily used to deliver content with low latency. Which AWS component does it leverage?

A. Availability Zones

B. AWS Regions

C. Edge Locations

D. VPCs

Answer: C

276. A company wants to run native AWS services, APIs, and tools locally in their own physical data center. Which service enables this hybrid cloud architecture?

A. Amazon Bedrock

B. AWS Elastic Beanstalk

C. AWS Outposts

D. Amazon RDS

Answer: C

277. A company is designing a disaster recovery strategy and wants the ability to instantly fail over to a secondary Region with near-zero data loss. Which approach is recommended?

A. Backup data manually to another Region every 24 hours

B. Implement Active-Active deployment across two Regions with real-time data replication

C. Keep a cold standby in another Region and restore from backups when needed

D. Use S3 cross-region replication only

Answer: B

278. Which disaster recovery approach requires the least RTO (Recovery Time Objective) and RPO (Recovery Point Objective)?

A. Backup and restore strategy

B. Pilot light approach

C. Warm standby approach

D. Active-Active approach (multi-region)

Answer: D

279. Which cloud computing service model allows users to control the operating system and applications while the cloud provider manages the physical infrastructure?

A. Software as a Service (SaaS)

B. Platform as a Service (PaaS)

C. Infrastructure as a Service (IaaS)

D. Network as a Service (NaaS)

Answer: C

280. A global e-commerce company wants to serve content to users with the lowest possible latency worldwide. Which combination of AWS services should they use?

A. Multiple Regions + Multi-AZ within each Region + CloudFront

B. Single Region + CloudFront

C. Multiple Regions only

D. Edge Locations only

Answer: A

281. Which AWS service enables organizations to intelligently route DNS queries to the optimal AWS Region based on the geographic location of users, health status of endpoints, and configurable routing policies?

A. Amazon Route 53

B. AWS Global Accelerator with anycast IP addresses

C. Amazon CloudFront distribution with origin failover configured

D. AWS Transit Gateway with inter-region peering connections

Answer: A

282. How does deploying resources across AWS Regions help organizations reduce their total cost of ownership when serving a globally distributed user base?

A. By reducing data transfer costs and latency charges through proximity to end users in each geographic market

B. By leveraging automated inter-Region replication to eliminate the need for backup storage and archival solutions across multiple locations

C. By reducing data transfer expenses through proximity to end users and enabling volume discounts on bandwidth consumption across Regions

D. By consolidating all workloads into a single Region with the lowest EC2 pricing tier to maximize compute savings

Answer: A

283. What benefit does AWS's distributed global infrastructure provide for organizations that need to maintain low-latency access for users in different continents?

A. Deployment of resources in a single strategic location that maximizes network backbone utilization to reduce round-trip time for global users

B. Deployment of resources closer to end users in multiple geographic locations to reduce network transmission time

C. Automatic replication of all data across every available Region to ensure users can access information from the nearest geographic location

D. Centralized routing through a single primary Region that optimizes bandwidth allocation and reduces network congestion for international users

Answer: B

284. Which deployment model requires fixed upfront costs and assigns the responsibility of system updates entirely to the enterprise's internal technical team?

A. Public Cloud

B. Hybrid Cloud

C. Serverless

D. On-Premises

Answer: D

285. In which cloud model do users only use the software while the provider manages everything else?

A. Infrastructure as a Service (IaaS)

B. Platform as a Service (PaaS)

C. Software as a Service (SaaS)

D. On-Premises

Answer: C

286. What is the term for a single component, node, or process in a system that, if it experiences an outage, causes the entire system to stop functioning or crash?

A. Single Point of Failure (SPOF)

B. Loose Coupling

C. Isolated Data Center

D. Monolithic Architecture

Answer: A

287. What is a disadvantage of on-premises infrastructure?

A. No control over infrastructure

B. Low upfront cost

C. High upfront cost for hardware and maintenance

D. Unlimited scalability

Answer: C

288. Which of the following is a cloud deployment model?

A. Private Cloud

B. Physical Cloud

C. Hardware Cloud

D. Network Cloud

Answer: A

289. Which deployment strategy do highly agile tech companies like Netflix, Spotify, and Airbnb primarily utilize to achieve massive economies of scale and global reach?

A. Pure On-Premises

B. Hybrid Cloud

C. Public Cloud

D. Isolated Data Centers

Answer: C

290. Which AWS Cloud advantage is demonstrated when a company can replace upfront fixed capital expenses with variable operational expenses?

A. High availability

B. Elasticity

C. Agility

D. Economies of scale

Answer: C

291. In a global disaster recovery architecture, to achieve "Traffic Routed Globally with Seamless Failover"when an AWS Region experiences a large-scale disaster, the system design must incorporate which of the following core mechanisms?

A. Utilizing Application Load Balancers (ALBs) to reroute global traffic and establishing an intra-region fiber network to replicate data synchronously between continents.

B. Grouping Regions into a single interdependent network to allow traffic to automatically spill over from one Region to another, while using local High Availability (HA) mechanisms to detect failures.

C. Use a Global Traffic Router (e.g., Route 53) to automatically detect errors and redirect users, combined with Asynchronous Global Data Replication between isolated Regions.

D. Configure a redundant Availability Zone (AZ) as a global switching point and require continuous data synchronization to ensure no latency across the global network.

Answer: C

292. An online gaming company wants to ensure that its application can survive a failure of a single data center with minimal downtime. Additionally, they want a separate plan to recover their entire operations in a different geographic area if a catastrophic regional event occurs. What is the most cost-effective architecture to meet both goals?

A. Deploy a Single-AZ architecture with frequent backups to S3

B. Use a Multi-AZ deployment for High Availability and a Multi-Region strategy for Disaster Recovery

C. Deploy the application solely in Edge Locations to ensure global reach

D. Use Multi-Region deployment for both goals as it covers all failure scenarios

Answer: B

293. In an Amazon RDS Multi-AZ deployment, what happens to the "Standby Instance"during normal operations and when a failure occurs on the "Primary Instance"?

A. The Standby Instance receives asynchronous data and automatically serves Read traffic when the Primary is overloaded

B. The Standby Instance is synchronously replicated and is promoted to the Primary Instance when a failure occurs

C. Users must manually reconfigure their connection strings to point to the Standby Instance during a failure

D. The Standby Instance is only created after the Primary Instance has completely stopped functioning

Answer: B

294. What is the primary benefit of deploying applications across multiple Availability Zones?

A. Increased application latency for better performance

B. High availability and fault tolerance

C. Reduced AWS pricing for the same workload

D. Automatic database replication without additional configuration

Answer: B

295. A company wants to host a legacy application in its corporate data center but use Amazon EC2 for its modern web servers. Which deployment model does this represent?

A. Public Cloud

B. Private Cloud

C. Hybrid Cloud

D. Multi-cloud

Answer: C

296. Which of the following services is designed with native fault tolerance, storing objects redundantly across multiple Availability Zones?

A. Amazon Redshift

B. AWS Snowball

C. Amazon RDS (when Multi-AZ enabled)

D. Amazon S3

Answer: D

297. A company's database administrator is concerned that a natural disaster could impact the local data center where their application is hosted. Which AWS infrastructure design would provide the HIGHEST level of protection against a large-scale regional disaster?
    A. Deploying the application across multiple Availability Zones in one Region.
    B. Deploying the application across multiple Edge Locations.
    C. Deploying the application across multiple AWS Regions
    D. Using a Hybrid Cloud model with an on-premises backup.

    Answer: C

298. A video processing application requires ultra-low latency for delivering static content to global end-users and needs centralized access management for its operations team worldwide. Which pair of services correctly meets these requirements based on their scope (Global/Regional)?
    A. EC2 (Global) to manage servers and CloudFront (Regional) to deliver content
    B. S3 (Regional) to store videos and Route 53 (Regional) to route users
    C. CloudFront (Global) to deliver content via Edge Locations and IAM (Global) to manage access
    D. VPC (Global) to set up the network and RDS (Global) to store video information

    Answer: C

299. An enterprise is currently operating an on-premises system and struggles to forecast resource demand. This leads to expensive idle resources during off-peak hours and system crashes during peak traffic. Which characteristic of Cloud Computing best addresses this issue while optimizing costs?
    A. High Availability
    B. Security
    C. Global Reach
    D. Elasticity

    Answer: D

300. Under the AWS Shared Responsibility Model, if a customer's Amazon EC2 instance is hacked because the guest operating system was not updated with the latest security patches, who is responsible?
    A. AWS, because they own the physical hardware.
    B. The Customer, because they are responsible for patching the guest OS.
    C. Both AWS and the Customer share the responsibility for OS patching.
    D. The Operating System vendor (e.g., Microsoft or Linux provider).

    Answer: B

301. When using an Infrastructure as a Service (IaaS) provider, which specific layers are managed strictly by the cloud provider rather than the customer?

A. Applications, Data, and Runtime

B. Operating System, Middleware, and Runtime

C. Servers, Storage, Networking, and Virtualization

D. Applications, OS, and Virtualization

Answer: C

302. A company wants to move to AWS to reduce their Total Cost of Ownership (TCO). Which of the following is an example of moving from Capital Expenditure (CAPEX) to Operational Expenditure (OPEX)?

A. Signing a 10-year lease for a new data center facility.

B. Buying high-end physical servers and networking gear upfront.

C. Paying only for the compute power used each month.

D. Hiring a large team of hardware maintenance engineers.

Answer: C

303. Startup wants to deploy an application without the burden of managing servers, operating systems, or underlying software stacks. They want to focus entirely on writing code and deploying their application. Which cloud service model is the most suitable?

A. Infrastructure as a Service (IaaS)

B. Platform as a Service (PaaS)

C. Software as a Service (SaaS)

D. On-premises

Answer: B

304. Which of the following components is primarily used to cache data and reduce latency for end users worldwide?

A. Availability Zone

B. AWS Region

C. Elastic Load Balancer

D. Edge Location

Answer: D

305. A company wants to ensure its critical application remains accessible even if an entire AWS Region experiences a catastrophic failure. Which architectural strategy should they adopt?

A. Multi-AZ deployment within a single Region.

B. Deploying multiple Elastic Load Balancers in one AZ.

C. Using Edge Locations to host the entire application backend.

D. Multi-Region active-active deployment.

Answer: D

306. Which Disaster Recovery strategy provides the lowest Recovery Time Objective (RTO) and Recovery Point Objective (RPO)?
    A. Pilot Light
    B. Multi-site Active/Active
    C. Warm Standby
    D. Backup and Restore

    Answer: B

307. What is one of the condition to prioritize to use Cloud rather than On-premise
    A. Stable workload
    B. Available operating team
    C. Seasonal workload
    D. High Internal Traffic

    Answer: C

308. Why does Multi-Az matter
    A. Trafficking automatically to healthy Az
    B. Preventing global downtime during a major regional failure
    C. Data replication around the globe allows rapid recovery
    D. Multi-Region state consistency

    Answer: A

309. Which of the following is not a primary drawback of Multi-AZ:
    A. Region outage risk
    B. Higher cost due to duplicated resources, inter-region data transfer
    C. Higher complexity for certain configurations
    D. Designed to prevent single data center failure from impacting service availability

    Answer: B

310. Which one is not an advantage of Outpost service
    A. Data Confidentiality
    B. Low Latency
    C. Cost
    D. Data policy compliance

    Answer: C

311. Which of the following best describes horizontal scaling in cloud computing?
    A. Increasing CPU and RAM of an existing server

B. Adding more servers to distribute workload

C. Moving workloads to a faster network

D. Using virtualization to divide a server

Answer: B

312. Which design principle suggests that you should break a system into small, independent components so that a failure in one does not cause the entire system to crash?

A. Vertical Scaling

B. Tight Coupling

C. Loose Coupling

D. Monolithic Architecture

Answer: C

313. A developer wants to use a service that allows them to upload their code, and the service handles the deployment, capacity provisioning, and load balancing automatically. They want to maintain some control over the underlying server settings if needed. Which service should they use?

A. Amazon EC2 (IaaS)

B. AWS Lambda (Serverless)

C. AWS Elastic Beanstalk (PaaS)

D. AWS CloudFormation

Answer: C

314. A small startup does not know how many users will visit their new app. They do not want to buy too many servers and waste money, or buy too few and have a slow app. Which advantage of cloud computing helps solve this problem?

A. Go global in minutes

B. Increase speed and agility

C. Stop guessing capacity and automatically scale resources up or down

D. Benefit from massive economies of scale

Answer: C

315. The AWS Well-Architected Framework helps you build good systems in the cloud. It is based on a set of pillars. How many pillars does the AWS Well-Architected Framework have?

A. 4

B. 5

C. 6

D. 7

Answer: C

316. A company currently pays large upfront costs (CapEx) to buy physical servers. They want to move to a model where they pay monthly based on actual usage (OpEx). Which advantage of cloud computing describes this change?

    A. Benefit from massive economies of scale

    B. Increase speed and agility

    C. Trade fixed expense for variable expense

    D. Stop spending money running and maintaining data centers

    Answer: C

317. What is the difference between "High Availability"and "Fault Tolerance"?

    A. High Availability means the system is always up; Fault Tolerance means the system can scale.

    B. High Availability ensures minimal downtime; Fault Tolerance ensures zero downtime or data loss by having redundant components.

    C. High Availability is a customer responsibility; Fault Tolerance is an AWS responsibility.

    D. They are the same thing.

    Answer: B

318. A fintech startup is legally required to ensure that all customer financial data physically remains within the borders of a specific country to comply with government regulations. Which AWS infrastructure concept is primarily used to address this data residency requirement?

    A. Edge Locations

    B. AWS Regions

    C. High Availability

    D. Amazon Route 53

    Answer: B

319. How are Availability Zones physically and logically connected to one another within a single AWS Region?

    A. Through the public internet using encrypted VPN tunnels

    B. Through distributed global Edge Locations

    C. With high-bandwidth, low-latency network links

    D. Using isolated, asynchronous satellite connections

    Answer: C

320. A company maintains a scaled-down but fully functional version of their production environment running at all times in an alternate location for disaster preparedness. Which Disaster Recovery (DR) approach does this describe?

    A. Backup and Restore

B. Pilot Light

C. Warm Standby

D. Multi-Site (Active-Active)

Answer: C

321. A development team is deploying a new application and needs to configure user permissions as well as launch virtual servers. Regarding how AWS services are scoped, which of the following is true?

A. Both user permissions and virtual servers must be configured separately for every single location

B. User permissions are managed centrally on a global scale, while virtual servers must be launched in specific geographic boundaries

C. Virtual servers are managed globally, while user permissions are strictly restricted to a single data center

D. Both user permissions and virtual servers operate globally and do not require location selection

Answer: B

322. A global media company wants to reduce latency for users accessing its website worldwide. Which AWS service is most appropriate?

A. Amazon VPC

B. AWS Global Accelerator

C. Amazon CloudFront

D. Amazon EC2 Auto Scaling

Answer: C

323. Which of the following statements is NOT true?

A. Multi-AZ deployment is a common best practice for improving availability of production workloads within a Region.

B. Multi-Region deployment is used only for global applications and is not relevant to disaster recovery planning.

C. RTO, RPO, business continuity, and cost tolerance are important factors in selecting a disaster recovery strategy

D. Multi-Region architectures usually provide stronger resilience than Multi-AZ architectures, but with greater cost and complexity.

Answer: B

324. A DevOps engineer launches an AWS virtual server, installs Ubuntu Linux, configures Python runtime, and deploys a web application. Which cloud service model does this represent?

A. SaaS

B. PaaS

C. IaaS

D. On-Premises

Answer: C

325. In IaaS, the responsibility boundary between the customer and AWS provider falls between which two layers?

A. Applications and Data

B. Runtime and Middleware

C. OS and Virtualization

D. Servers and Storage

Answer: C

326. A startup needs to launch a web server in 5 minutes, scale from 2 to 50 servers during a flash sale, then scale back down immediately after. Which AWS service and feature best supports this need?

A. On-Premises servers with manual scaling

B. Amazon S3 with static hosting

C. Amazon EC2 with Elastic scaling

D. AWS Lambda with fixed capacity

Answer: C

327. According to the IaaS responsibility model, which of the following is the customer's responsibility, NOT AWS's?

A. Managing physical servers in the data center

B. Maintaining network cables and routers

C. Providing and maintaining the hypervisor

D. Installing and patching the operating system

Answer: D

328. A company is comparing cloud service models. They want a solution where they only manage their application code and data, while the cloud provider handles everything else including the runtime environment and operating system.

A. On-Premises

B. IaaS

C. PaaS

D. Hybrid Cloud

Answer: C

329. Which statement is TRUE regarding the relationship between Regions and Availability Zones?

A. A Region consists of at least two Availability Zones.

B. Availability Zones are connected to each other via the public internet.

C. Each Region contains multiple, isolated, and physically separate Availability Zones.

D. An Availability Zone can span across multiple Regions.

Answer: C

330. Which of the following is FALSE?

A. Cached content is delivered to users with higher latency when cache hits occur.

B. Leveraging edge use cases can optimize performance, increase availability, and enhance security for globally distributed applications.

C. In an edge model using a CDN, origin content is fetched only when needed, such as on a cache miss or revalidation.

D. Edge locations are AWS sites positioned close to users to reduce latency.

Answer: A

331. "1.An ecommerce company has migrated its IT infrastructure from an on-premises data center to the AWS Cloud. Which cost is the company's direct responsibility?

A. Cost of application software licenses

B. Cost of power for the AWS servers

C. Cost of the hardware infrastructure on AWS

D. Cost of physical security for the AWS data center

Answer: A

332. Which option is a physical location of the AWS global infrastructure?

A. AWS DataSync

B. Amazon Connect

C. AWS Region

D. AWS Organizations

Answer: C

333. An AI startup is training deep learning models with continuously changing data and needs to experiment with different types of hardware (A100, H100 GPUs) in a short period of time. Based on the characteristics of cloud computing, why should they choose the cloud instead of building their own internal GPU farm?

A. Because cloud computing provides full control over both the physical and virtualization layers for engineering teams.

B. To optimize CAPEX costs by making long-term investments in hardware.

C. To ensure that data is always stored on-premise in order to comply with the strictest security regulations.

D. To leverage Agility and Stop Guessing Capacity so they can instantly change hardware configurations according to their experimentation needs.

Answer: D

334. A large financial corporation has strict regulations requiring that sensitive data must not leave national borders, but it still wants to use advanced AI services from AWS to analyze data on-site. Which architecture solution is the most optimal?

A. Build a private cloud that is completely isolated from the Internet.

B. Migrate the entire system to the public cloud and encrypt the data.

C. Use AWS Outposts to run AWS services directly within their internal data center.

D. Use a SaaS model for all financial applications.

Answer: C

335. Which statement is NOT typically considered a cloud computing advantage?

A. Rapid elasticity allows capacity to expand and shrink with workload demand.

B. Measured service supports usage visibility and pay-for-use billing

C. Cloud computing is stronger for full control, customization and stable workloads - these are all situations where on-premise is stronger

D. Broad network access allows services to be consumed from multiple locations and device types.

Answer: C

336. Which section that we develop our-self in IaaS structure ?

A. Applications, Data, Runtime, Middleware, OS, Servers

B. Applications, Data

C. Applications, Data, Runtime, Middleware, OS

D. Applications, Data, Runtime, Middleware, OS, Servers, Storage, Networking

Answer: C

337. A Solutions Architect is designing a web application that will run on a fleet of Amazon EC2 instances. To ensure the application remains available even if some instances fail, the Architect needs a service that can automatically detect unhealthy instances and redirect traffic to the healthy ones.

Which AWS service should the Architect use to meet this requirement?

A. AWS Step Functions

B. Amazon Route 53

C. Elastic Load Balancing (ELB)

D. Amazon CloudFront

Answer: C

338. The marketing department of a technology company lacks expertise in coding or application maintenance. However, the company wants them to leverage cloud computing to align with the organization's existing technology stack. Which cloud computing model is most suitable for them?

    A. Infrastructure as a Service (IaaS).

    B. Networking as a Service (NaaS).

    C. Platform as a Service (PaaS).

    D. Software as a Service (SaaS).

    Answer: D

339. Which of the following are benefits of using a Load Balancer in a Cloud workload? (Choose all that apply)

    A. Improved Performance and Reduced Latency

    B. Enhanced Security (SSL Offloading, IP Masking) and Seamless Scalability

    C. High Availability through Health Checks

    D. All

    Answer: D

340. An insurance company want to scale globally while ensuring strict data compliance for customer information. They also require low latency and rapid deployment for their web applications Which infrastructure model is the best fit for this company?

    A. On-premises

    B. Hybrid Cloud

    C. Public Cloud

    D. All of them

    Answer: B

341. The 'Stop guessing capacity' benefit of the Cloud helps businesses avoid the 'Under-provisioning' scenario. What is the main consequence of 'Under-provisioning'?

    A. Increased time-to-market for application deployments.

    B. Increased time-to-market for application deployments.

    C. The system becomes overloaded, leading to degraded performance or service downtime when traffic spikes.

    D. Having to hire more data center operations staff.

    Answer: C